



US 20250283104A1

(19) **United States**

(12) **Patent Application Publication**
Stacey et al.

(10) **Pub. No.: US 2025/0283104 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **METHODS AND COMPOSITIONS FOR
INCREASING AMINO ACID AND PROTEIN
CONTENT IN PLANTS**

C07K 14/415 (2006.01)

C12N 9/10 (2006.01)

C12N 9/22 (2006.01)

C12Q 1/6895 (2018.01)

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(52) **U.S. Cl.**

CPC *C12N 15/8251* (2013.01); *A23K 10/30*

(2016.05); *C07K 14/415* (2013.01); *C12N*

9/1085 (2013.01); *C12N 9/222* (2025.05);

C12N 15/8201 (2013.01); *C12Q 1/6895*

(2013.01); *C12N 23/10/20* (2017.05); *C12Q*

2600/13 (2013.01); *C12Y 205/01048* (2013.01)

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(21) Appl. No.: **18/861,126**

(22) PCT Filed: **May 2, 2023**

(86) PCT No.: **PCT/US2023/066484**

§ 371 (c)(1),

(2) Date: **Oct. 28, 2024**

Related U.S. Application Data

(60) Provisional application No. 63/363,988, filed on May
2, 2022.

Publication Classification

(51) **Int. Cl.**

C12N 15/82 (2006.01)

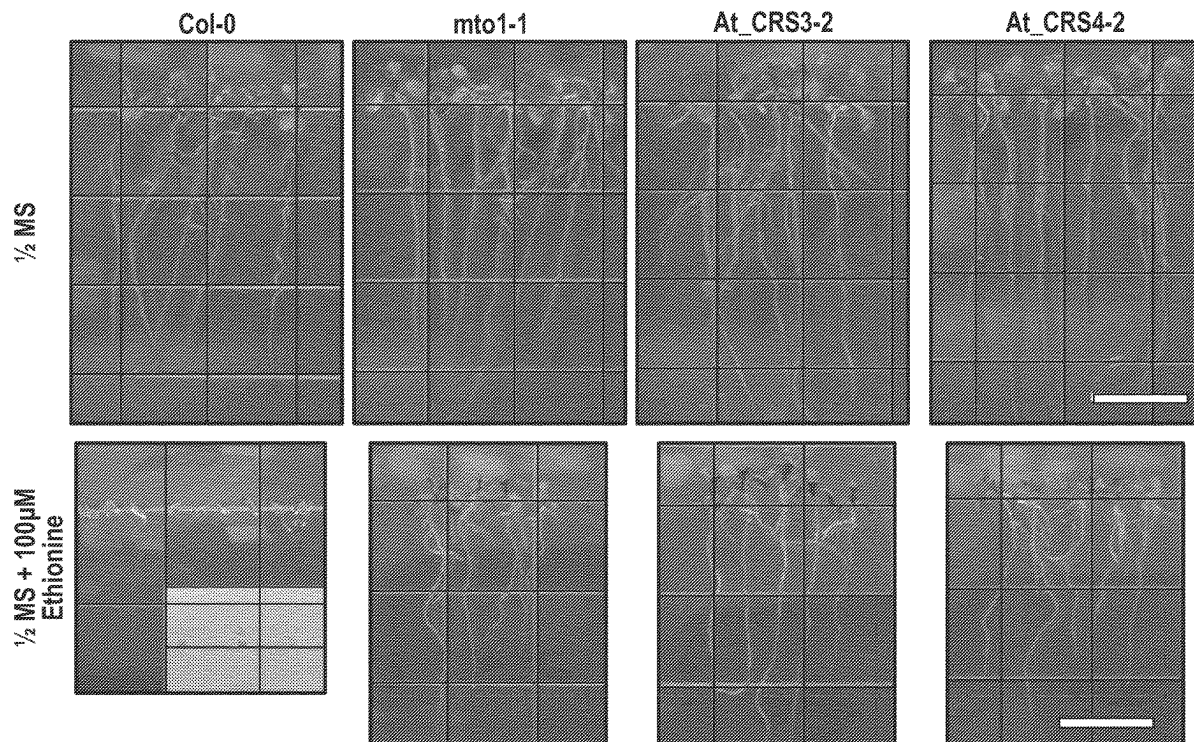
A23K 10/30 (2016.01)

(57)

ABSTRACT

Plant seeds with increased protein and methionine content and having a modified expression of a cystathionine γ -synthase (CGS) polypeptide, modified expression of a low methionine content (< about 1.0% methionine) storage protein polypeptide, or modified expression of both are provided. Methods for modifying expression of CGS polypeptides and polynucleotides and low methionine content seed storage polypeptides and polynucleotides include genome editing to modify the MTO1 regulatory region of CGS to create feedback insensitive methionine production and low methionine content polypeptides to create proteome rebalancing toward high methionine content storage protein production, and transformation with recombinant DNA constructs to enhance or suppress expression are disclosed herein.

Specification includes a Sequence Listing.



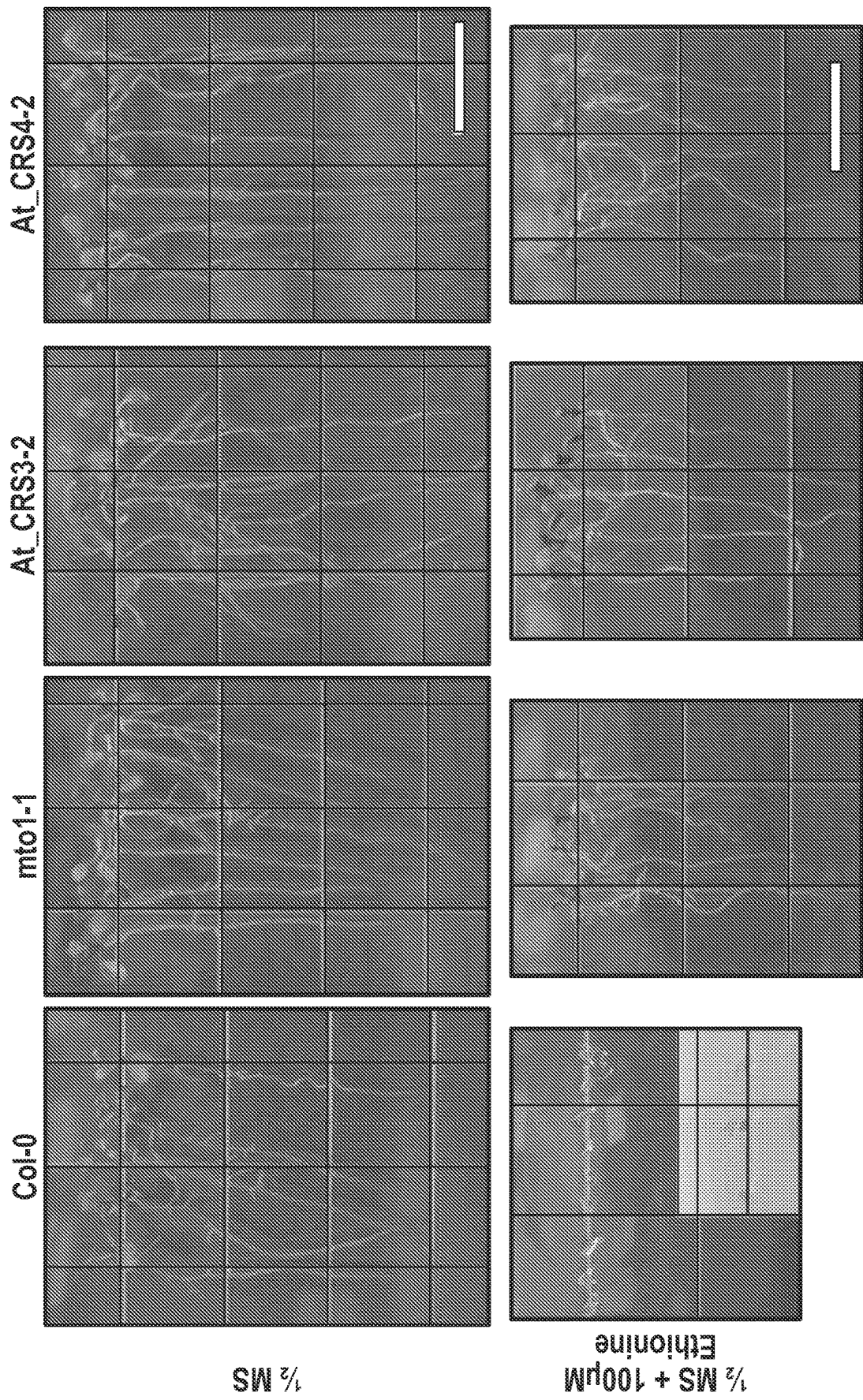


FIG. 1A

ATCGS1	AGA AAC TGT AGC AAC ATC GGT GTT GCA CAG ATC GTG GCG GCT AAG TGG TCC AAC AAC CCA TCC	
	R N C S N I G V A Q I V A A K W S N N P S	
MOTO1-1	AGA AAC TGT AGC AAC ATC AGT GTT GCA CAG ATC GTG GCG GCT AAG TGG TCC AAC AAC CCA TCC	
	R N C S N I S V A Q I V A A K W S N N P S	
AT_CRS3-2	AGA AAC TGT AGC ---	△ -39BPS
AT_CRS4-2	AGA AAC TGT AGC ---	△ -42BPS
	R N C S	N N P S

FIG. 1B

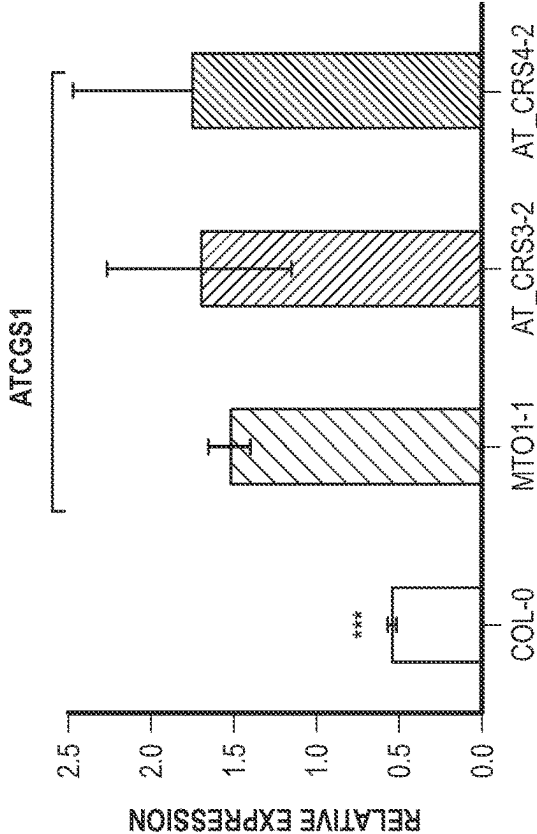
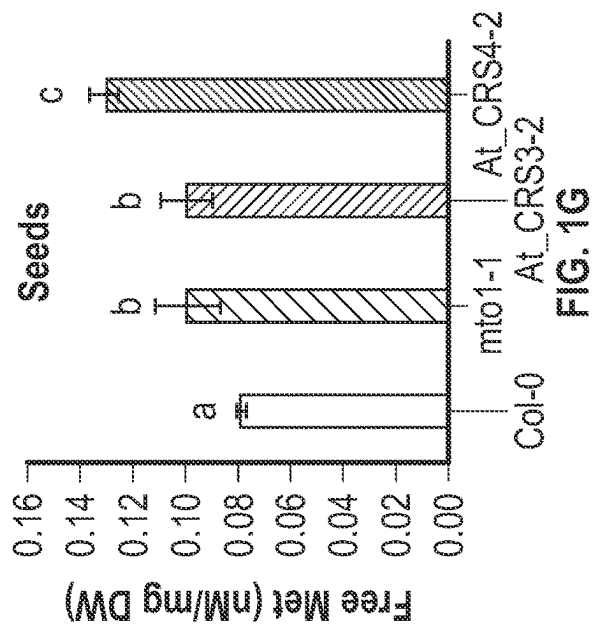
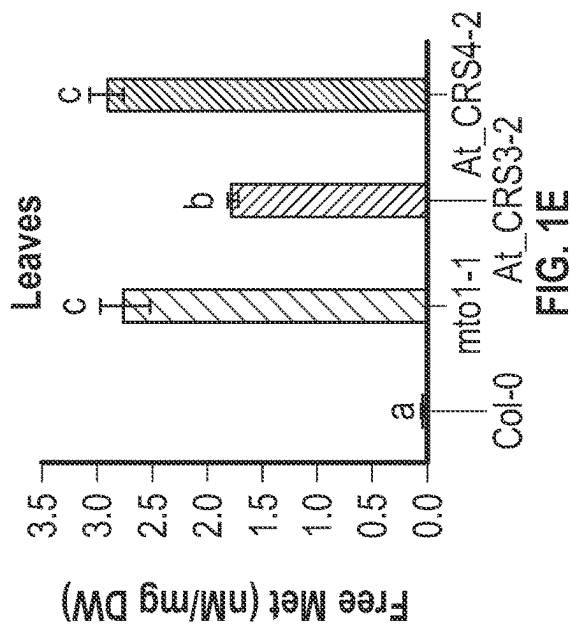
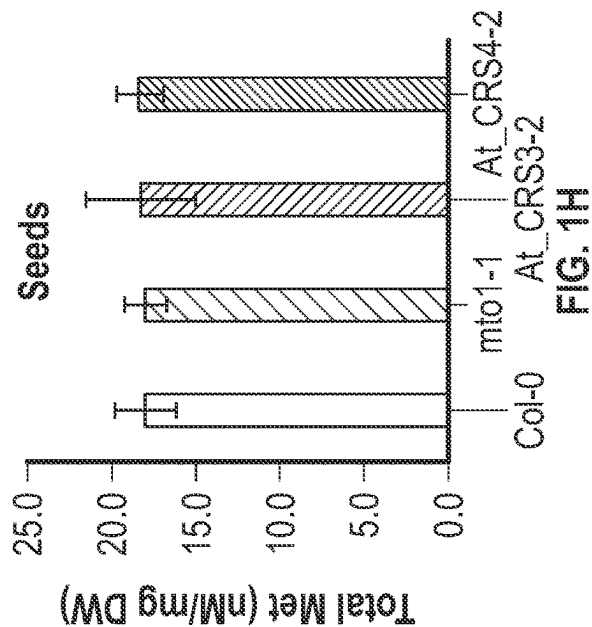
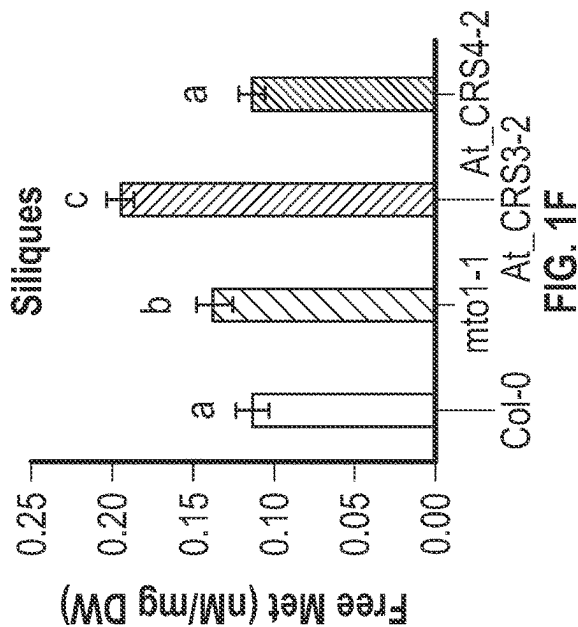


FIG. 1D



FIG. 1C



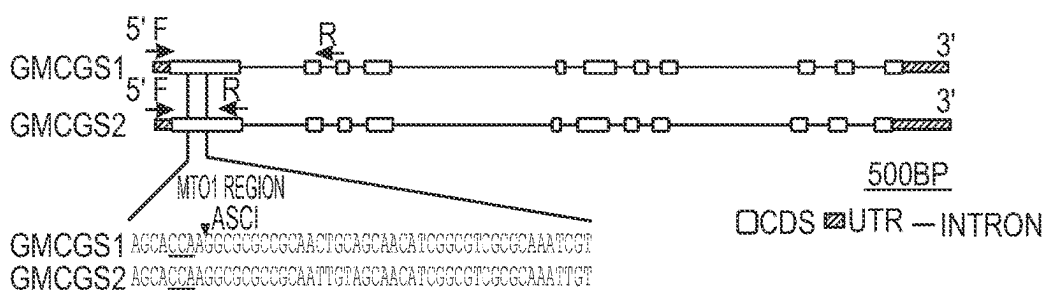


FIG. 2A

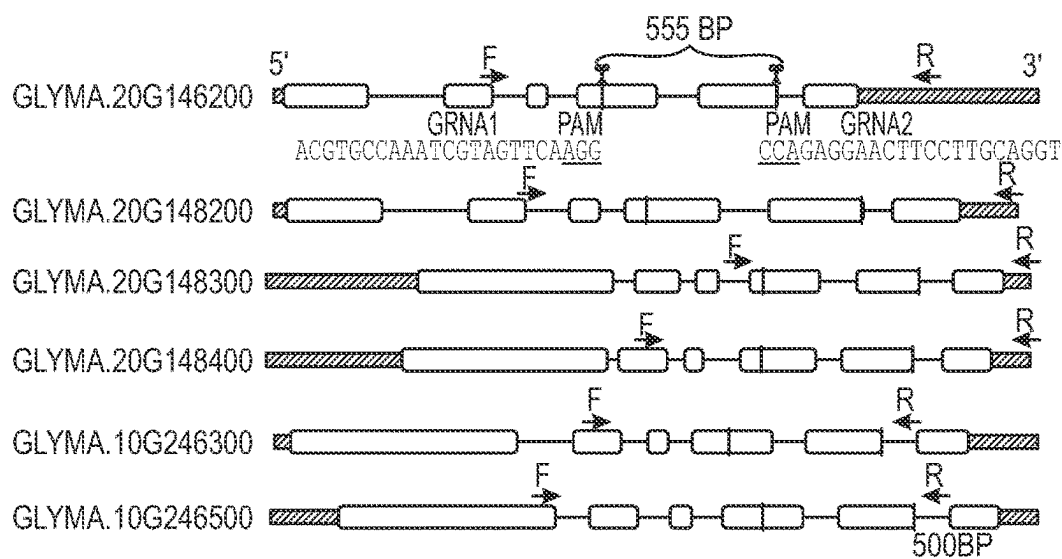


FIG. 2B

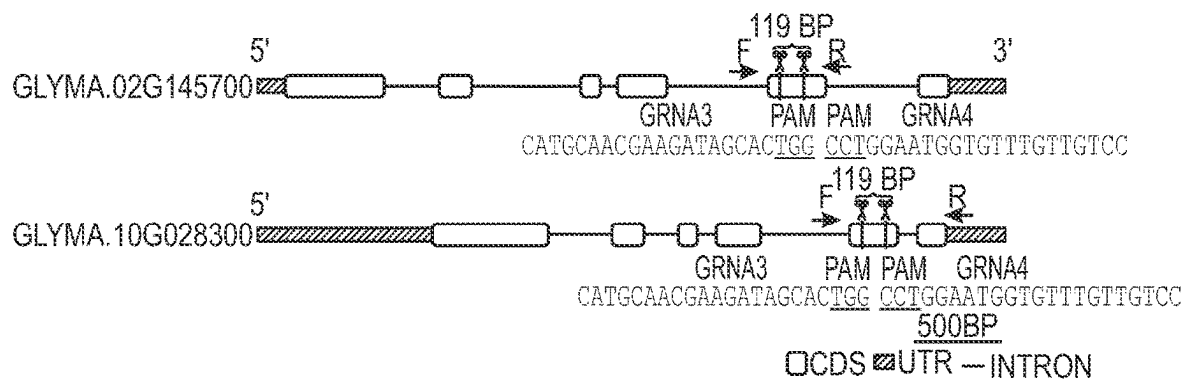


FIG. 2C

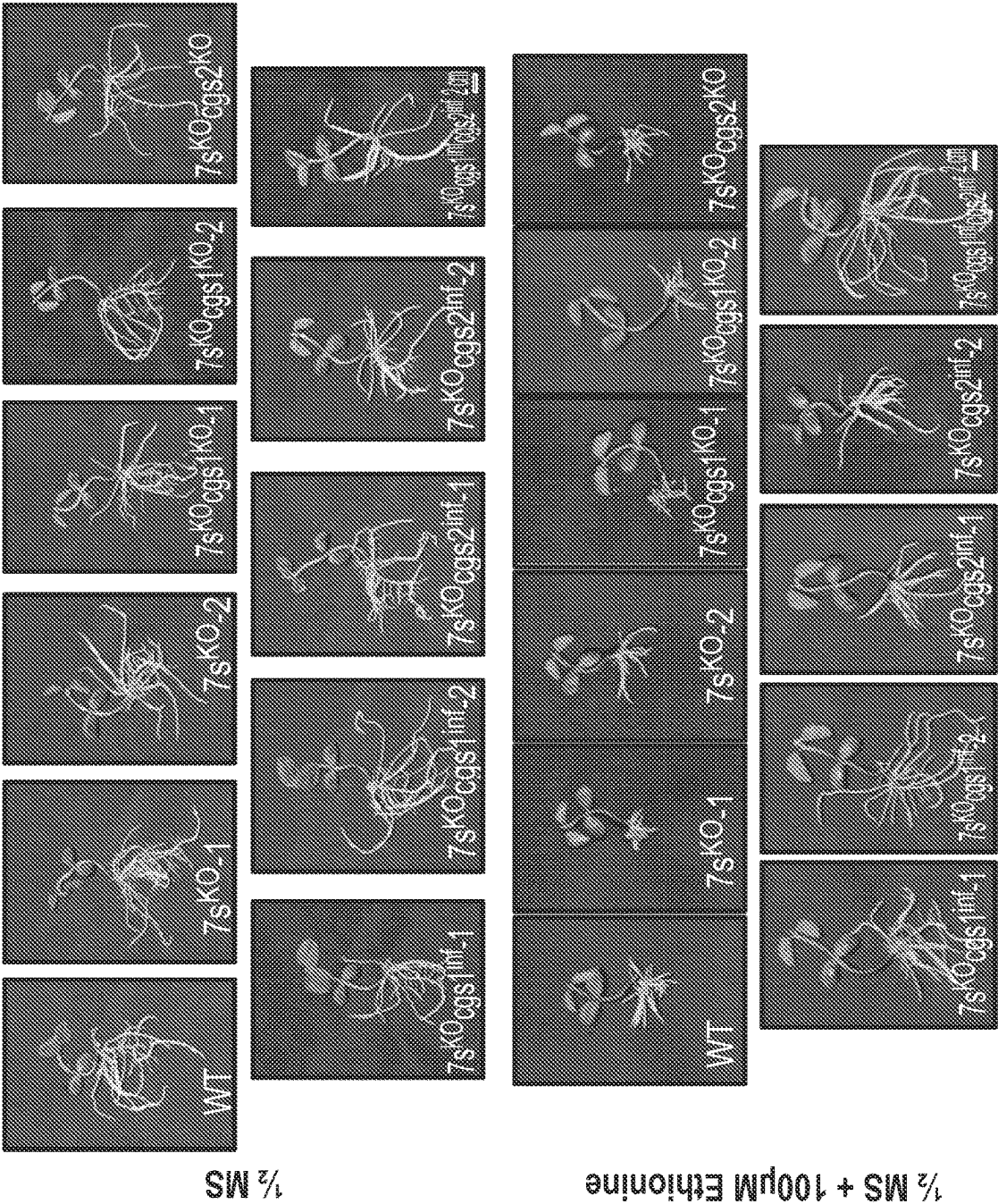


FIG. 3A

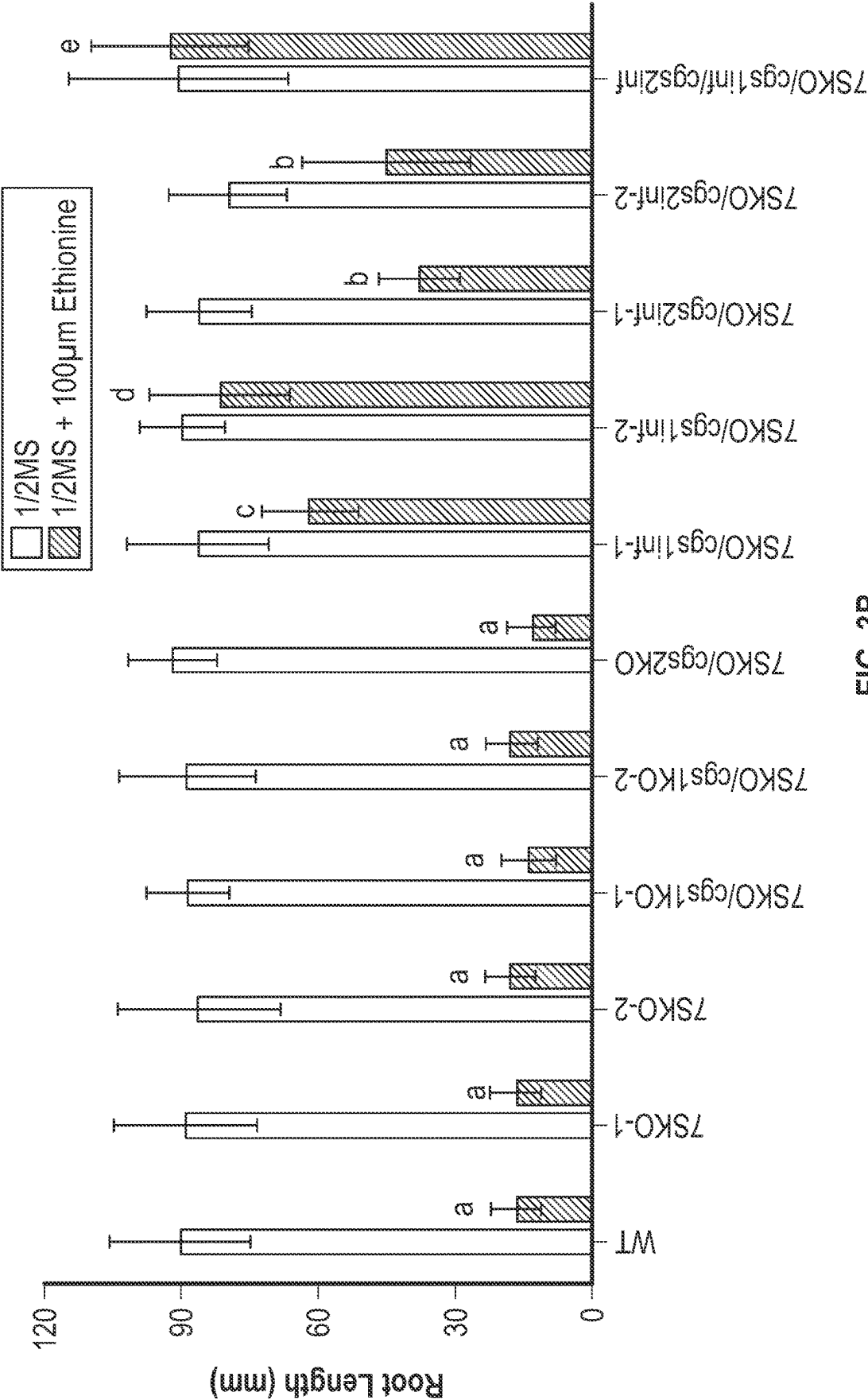


FIG. 3B

Line 7S-CGS-4	Asc/ V	Line 7S-CGS-6	
GmCGS1_WT	CAGCTAAGCACCAGGCGCGCGGCAACTGCAGCAACAT	GmCGS1_WT	CAGCTAAGCACCAGGCGCGCGGCAACTGCAGCAACAT
7S-CGS-4_allele1	CAGCTAAGCACCAGGCGCGCGGCAACTGCAGCAACAT	7S-CGS-6_allele1	CAGCTAAGCACCAGGCGCGCGGCAACTGCAGCAACAT
7S-CGS-4_allele2	CAGCTAAGCACCAGG-GCGCGGCAACTGCAGCAACAT -1bp	7S-CGS-6_allele2	CAGCTAAGCACCAGG---GCGGCAACTGCAGCAACAT -3bp
	*****		*****
GmCGS2_WT	Asc/ V	GmCGS2_WT	CAGCTCAGCAGCAAGG-CGGCGGCAATT-GTAGCAACAT
7S-CGS-4_allele1	CAGCTCAGCAGCAAGGCGCGCGGCAATTGTAGCAACAT	7S-CGS-6_allele1	CAGCTCAGCAGCAAGG-CGGCGGCAATT-GTAGCAACAT
7S-CGS-4_allele2	CAGCTCAGCAGCAAGGCGCGCGGCAATTGTAGCAACAT	7S-CGS-6_allele2	CAGCTCAGCAGCAAGGTGGCGCGGCAATTGGTAGCAACAT +2bp
	CAGCT -----CAGCAT -27bp		*****
	*****		*****
Line 7S-CGS-13		Line 7S-CGS-36	
GmCGS1_WT	CAGCTAAGCACCAGGCGCGCGGCAACTGCAGCAACAT	GmCGS1_WT	CAGCTAAGCACCAGGCGCGCGGCAACTGCAGCAACAT
7S-CGS-13_allele1	CAGCTAAGCACCAGGCGCGCGGCAACTGCAGCAACAT	7S-CGS-36_allele1	CAGCTAAGCACCAGGCGCGCGGCAACTGCAGCAACAT
7S-CGS-13_allele2	CAGCTAAGCACCAGG----GCGGCAACTGCAGCAACAT -4bp	7S-CGS-36_allele2	CAGCTAAGCACCAGG---GCGGCAACTGCAGCAACAT -3bp
	*****		*****
GmCGS2_WT	CAGCTCAGCAGCAGGCGCGGCAATTGTAGCAACAT	GmCGS2_WT	CAGCTCAGCAGCAGG-CGGCGGCAATTGTAGCAACAT
7S-CGS-13_allele1	CAGCTCAGCAGCAGGCGCGGCAATTGTAGCAACAT	7S-CGS-36_allele1	CAGCTCAGCAGCAGG-CGGCGGCAATTGTAGCAACAT
7S-CGS-13_allele2	CAGCTCAGCAGCAA-----TTGTAGCAACAT -12bp	7S-CGS-36_allele2	CAGCTCAGCAGCAAGGTGGCGCGGCAATTGTAGCAACAT +1bp
	*****		*****

FIG. 3C

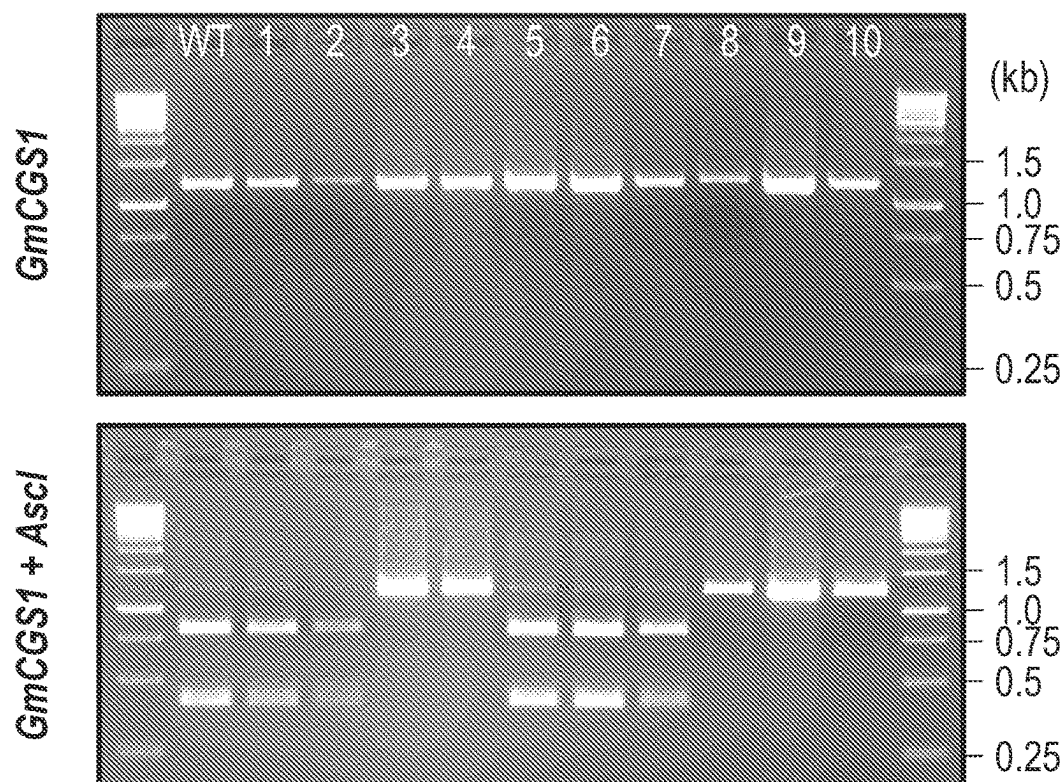


FIG. 3D

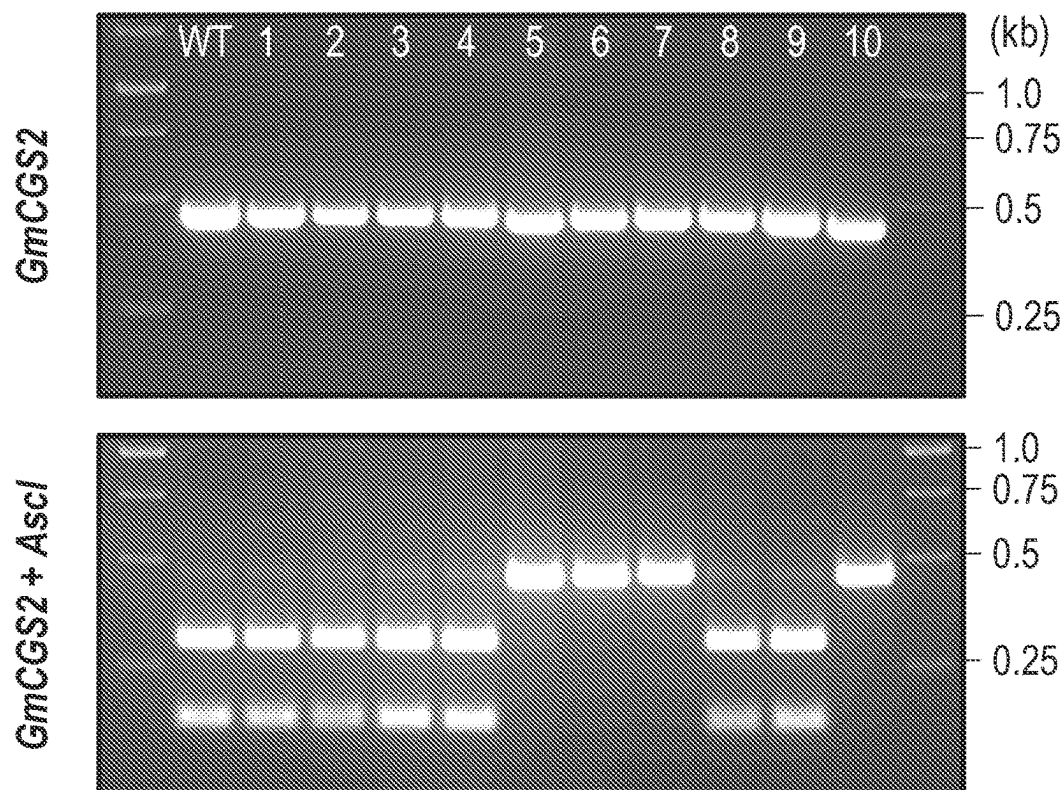
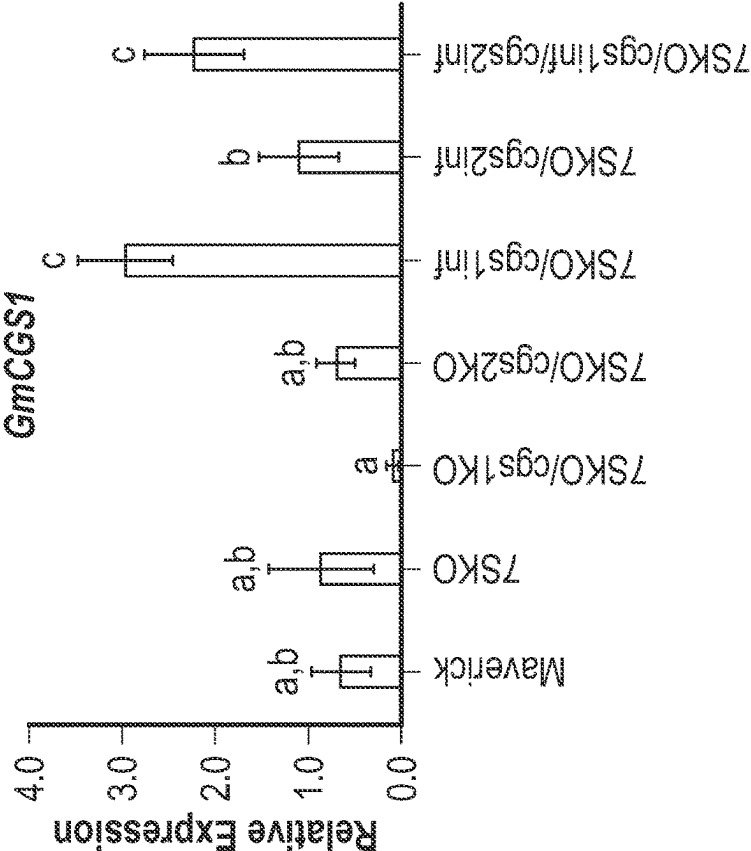
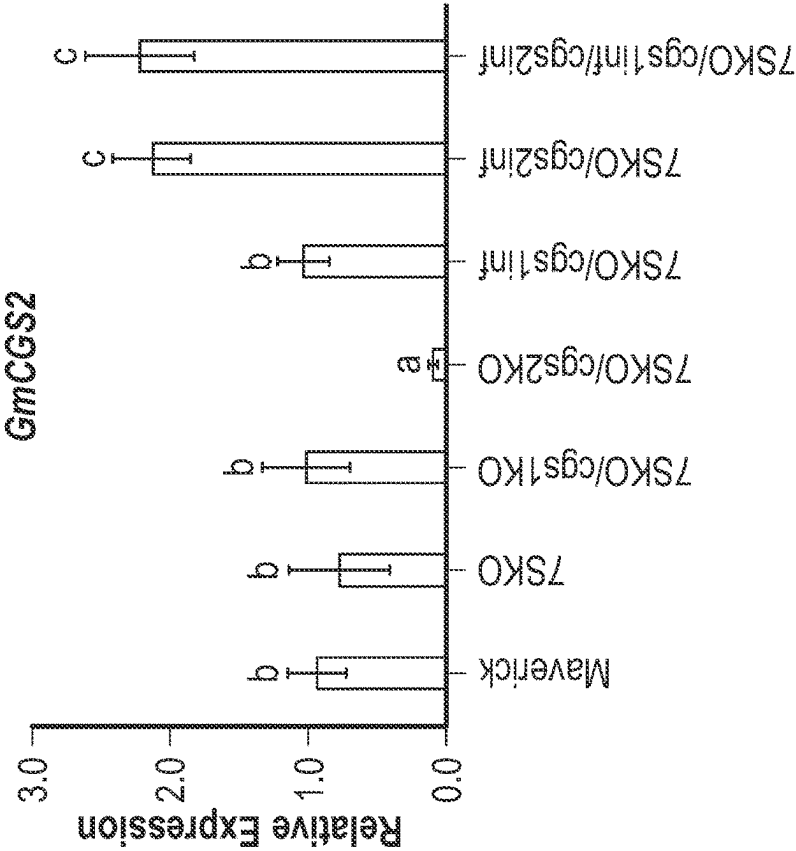
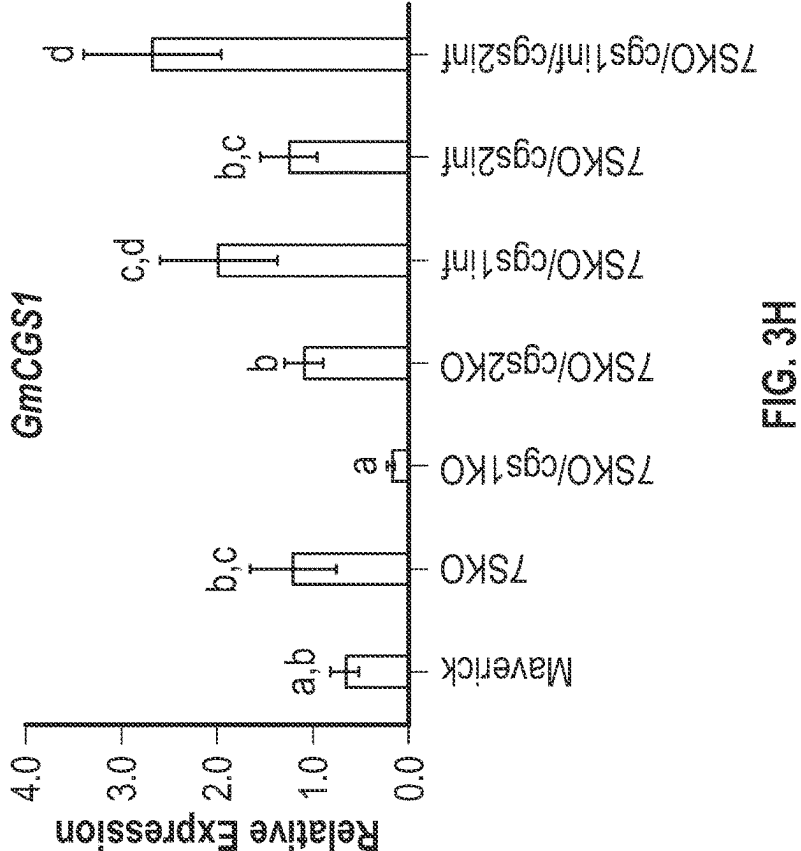
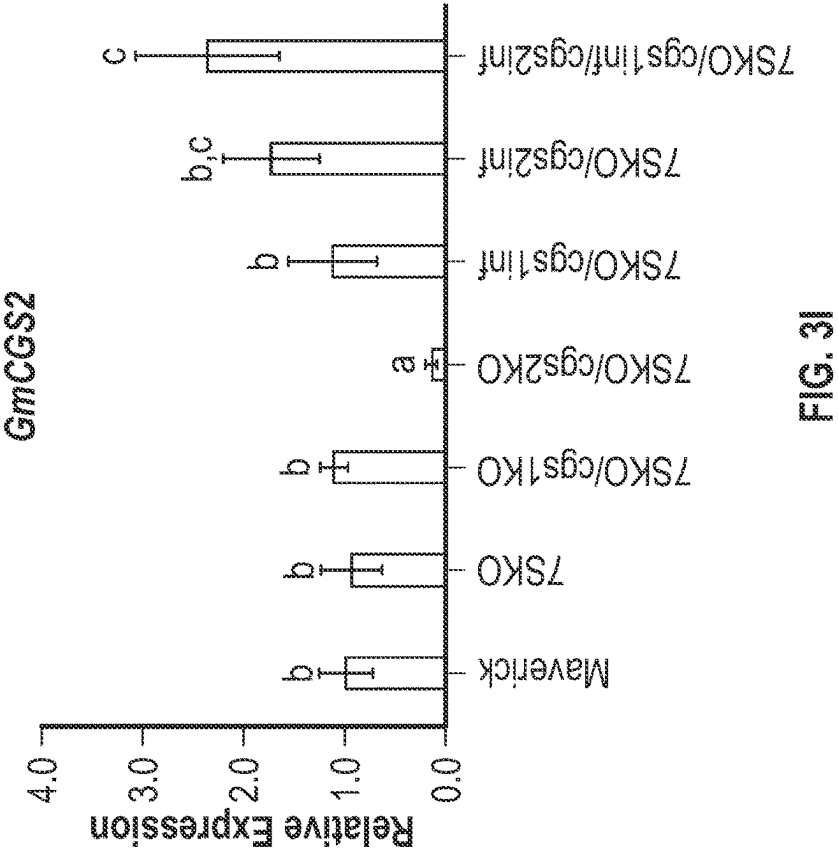
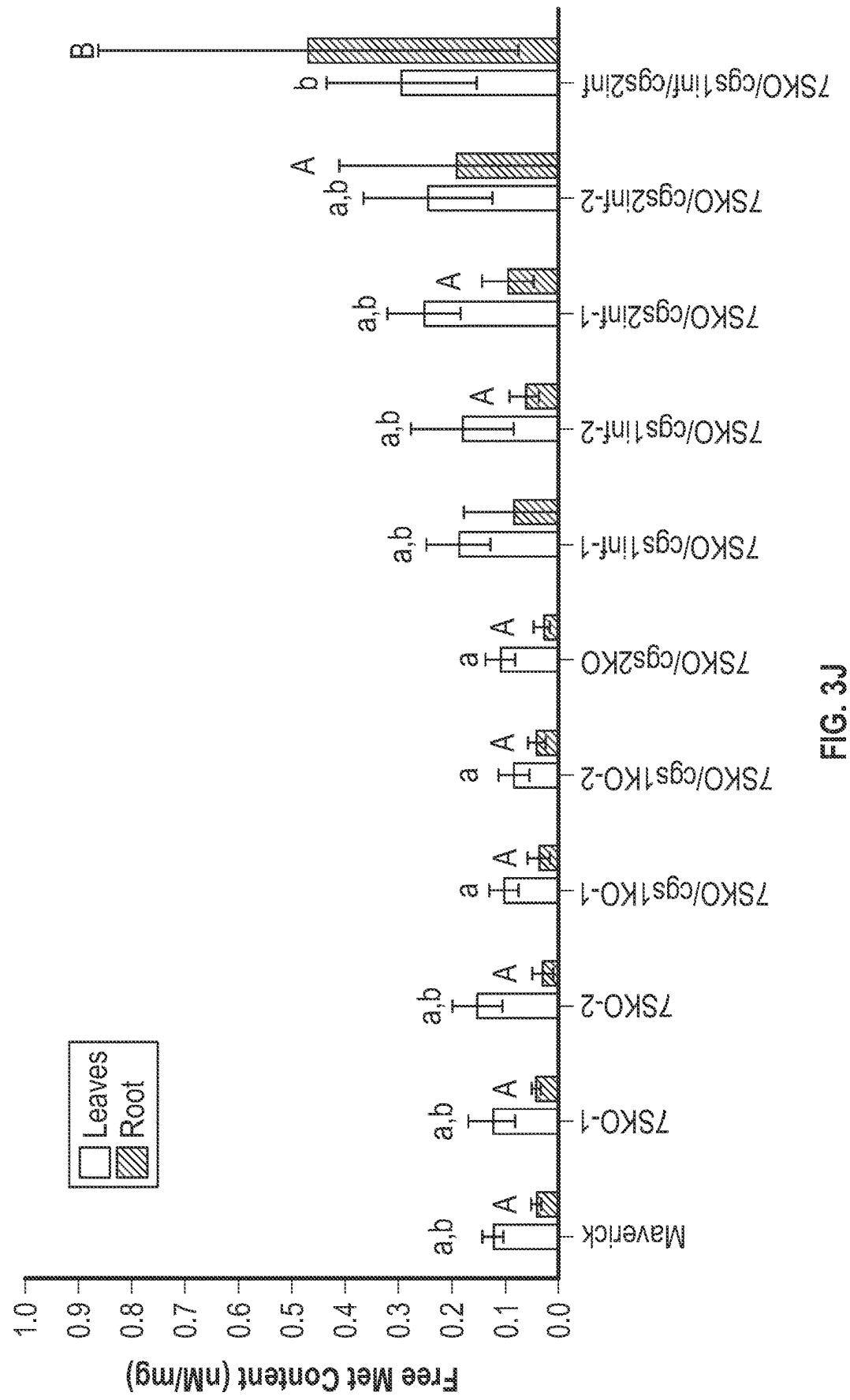
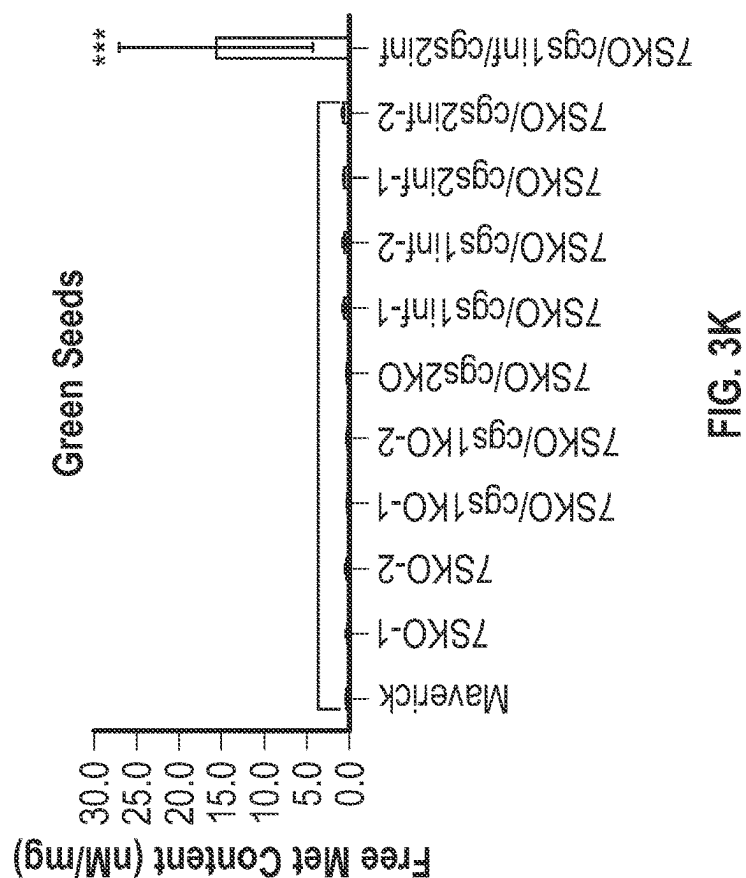
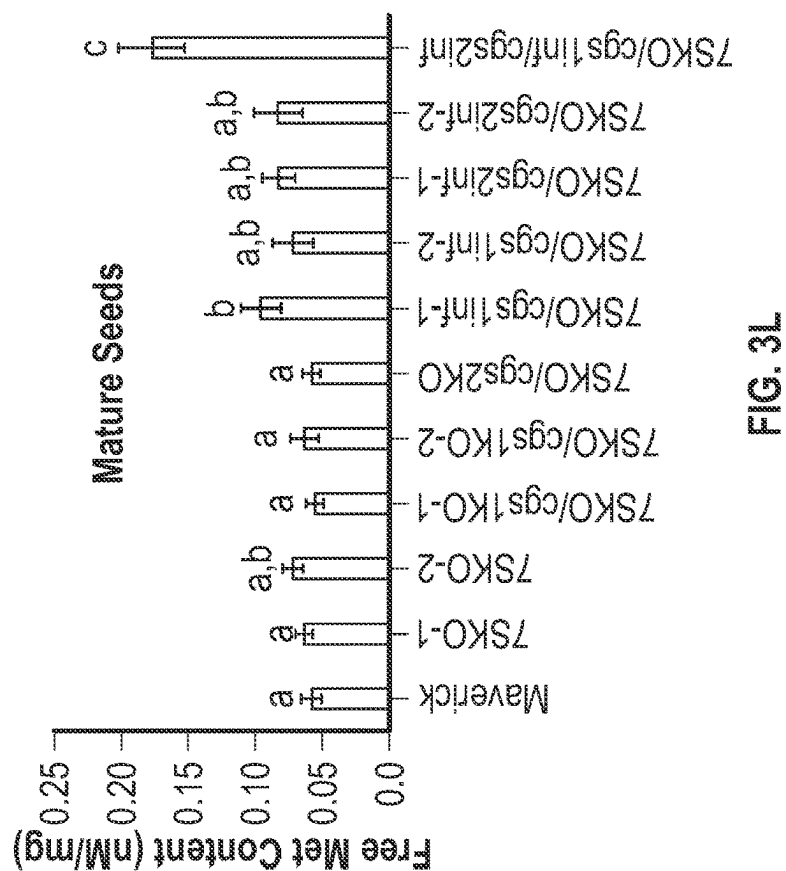


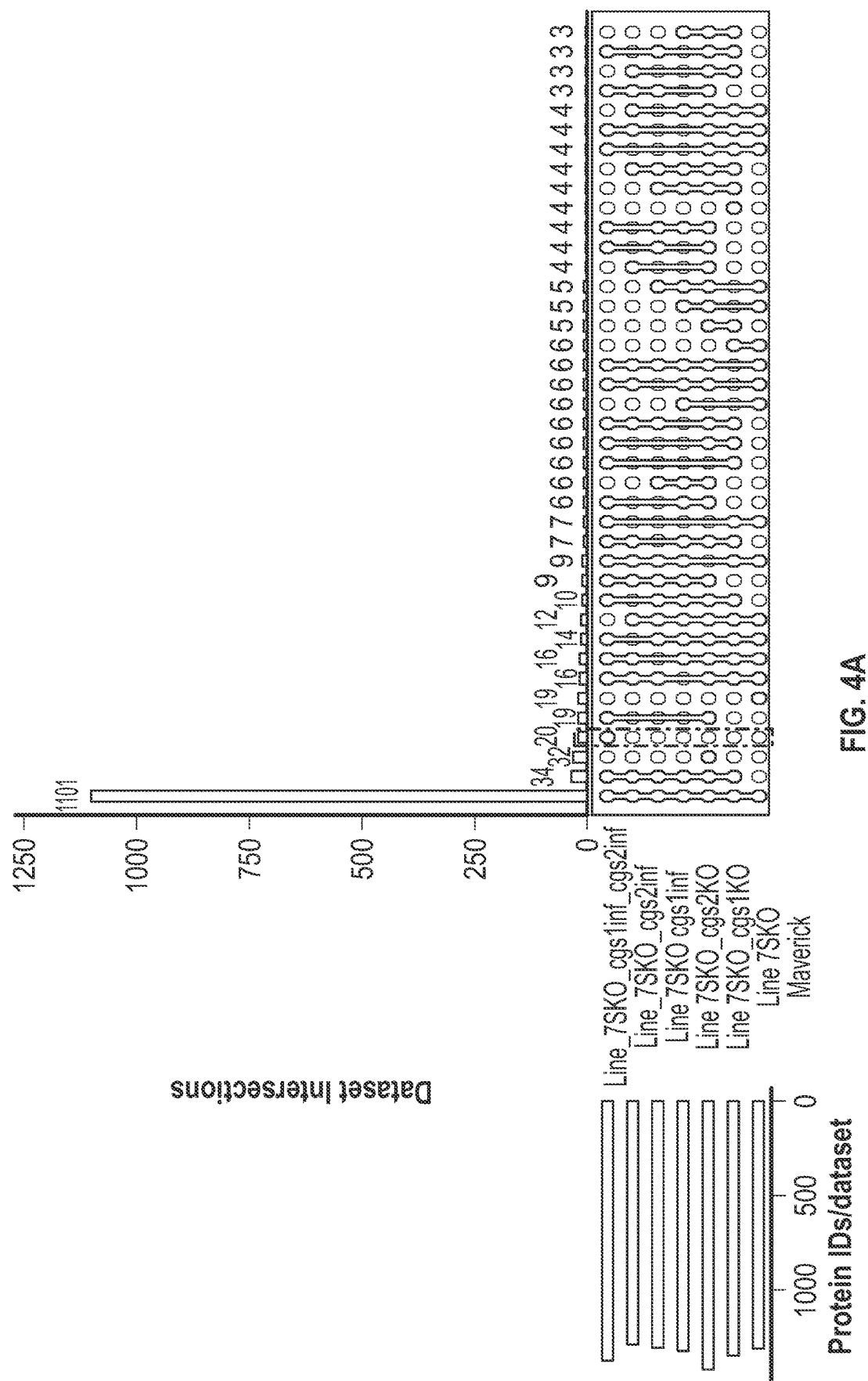
FIG. 3E

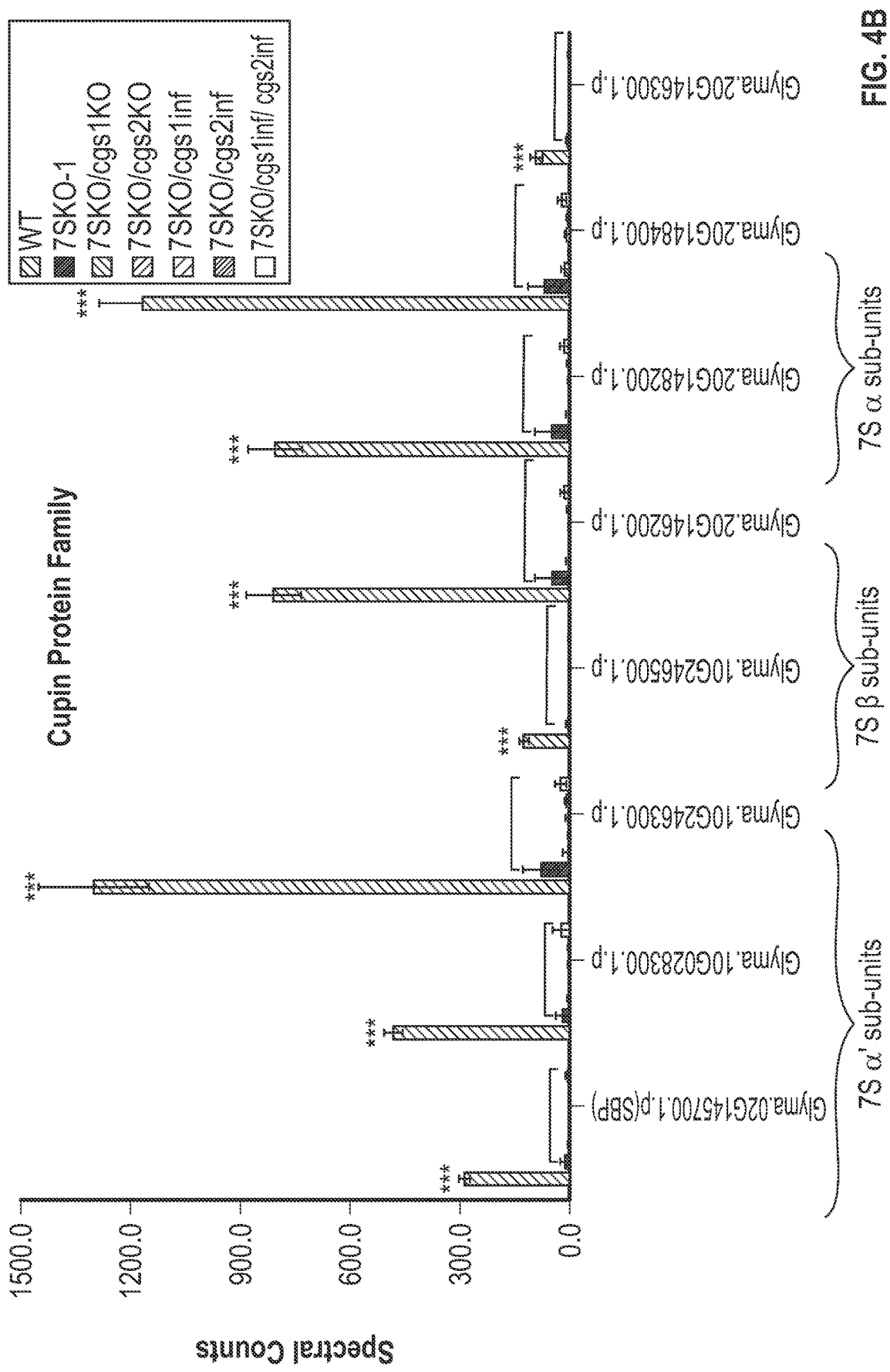












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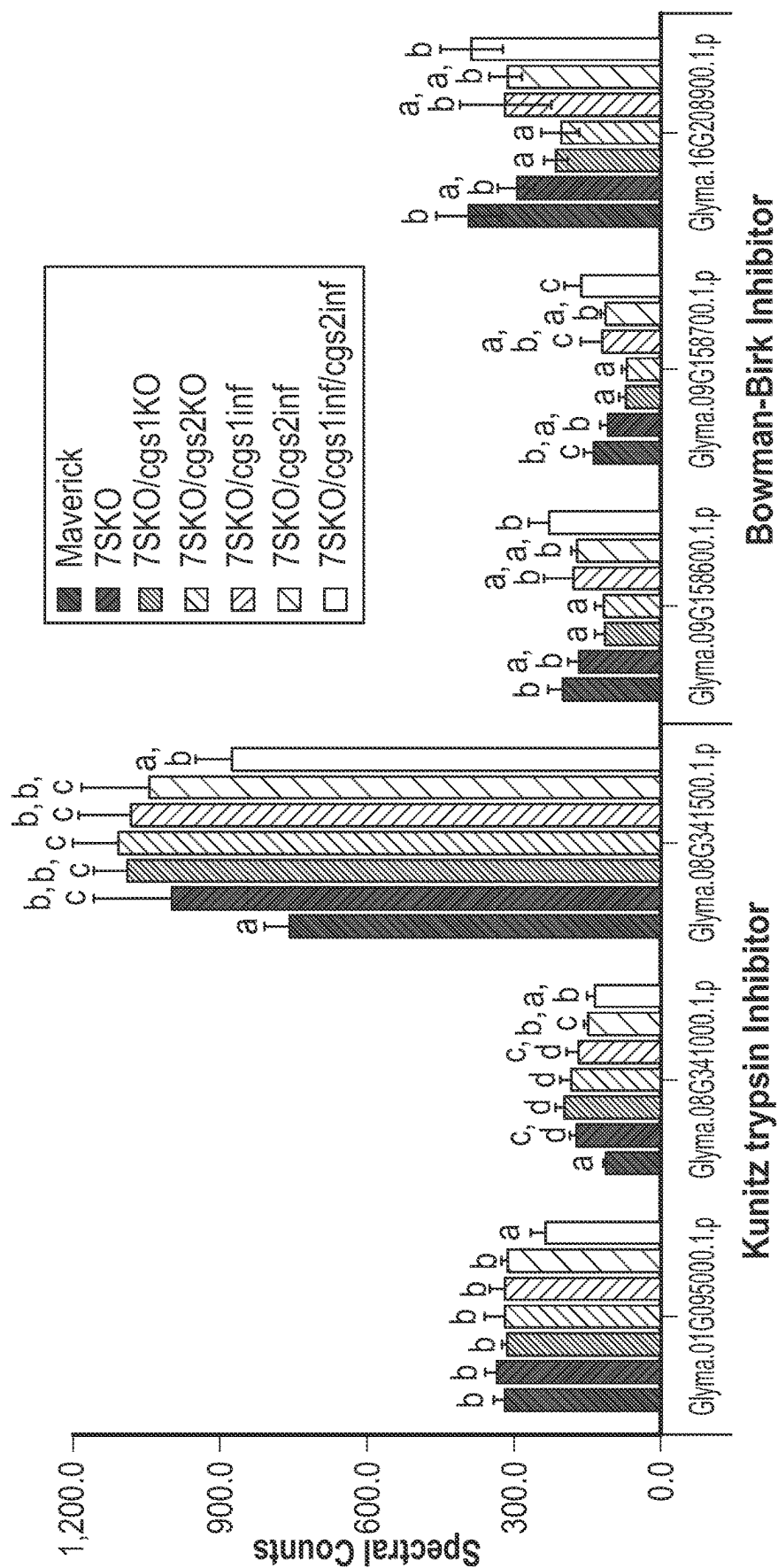
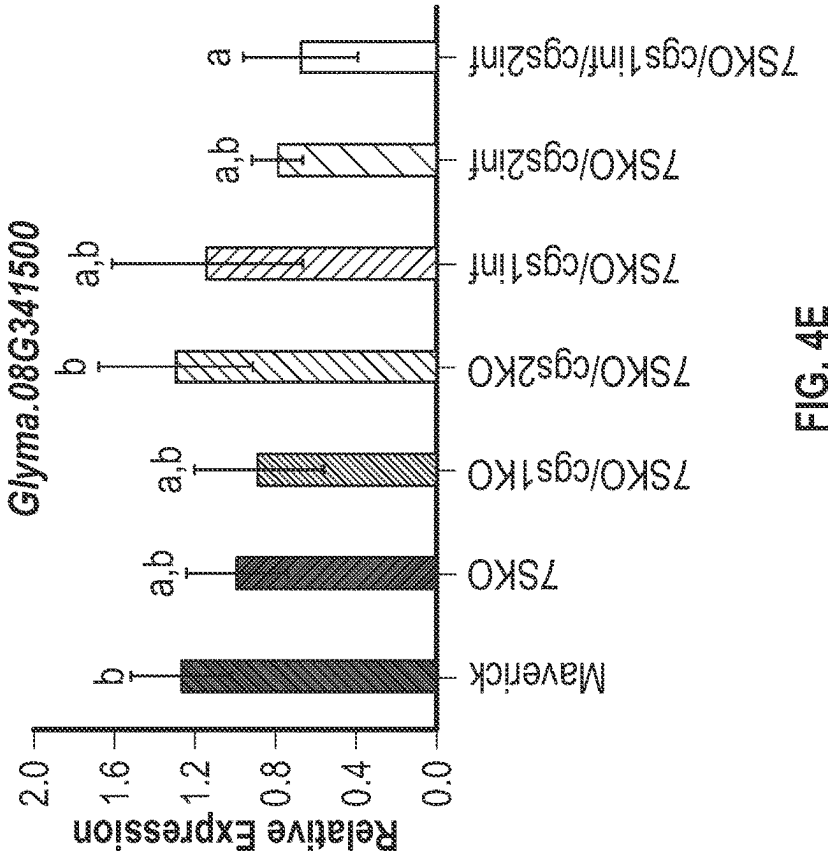
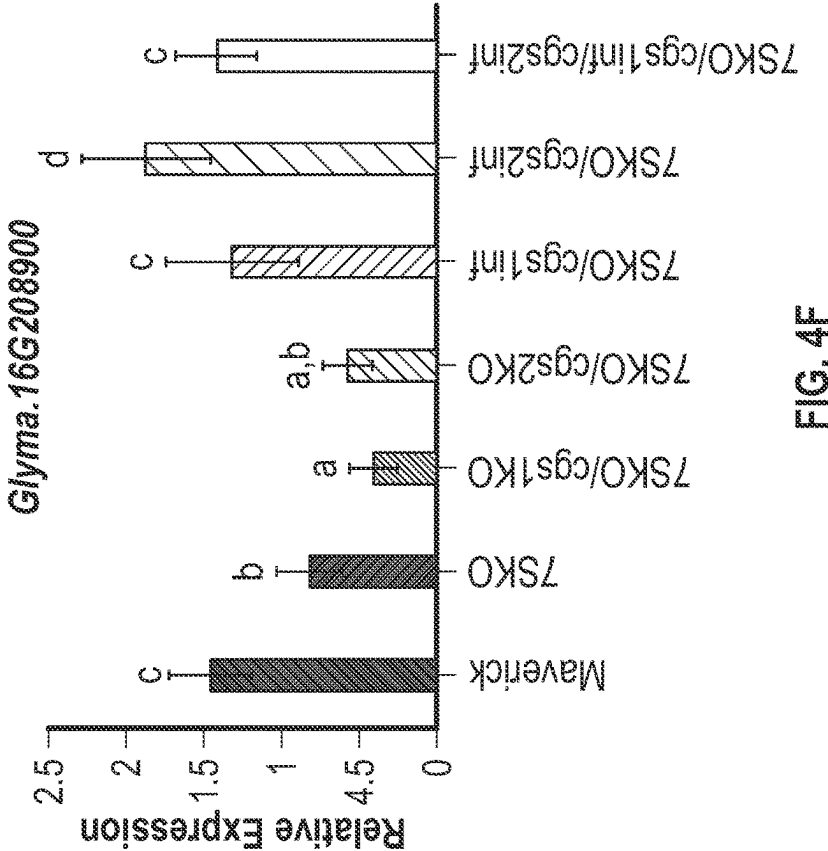


FIG. 4D



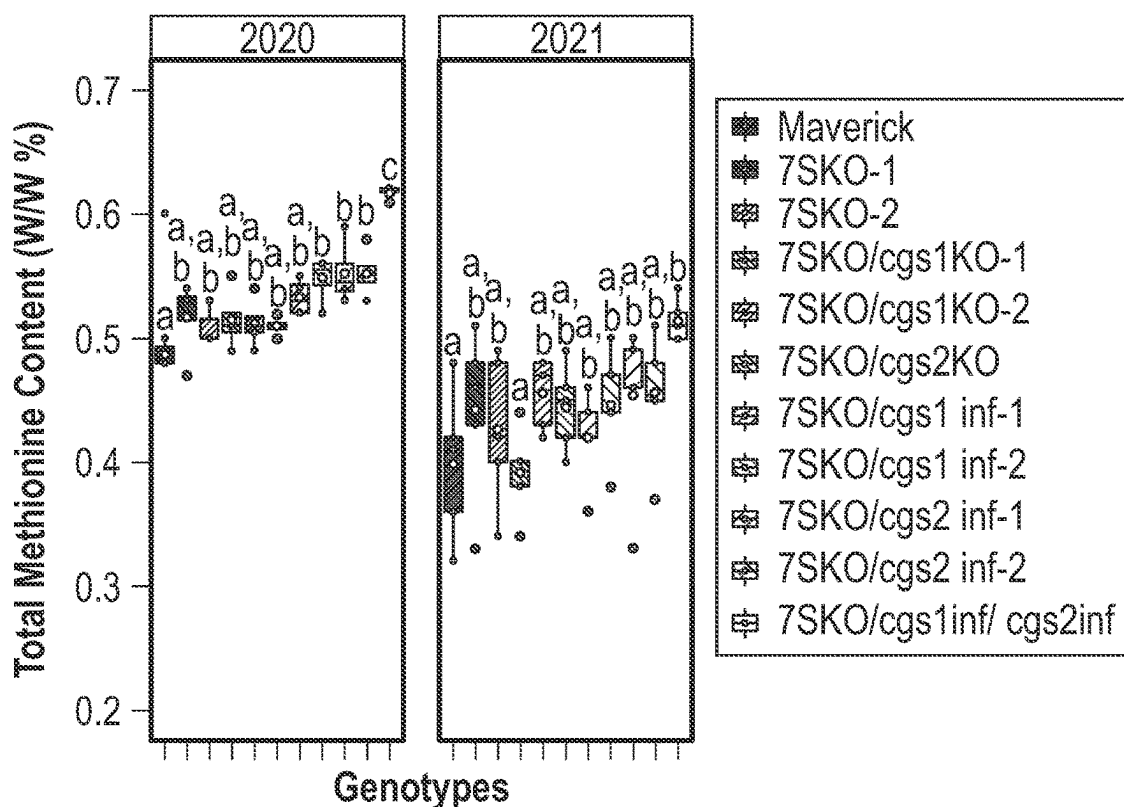


FIG. 5A

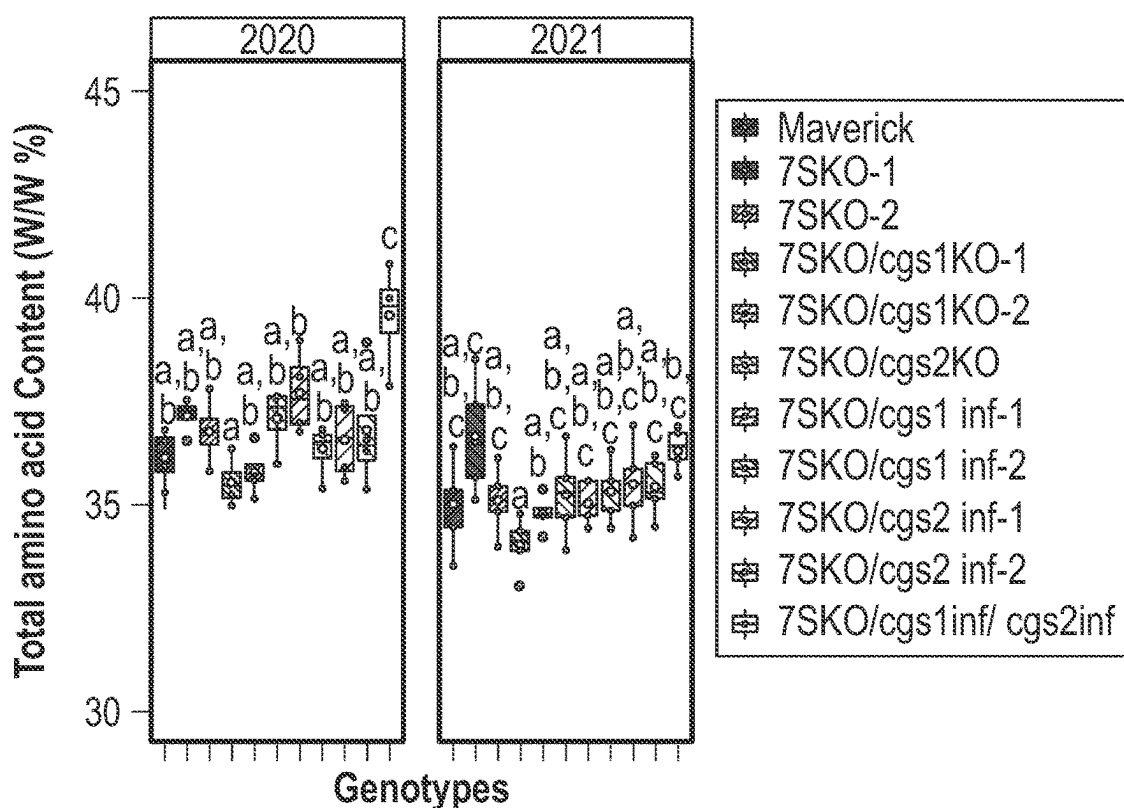


FIG. 5B

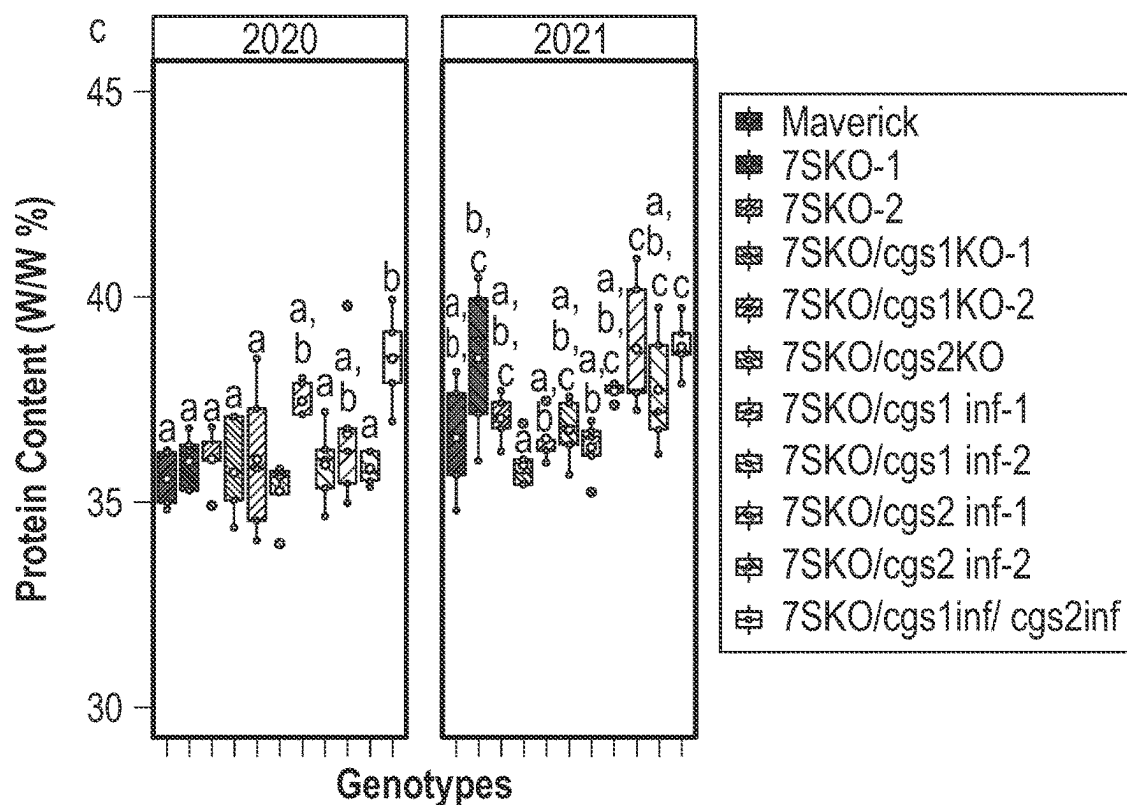


FIG. 5C

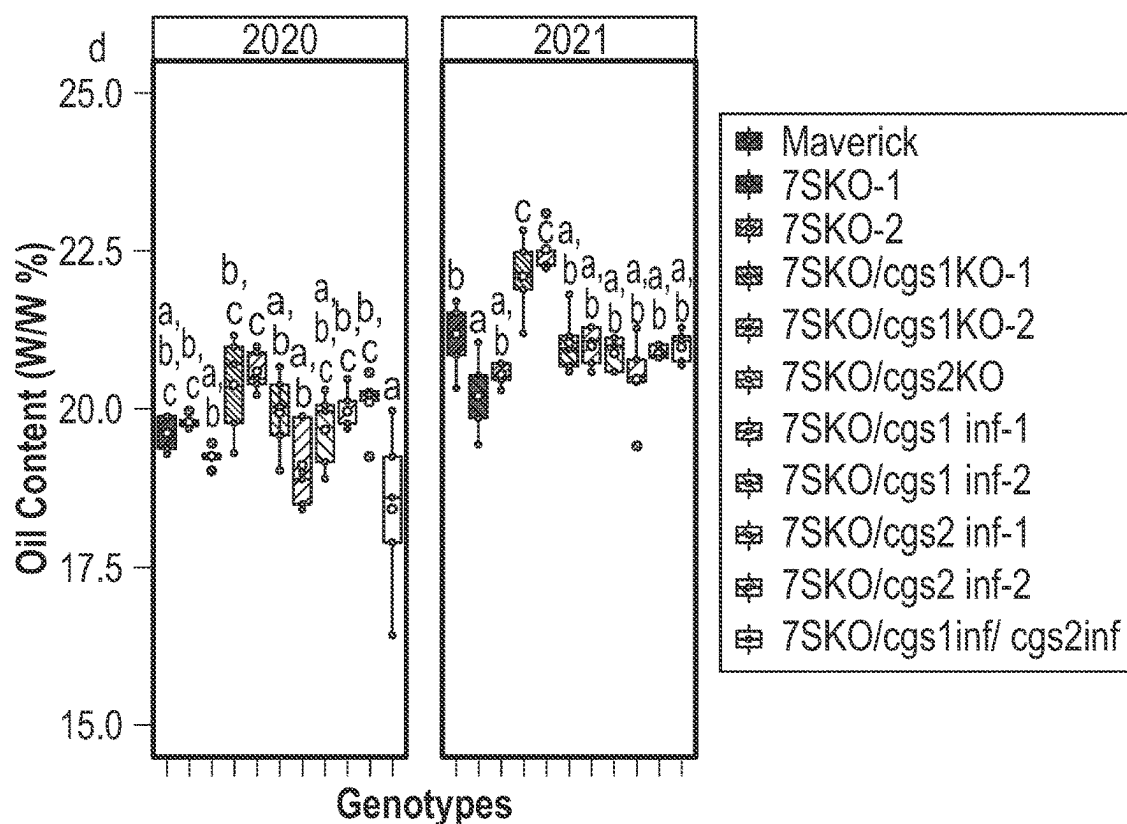


FIG. 5D

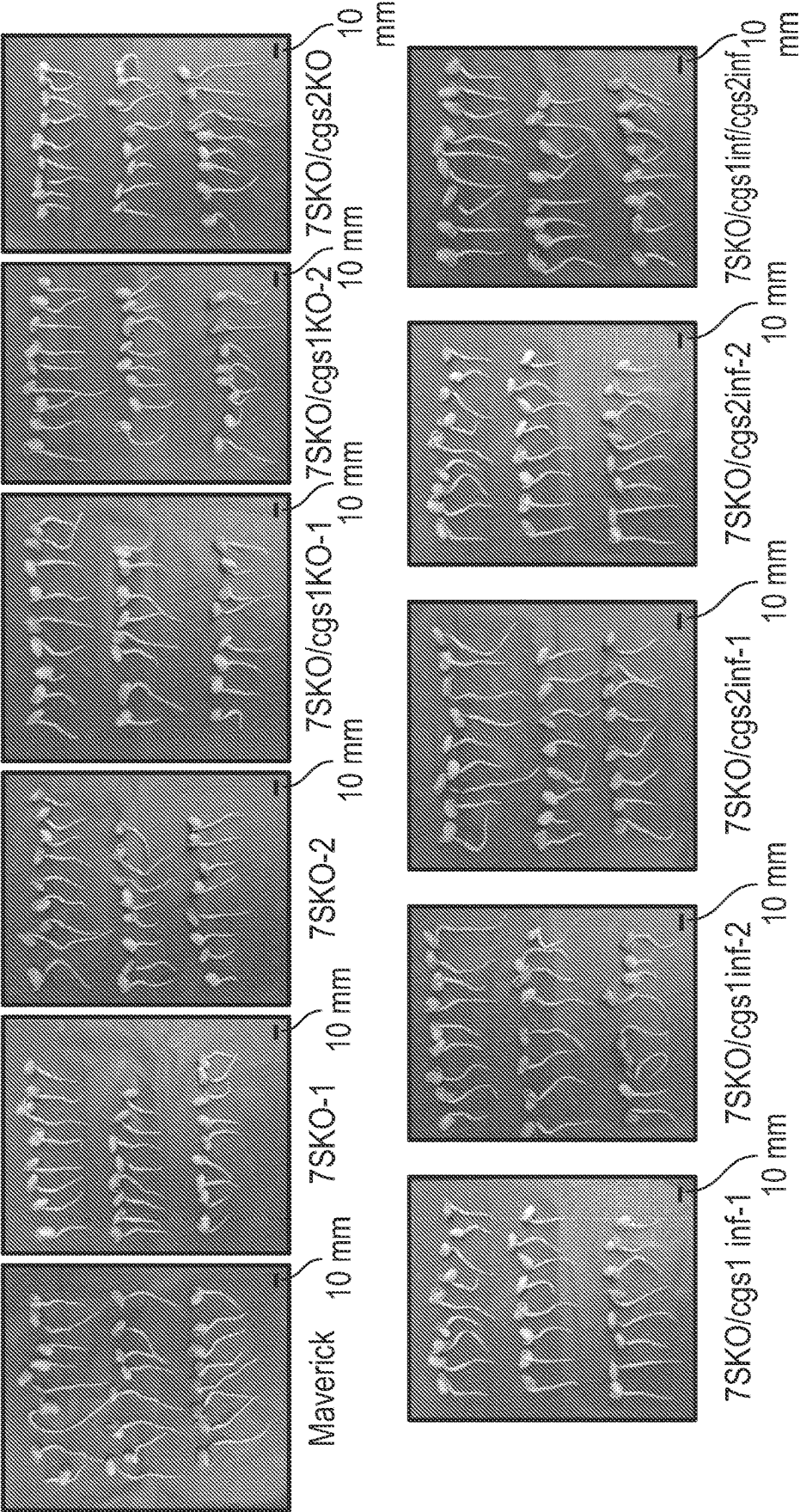
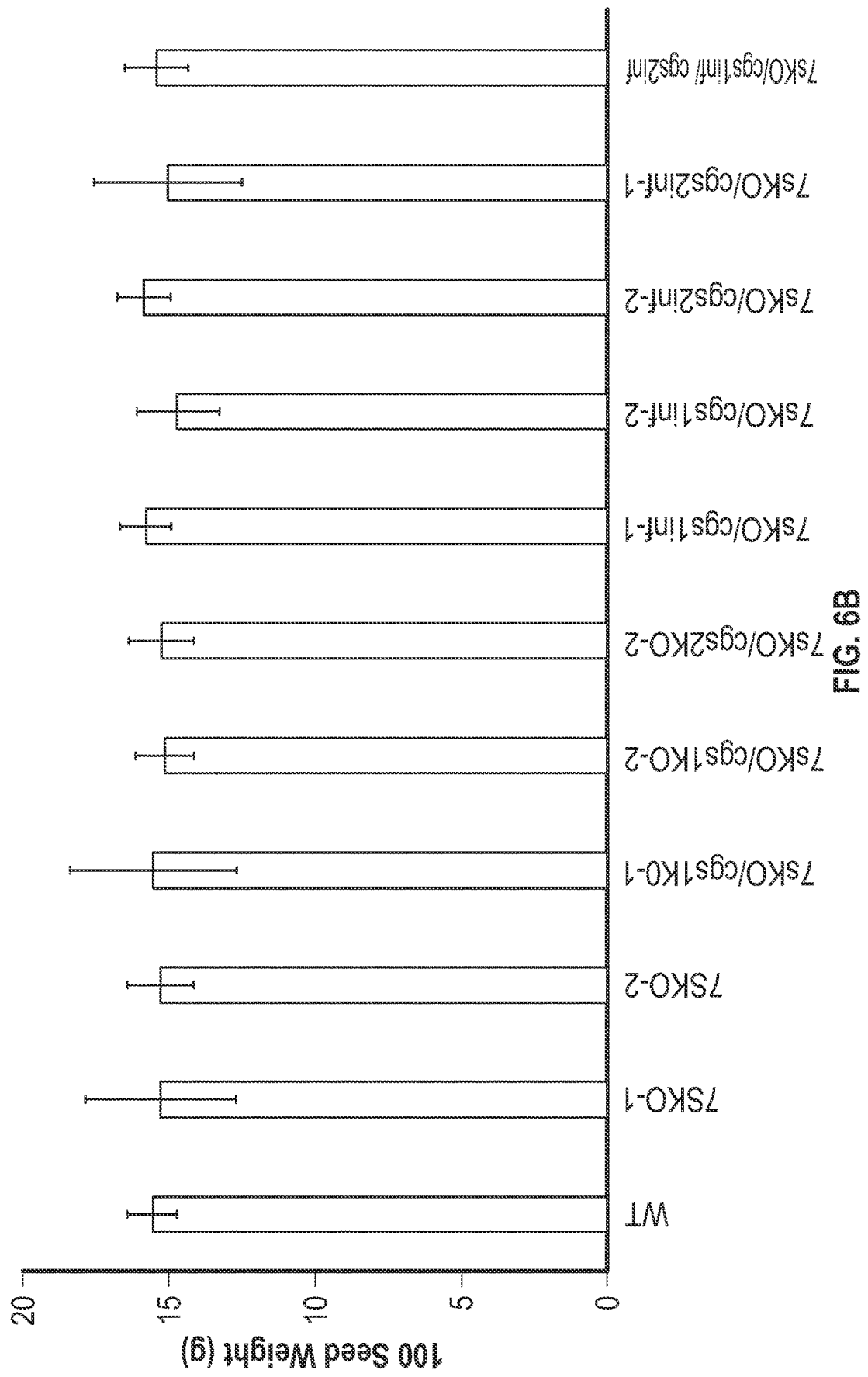
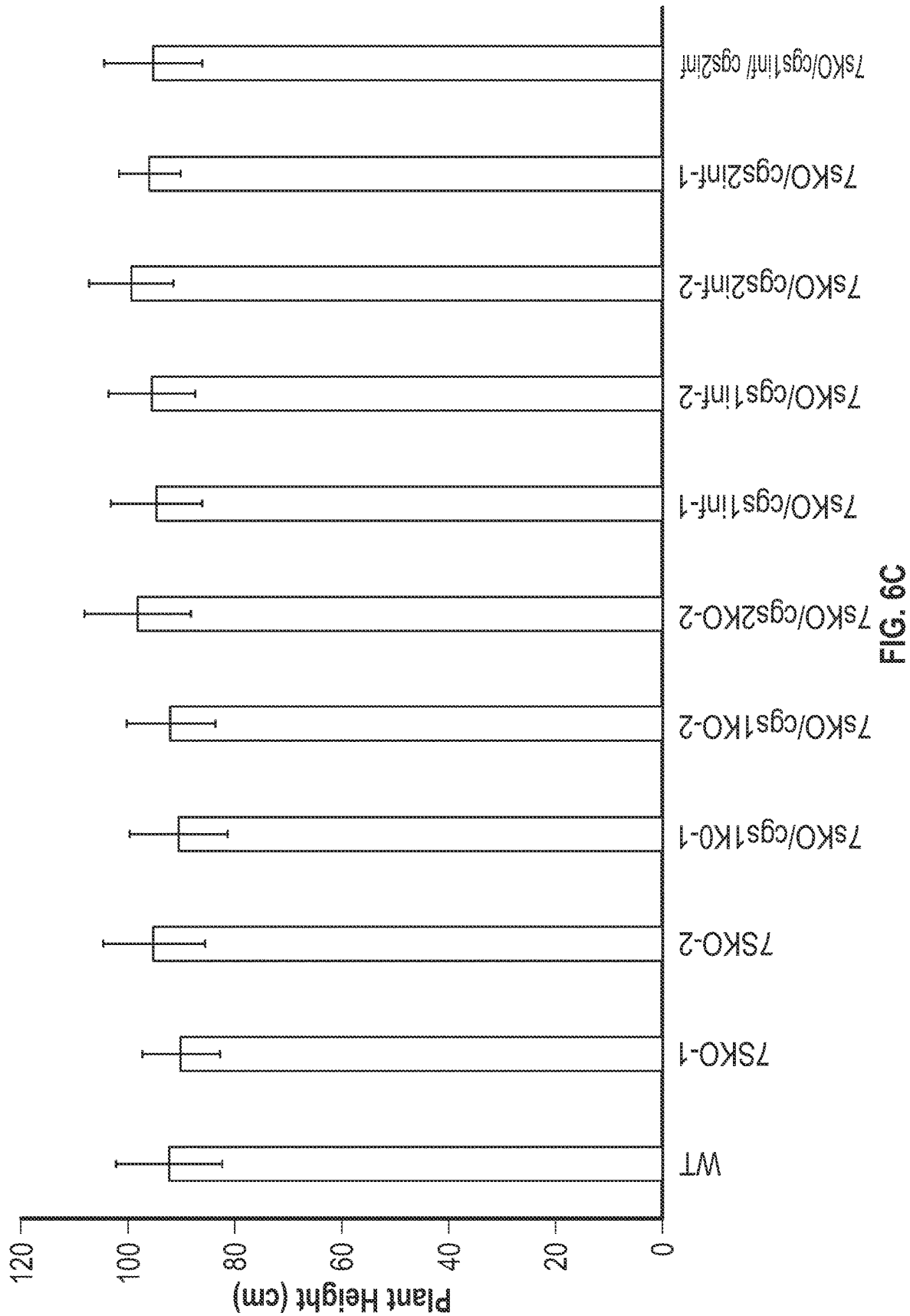
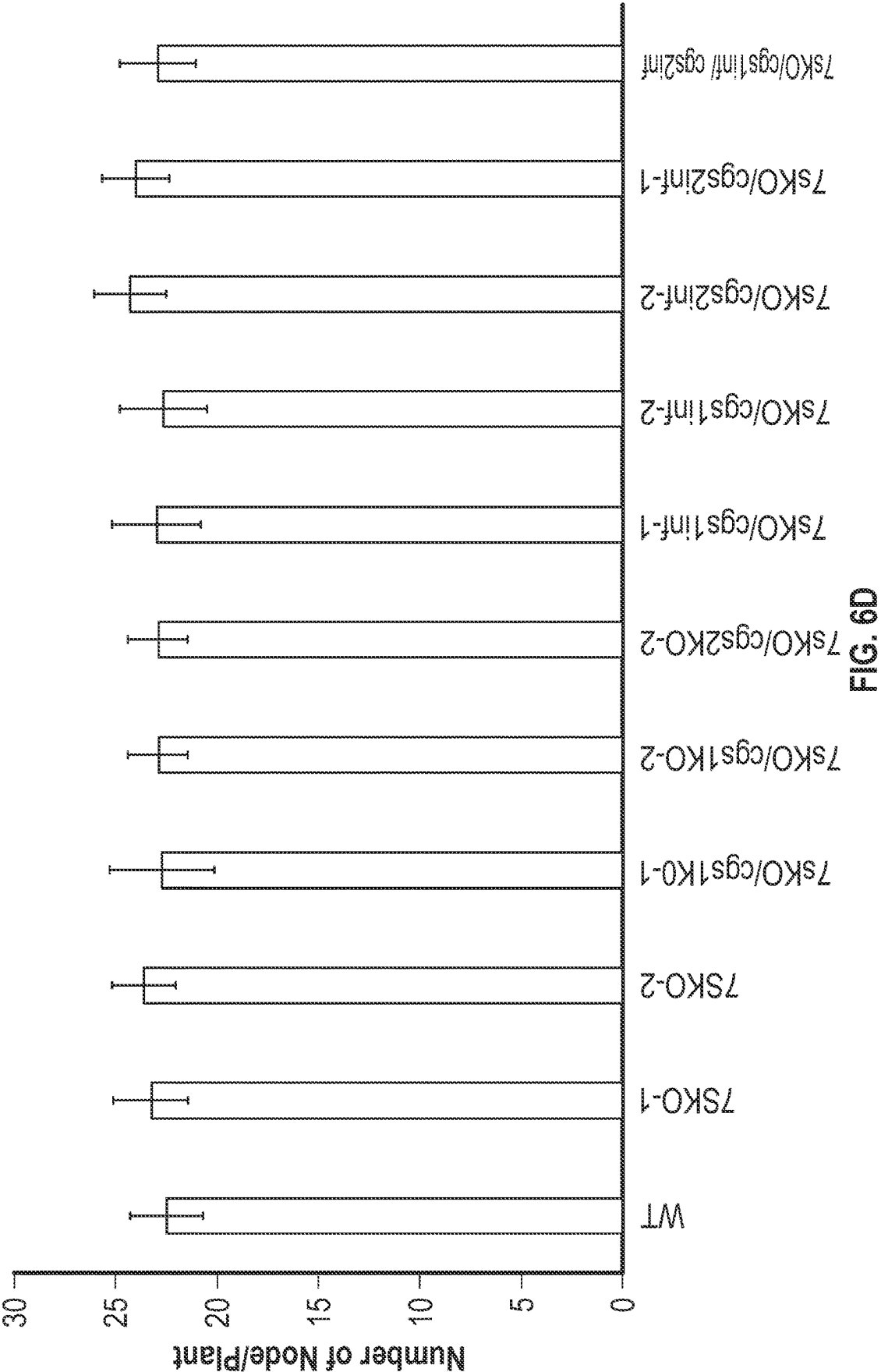


FIG. 6A







METHODS AND COMPOSITIONS FOR INCREASING AMINO ACID AND PROTEIN CONTENT IN PLANTS

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to provisional application U.S. Ser. No. 63/363,988, filed May 2, 2022, which is incorporated herein by reference in its entirety.

SEQUENCE LISTING XML

[0002] The instant application contains a sequence listing, which has been submitted in XML file format by electronic submission and is hereby incorporated by reference in its entirety. The XML file, created on Apr. 28, 2023, is named P13867WO00.xml and is 365,279 bytes in size.

TECHNICAL FIELD

[0003] The present disclosure generally relates to the field of biotechnology. More specifically, the present disclosure relates to increasing amino acid content in plants, particularly methionine.

BACKGROUND

[0004] Human and monogastric mammals are unable to synthesize essential amino acids that are required for survival, such as methionine. Hence, they have to obtain them from their diets, mainly from plant-based resources. Deficiency of these amino acids can lead to protein-energy malnutrition (PEM) syndrome which shows some nonspecific signs of protein deficiencies such as lowered resistance to diseases, decreased blood proteins, and retarded mental and physical development.

SUMMARY

[0005] Provided are plants, seeds, plant parts, and plant cells and descendants thereof that have a genomic modification that inhibits the feedback regulation cystathionine γ -synthase (CGS) catalyzed methionine production and/or a genomic modification that decreases expression of low methionine storage proteins. The genomic modification can be a deletion, insertion, or substitution of one or more nucleotides in a genomic sequence encoding a CGS or a low methionine content storage protein which modification suppresses the activity of the feedback regulation by MTO1 of the CGS and/or that suppresses the production of low methionine content storage proteins to cause proteome rebalancing favoring production of high methionine content storage proteins.

[0006] In some embodiments, the low methionine content storage protein is a beta-conglycinin and the high methionine storage protein is a glycinin. In some embodiments, the genomic modification suppresses the activity or production of 7S storage protein (e.g., an α' , α , or β subunit of the 7S storage protein). In some embodiments, the modification in the plant comprises a deletion or insertion in a coding sequence of the polypeptide, which may result in a frameshift of the sequence encoding the low methionine storage polypeptide. In some embodiments, the modification comprises a deletion, insertion, or substitution which may

result in an in-frame modification of the sequence encoding the MTO1 feedback regulatory region of CGS which retains enzymatic activity.

[0007] Plants producing seeds and seeds are provided, which have an increased total methionine and protein content of up to 30% and 3%, respectively, in seeds compared to seeds not comprising the modification.

[0008] In some embodiments, plants are provided which further include a heterologous nucleic acid sequence such as one or more of a reporter gene, a selection marker, a disease resistance gene, a herbicide resistance gene, an insect resistance gene; a gene involved in carbohydrate metabolism, a gene involved in fatty acid metabolism, a gene involved in amino acid metabolism, a gene involved in plant development, a gene involved in plant growth regulation, a gene involved in yield improvement, a gene involved in drought resistance, a gene involved in increasing nutrient utilization efficiency, a gene involved in cold resistance, a gene involved in heat resistance and a gene involved in salt resistance in plants.

[0009] Methods are provided for increasing protein content in the seed of a plant, by introducing a modification into a CGS gene within or near the MTO1 region and/or a low methionine content storage polypeptide gene, or both genes in a plant. Examples of modifications include a deletion, insertion, or substitution of one or more nucleotides in a genomic sequence encoding a CGS feedback regulatory region and/or a low methionine storage polypeptide comprising an amino acid sequence as disclosed herein to produce a modified plant. The modified plant can be grown to produce a modified seed, which has an increased protein content or increased methionine content compared to seed of a control plant not comprising the modification.

[0010] In some embodiments, methods of plant breeding include crossing the plants provided with a second plant to produce progeny seed. Progeny seed produced by these methods may comprise the modification and have increased protein content or increased methionine content relative to a control seed not comprising the modification.

[0011] In some embodiments, the modification comprises a deletion or insertion in a coding sequence of the polypeptide, which may result in a frameshift of the sequence encoding the low methionine storage polypeptide. In some embodiments, the modification comprises a deletion or insertion in a coding sequence of the CGS feedback regulatory region of the CGS polypeptide, which may result in an in-frame modification that inhibits feedback regulation of the CGS polypeptide. The deletion or insertion may be introduced through targeted DNA breaks.

[0012] Provided are methods for producing a plant producing a seed comprising increased protein content or increased methionine content, by introducing into a plant a recombinant DNA construct comprising a heterologous polynucleotide, the polynucleotide comprising a sequence that results in reduced expression of a low methionine content seed storage and/or a reduced feedback regulation of CGS activity, or a combination thereof and regenerating a plant producing a seed comprising increased protein content or increased methionine content. In some embodiments, the reduced expression of the polypeptide is achieved through gene silencing or RNAi. Plants producing seeds and seeds produced by this method are provided, in which the plant or seed comprises the heterologous polynucleotide stably incorporated into its genome, and the seed has an increased

protein content or increased methionine content compared to control seeds not comprising the heterologous polynucleotide. In some embodiments, the seed or progeny seed produced by the plant has seed methionine content that is increased by about 30% compared to a control seed not comprising the heterologous polynucleotide.

[0013] Provided are guide RNA sequences, and recombinant DNA constructs expressing the guide RNAs, that target a genomic locus of a plant cell. Soybean plant cells, plants and seeds comprising the guide RNA sequence are provided. In some embodiments, the guide RNA comprises the nucleotide sequence of SEQ ID NO: 109, 110, 111, 112, 113, or 114. Seeds, such as produced by the plant or regenerated from the plant cell have an increased protein content or increased methionine content compared to control seeds not comprising the guide RNA sequence.

[0014] Provided are methods of detecting the presence of a polynucleotide comprising a sequence disclosed herein, by contacting a DNA sample obtained from a plant which comprises the modifications disclosed herein with a first primer molecule which binds to a genomic region upstream of the modification or at least partially contained in the modification and a second primer molecule which binds to a genomic region downstream of modification or at least partially contained in the same. A nucleic acid amplification reaction condition is provided and performed to produce a DNA amplicon molecule, such as 48 to 5000 nucleotides in length, which indicates the presence of the modification and which is detected.

[0015] While multiple embodiments are disclosed, still other embodiments will become apparent to those skilled in the art from the following detailed description, which shows and describes illustrative embodiments. Accordingly, the figures and detailed description are to be regarded as illustrative in nature and not restrictive.

BRIEF DESCRIPTION OF THE FIGURES

[0016] The following drawings form part of the specification and are included to further demonstrate certain embodiments. In some instances, embodiments can be best understood by referring to the accompanying figures in combination with the detailed description presented herein. The description and accompanying figures may highlight a certain specific example, or a certain embodiment. However, one skilled in the art will understand that portions of the example or embodiment may be used in combination with other examples or embodiments.

[0017] FIG. 1A-H shows increased free Met in leaves and seed of the *Arabidopsis* in-frame MTO1 CRISPR mutant lines. FIG. 1A are photographs showing the growth of 10-day-old seedling of wild-type (Col-0), mto1-1, and two CRISPR lines (At_CRS3-2 and At_CRS4-2) on ½ MS media (upper panel) and ½ MS media+100 µM methionine (lower panel). Scale bar=1 cm. FIG. 1B shows an alignment of DNA sequences of the MTO1 region in wild-type, mto1-1, and the two CRISPR lines (SEQ ID NOs: 31-36). The PAM sequence is underlined, and the point mutation of mto1-1 is indicated. Mutant amplicons showed -39 bp and -42 bp deletions due to excision of DNA sequences near the PAM site. FIG. 1C shows PCR-based genotyping of wild-type (Col-0), mto1-1, and the two CRISPR lines. Higher mobility amplicons indicate CRISPR/Cas9-induced deletion in AtCGS1 in the two CRISPR mutant lines. FIG. 1D shows the relative expression level of AtCGS1 from leaves of two

weeks old seedling on ½ MS media. Data represents mean±SD, n=4. Statistical analysis was performed using one-way ANOVA followed by a post-hoc Tukey's multiple range test. Different letters denote significant differences at P<0.001. FIG. 1E-G shows free Met content in leaves, siliques, and seed; FIG. 1H shows total Met content in seeds. Data represents #SD, n=5. Statistical analysis was performed using one-way ANOVA followed by a post-hoc Tukey's multiple range test. Different letters denote significant differences at P<0.01.

[0018] FIG. 2A-C is a schematic of the CRISPR/Cas9 targets in soybean. FIG. 2A shows the location of the target in the MTO1 region of GmCGS1 and GmCGS2 (SEQ ID NOs: 37-38). Gene-specific primers used for polymerase chain reaction (PCR) genotyping are indicated. The PAM sites are underlined, and the *AscI* restriction sites are shown as an arrow. FIG. 2B-C shows the location of targets in the 7S genes (SEQ ID NOs: 39-42). Gene-specific primers used for polymerase chain reaction (PCR) genotyping are indicated and the PAM sites are underlined.

[0019] FIG. 3A-L shows mutations within the MTO1 region of the GmCGS genes lead to changing free Met in leaves and seed in soybean. FIG. 3A are photographs showing the growth of 10 days old seedling of wild-type (Maverick), and CRISPR mutant lines 7SKO-1, 7SKO-2, 7SKO/cgs1KO-1, 7SKO/cgs1KO-2, 7SKO/cgs2KO, 7SKO/cgs1inf-1, 7SKO/cgs1inf-2, 7SKO/cgs2inf-1, 7SKO/cgs2inf-2 and 7SKO/cgs1inf/cgs2inf on ½ MS media (upper panel) and ½ MS media+100 µM methionine (lower panel). FIG. 3B shows quantification of root length of wild-type and the CRISPR mutant lines. Data represents mean±SD, n>15 and n>22 for control and methionine treatment, respectively. Statistical analysis was performed using one-way ANOVA using genotype as a factor followed by a post-hoc Tukey's multiple range test. Different letters denote significant differences at p<0.001 among genotypes under the same treatment. FIG. 3C shows DNA sequence alignments of MTO1 region within the GmCGS1 (Glyma.09g235400) and GmCGS2 (Glyma.18g261600) genes in wild-type and four T0 transgenic lines: 7S-CGS-4, 7S-CGS-6, 7S-CGS-13, 7S-CGS-36 (SEQ ID NOs: 42-51). The PAM sequence is underlined. The sequence was cleaved by *AscI* (arrows indicate the cutting site). Mutant amplicons showed deletions (-) or insertions (+) near the PAM site. FIG. 3D-E shows CAPS (Cleaved Amplified Polymorphic Sequences) marker genotyping of wild-type (WT) and the T6 CRISPR mutant lines [7SKO-1 (1), 7SKO-2 (2), 7SKO/cgs1KO-1 (3), 7SKO/cgs1KO-2 (4), 7SKO/cgs2KO (5), 7SKO/cgs1inf-1 (6), 7SKO/cgs1inf-2 (7), 7SKO/cgs2inf-1 (8), 7SKO/cgs2inf-2 (9), and 7SKO/cgs1inf/cgs2 (10)] to detect mutations in GmCGS1 and GmCGS2 before (upper panel) and after (lower panel) *AscI* digestion. FIG. 3F-I shows relative expression level of GmCGS1 and GmCGS2 from leaves (FIG. 3F-G) of four weeks old plants and green seeds (FIG. 3H-I). Data represents mean±SD, n=4. Statistical analysis was performed using one-way ANOVA followed by a post-hoc Tukey's multiple range test. Different letters denote significant differences at P<0.01.

[0020] FIG. 3J-L shows free Met content in roots, leaves, green seeds, and mature seed. Data represents mean±SD, n=5. Statistical analysis was performed using one-way ANOVA followed by a post-hoc Tukey's multiple range test. Different letters denote significant differences at P<0.01.

[0021] FIG. 4A-F shows quantitative proteomic analysis of soybean seeds of different genotypes. FIG. 4A shows qualitative overview and intersections of quantified proteins from the mature seed proteome of six soybean CRISPR/Cas9 lines and Maverick wild-type. Horizontal bars show the number of proteins found in the genotypes. Vertical bars display intersects between comparisons as indicated in the matrix below the graph. FIG. 4B-F shows protein abundance of 7S, 11S (glycinin), KTI, and BBI in the mature soybean seeds among seven genotypes. Data represents mean $\pm$ SD, n=5. Statistical analysis was performed using one-way ANOVA followed by a post-hoc Tukey's multiple range test. Different letters denote significant differences at P<0.01.

[0022] FIG. 5A-D shows the increase in total methionine, amino acid, and protein in seed of the field grown soybean CRISPR/Cas9 lines. FIG. 5A-B shows total methionine and total amino acid content in dry mature seeds of the wild-type and CRISPR lines, planted in the Bradford Research and Experiment Center, University of Missouri in 2020 and 2021. Quantitative analysis of amino acids were performed with standard protein hydrolysate at the Agricultural Experiment Station Chemical Laboratories (ESCL), University of Missouri, Columbia. FIG. 5C-D shows protein content, measured by Combustion method (LECO) and oil content, measured by the Soxhlet method in wild-type (Maverick) and CRISPR/Cas9 lines. For all boxplots, the box represents the interquartile range, the horizontal line in the box is the median, red dot is the mean, n=5. Statistical analysis was performed using one-way ANOVA followed by a post-hoc Tukey's multiple range test. Different letters denote significant differences at P<0.01.

[0023] FIG. 6A-D shows germination rate and plant performance of the soybean lines. FIG. 6A are photographs showing soybean seed germination 5 days after sowing. The experiments were carried out in an incubator at 25° C., 75% humidity FIG. 6B-D shows 100 seed weight, plant height, and number of nodes per plant measured from field grown soybeans. Data represents mean $\pm$ SD, n>15. Statistical analysis was performed using one-way ANOVA followed by a post-hoc Tukey's multiple range test. P<0.01.

DETAILED DESCRIPTION

[0024] So that the present invention may be more readily understood, certain terms are first defined. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which embodiments of the invention pertain. Many methods and materials similar, modified, or equivalent to those described herein can be used in the practice of the embodiments of the present invention without undue experimentation, the preferred materials and methods are described herein. In describing and claiming the embodiments of the present invention, the following terminology will be used in accordance with the definitions set out herein.

[0025] It is to be understood that all terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting in any manner or scope. For example, as used in this specification and the appended claims, the singular forms "a," "an" and "the" can include plural referents unless the content clearly indicates otherwise. Similarly, the word "or" is intended to include "and" unless the context clearly indicate otherwise. The word "or" means any one member of a particular list and

includes any combination of members of that list. Further, all units, prefixes, and symbols may be denoted in its SI accepted form.

[0026] Compositions and methods related to modified plants producing seeds high in protein content are provided. Plants that have been modified using genomic editing techniques, transformation, or mutagenesis to produce seeds having increased protein or methionine are provided. Suitable plants include oil-seed plants, such as palm, canola, sunflower, and soybean as well as, without limitation, rice, cotton, sorghum, wheat, maize, alfalfa, and barley. Modifying expression of a CGS polypeptide and/or a low methionine content seed storage polypeptide in a plant, results in a seed with high protein relative to a comparable seed not comprising the modification. The modification can be introduced using genomic editing technology, transformation, or mutagenesis, such as described herein. Plants, such as soybean plants, that show reduced feedback regulation of CGS and/or reduced expression of a low methionine content seed storage protein or a combination thereof, do not show any detrimental plant growth or plant morphology characteristics are robust, high-yielding, and produce seeds containing increased protein are provided.

[0027] Unless specified otherwise, protein, fiber, stachyose, sucrosyl-oligosaccharide and other components are measured by weight at or adjusted to a 13% moisture basis in the seeds. Seeds, plants (or plant parts thereof) producing seeds, and methods of making or using the seeds and plants (or plant parts thereof) and having the seed compositions described herein are provided.

[0028] Provided are modified plant seeds and plants producing such seeds, as described herein, containing a substantially similar or increased protein content compared with a comparable unmodified, control, null or wild-type seed. The total content of methionine of the modified plant seed may be at least or at least about 15%, 16%, 17%, 18%, 19%, 19.5%, 20%, 20.5%, 21%, 21.5%, 22%, 22.5%, 23%, 23.5%, 24%, 24.5%, or 25% 26%, 27%, 28%, 29, or 30% greater than the comparable unmodified, control, null or wild-type plant. The seed protein content may be 0.1, 0.2, 0.3, 0.4, 0.5, 0.5, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, or 3.0 percent greater when compared to the amount of protein in a comparable unmodified, control, null or wild-type seed.

[0029] The seeds can be efficiently processed to produce meal (either high-protein meal produced from dehulled seeds or conventional meal produced from whole seeds) having a high protein content compared with comparable meal produced from comparable seeds that do not contain the modification. In some embodiments, meal is provided which has a protein content that is increased by up to 3% by weight compared to meal prepared from a control soybean seed not comprising the modification, such as a null, unmodified or wild-type soybean seed. In some embodiments, meal is provided which has a total methionine content that is increased by up to 30% by weight compared to meal prepared from a control soybean seed not comprising the modification, such as a null, unmodified or wild-type soybean seed. The meal may be prepared from a plant comprising the modification and may comprise a modified polynucleotide described herein.

[0030] Provided are modified soybean seeds and plants, plant parts and plant cells which have an increased protein or methionine content and at least a comparable or increased

yield, such as described herein, relative to a comparable control unmodified seed and plant, plant part or plant cell not comprising the modification.

[0031] The modified polypeptides and polynucleotides described herein include or encode feedback insensitive CGS polypeptides or low methionine content storage polypeptides, or a combination thereof. CGS polypeptides are a group of proteins that catalyze the first committed step of methionine biosynthesis in plants. The N-terminal extension of plant CGS possess a chloroplast transit peptide (cTP) and a MTO1 region for negative feedback regulation of CGS gene expression. Glycinins (11S) and β -conglycinins (7S) are two major classes of seed storage proteins which constitute approximately 33-40% and 21-30% of total seed protein, respectively. Met and Cys comprise 3.0-4.5% of glycinins amino acid residues; whereas less than 1.0% of Met and Cys is found in the 7S β -conglycinin fractions. Suppressing genes encoding for methionine-poor storage seed proteins showed accumulation of methionine-rich proteins through proteome rebalancing. Plants with higher 11S: 7S ratio were found to have higher total amino acid content in seed.

[0032] Table 1 provides a summary of wild-type soybean CGS polypeptide and low methionine content storage polypeptide sequences.

TABLE 1

	Genomic sequence	Coding sequence	Amino acid sequence
CGS1	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 3
CGS2	SEQ ID NO: 4	SEQ ID NO: 5	SEQ ID NO: 6
7S α subunit	SEQ ID NO: 7	SEQ ID NO: 8	SEQ ID NO: 9
7S α subunit	SEQ ID NO: 10	SEQ ID NO: 11	SEQ ID NO: 12
7S α' subunit	SEQ ID NO: 13	SEQ ID NO: 14	SEQ ID NO: 15
7S α' subunit	SEQ ID NO: 16	SEQ ID NO: 17	SEQ ID NO: 18
7S α' subunit	SEQ ID NO: 19	SEQ ID NO: 20	SEQ ID NO: 21
7S β -subunit	SEQ ID NO: 22	SEQ ID NO: 23	SEQ ID NO: 24
7S β -subunit	SEQ ID NO: 25	SEQ ID NO: 26	SEQ ID NO: 27
SBP1 (Sucrose Binding Protein 1)	SEQ ID NO: 28	SEQ ID NO: 29	SEQ ID NO: 30

[0033] Provided are plants, plant cell, plant parts and seeds which have had expression of a polypeptide or polynucleotide sequence that encodes the polypeptide suppressed, knocked out, decreased, or inhibited. Examples of polypeptides include those depicted herein. In some embodiments, soybean plants, seeds, plant cells and methods are provided in which expression of both the CGS feedback regulatory function and the low methionine content polypeptide is reduced or suppressed.

[0034] In some embodiments, the modification results in the suppression of the native CGS or low methionine content storage polypeptide or both. The genome is modified to knock-out, silence, reduce or suppress expression of the native gene or the native feedback regulatory region of the CGS or both, such as by disrupting the reading frame through insertion or deletion of one or more single bases or short or long sequences, introducing a sufficient number of SNPs to disrupt function or by modifying a transcription regulatory sequence in the transcription regulatory region to include for example repressor elements, repressor binding elements or disrupted promotor enhancer elements to reduce or prevent expression of the same. In the case of CGS the insertion or deletion is an in-frame insertion or deletion that

suppresses the feedback regulation of the gene but not the catalytic activity of the protein encoded thereby through, for example, and an in-frame modification.

[0035] In some embodiments, the expression level of the seed storage polynucleotide or polypeptide in a tissue or organ of interest, such as the seed, seed endosperm, embryo, leaf, root or stalk, is less than 95, 90, 85, 80, 75, 70, 65, 60, 55, 50, 45, 40, 35, 30, 25, 20, 15, 10, 5, 4, 3, 2, or 1% of the expression level of the polynucleotide or polypeptide in a comparable control, unmodified or null tissue or organ of interest. In some embodiments, the expression level of the CGS polynucleotide or polypeptide in a tissue or organ of interest, such as the seed, seed endosperm, embryo, leaf, root or stalk, is greater than 95, 90, 85, 80, 75, 70, 65, 60, 55, 50, 45, 40, 35, 30, 25, 20, 15, 10, 5, 4, 3, 2, or 1% of the expression level of the polynucleotide or polypeptide in a comparable control, unmodified or null tissue or organ of interest. Plants producing seeds with increased protein as described herein are obtained.

[0036] In certain embodiments, plants, plant parts, and seeds comprising a loss-of-function mutation (e.g., a frameshift mutation) in at least one subunit of the 7S storage protein (beta-conglycinin) are provided. In certain embodiments, the loss-of-function mutation (e.g., a frameshift mutation) is in the coding sequence of one or more alpha' subunits of the 7S storage protein, one or more alpha subunits of the 7S storage protein, or one or more beta subunits of the 7S storage protein. In certain embodiments, the loss-of-function mutation (e.g., a frameshift mutation) is in the coding sequence of one or more alpha' subunits of the 7S storage protein, one or more alpha subunits of the 7S storage protein, and one or more beta subunits of the 7S storage protein. In certain embodiments, the loss-of-function mutation (e.g., a frameshift mutation) is in the coding sequence of each alpha' subunit of the 7S storage protein, each alpha subunit of the 7S storage protein, or each beta subunit of the 7S storage protein. In certain embodiments, the loss-of-function mutation (e.g., a frameshift mutation) is in the coding sequence of each alpha' subunit of the 7S storage protein, each alpha subunit of the 7S storage protein, and each beta subunit of the 7S storage protein. In certain embodiments, the plants, plant parts, and seeds comprise (i) a loss-of-function mutation (e.g., a frameshift mutation) in the coding sequence of each alpha' subunit of the 7S storage protein, each alpha subunit of the 7S storage protein, and each beta subunit of the 7S storage protein; and (ii) a modification (e.g., an in-frame mutation) in the coding sequence of a cystathionine gamma-synthase that suppresses feedback regulation and retains enzymatic activity.

[0037] In some embodiments, the plant, plant cell, plant part or seed includes or expresses the sequences shown in SEQ ID NOs herein or any combination thereof, or sequences sharing a percent identity with such sequences. In some embodiments, the plant, plant cell, plant part or seed includes a recombinant DNA construct or molecule or suppression construct described herein which suppresses or reduces expression of the polypeptide. Transformation methods for producing such soybean plants, plant cells, plant parts or seeds are provided.

[0038] In some embodiments, the plant further includes a heterologous nucleic acid sequence selected from the group of: a reporter gene, a selection marker, a disease resistance gene, a herbicide resistance gene, an insect resistance gene; a gene involved in carbohydrate metabolism, a gene

involved in fatty acid metabolism, a gene involved in amino acid metabolism, a gene involved in plant development, a gene involved in plant growth regulation, a gene involved in yield improvement, a gene involved in drought resistance, a gene involved in increasing nutrient utilization efficiency, a gene involved in cold resistance, a gene involved in heat resistance and a gene involved in salt resistance in plants. The heterologous nucleic acid may be introduced by back-crossing or transformation.

[0039] Provided are polynucleotides that have at least about or at least 40%, 45%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or greater sequence identity compared to a reference nucleotide sequence, such as a nucleotide sequence disclosed in the sequence listing herein, using one of the alignment programs described herein using standard parameters, as well as nucleotide substitutions, deletions, insertions, fragments thereof, and combinations thereof.

[0040] An “isolated polynucleotide” generally refers to a polymer of ribonucleotides (RNA) or deoxyribonucleotides (DNA) that is single- or double-stranded, optionally containing synthetic, non-natural or altered nucleotide bases, that is no longer in its natural environment and have been placed in a difference environment by the hand of man, for example in vitro. An isolated polynucleotide in the form of DNA may be comprised of one or more segments of cDNA, genomic DNA or synthetic DNA.

[0041] A “recombinant” nucleic acid molecule (or DNA) is used herein to refer to a nucleic acid sequence (or DNA) that is in a recombinant plant host cell. In some embodiments, an “isolated” or “recombinant” nucleic acid is free of sequences (preferably protein encoding sequences) that naturally flank the nucleic acid (i.e., sequences located at the 5' and 3' ends of the nucleic acid) in the genomic DNA of the organism from which the nucleic acid is derived.

[0042] The terms “polynucleotide”, “polynucleotide sequence”, “nucleic acid sequence”, “nucleic acid fragment”, and “isolated nucleic acid fragment” are used interchangeably herein. These terms encompass nucleotide sequences and the like. A polynucleotide may be a polymer of RNA or DNA that is single- or double-stranded, that optionally contains synthetic, non-natural or altered nucleotide bases. A polynucleotide in the form of a polymer of DNA may be comprised of one or more segments of cDNA, genomic DNA, synthetic DNA, or mixtures thereof. Nucleotides (usually found in their 5'-monophosphate form) are referred to by a single letter designation as follows: “A” for adenylate or deoxyadenylate (for RNA or DNA, respectively), “C” for cytidylate or deoxycytidylate, “G” for guanylate or deoxyguanylate, “U” for uridylylate, “T” for deoxythymidylate, “R” for purines (A or G), “Y” for pyrimidines (C or T), “K” for G or T, “H” for A or C or T, “I” for inosine, and “N” for any nucleotide.

[0043] A transcription regulatory element or sequence, or a regulatory element or sequence generally refers to a transcriptional regulatory element involved in regulating the transcription of a nucleic acid molecule such as a gene or a target gene. The regulatory element is a nucleic acid and may include a promoter, an enhancer, an intron, a 5'-untranslated region (5'-UTR, also known as a leader sequence),

or a 3'-UTR or a combination thereof. A regulatory element may act in “cis” or “trans”, and generally it acts in “cis”, i.e. it activates expression of genes located on the same nucleic acid molecule, e.g. a chromosome, where the regulatory element is located. The nucleic acid molecule regulated by a regulatory element does not necessarily have to encode a functional peptide or polypeptide, e.g., the regulatory element can modulate the expression of a short interfering RNA or an anti-sense RNA.

[0044] In some embodiments, the modified polynucleotide includes a modified transcriptional enhancer sequence. An enhancer element is any nucleic acid molecule that increases transcription of a nucleic acid molecule when functionally linked to a promoter regardless of its relative position. An enhancer may be an innate element of the promoter or a heterologous element inserted to enhance the amount of promoter activity or tissue-specificity of a promoter.

[0045] Various enhancers may be used including introns with gene expression enhancing properties in plants (US Patent Application Publication Number 2009/0144863), the ubiquitin intron (i.e., the maize ubiquitin intron 1 (see, for example, NCBI sequence S94464)), the omega enhancer or the omega prime enhancer (Gallie, et al., (1989) Molecular Biology of RNA ed. Cech (Liss, New York) 237-256 and Gallie, et al., (1987) Gene 60:217-25), the CaMV 35S enhancer (see, e.g., Benfey, et al., (1990) EMBO J. 9:1685-96) and the enhancers of U.S. Pat. No. 7,803,992 may also be used, each of which is incorporated by reference. The above list of transcriptional enhancers is not meant to be limiting. Any appropriate transcriptional enhancer can be used in the embodiments.

[0046] A repressor (also sometimes called herein silencer, repressor element, or repressor binding element) is defined as any nucleic acid molecule which inhibits the transcription when functionally linked to a promoter regardless of relative position.

[0047] “Promoter” generally refers to a nucleic acid fragment capable of controlling transcription of another nucleic acid fragment. A promoter generally includes a core promoter (also known as minimal promoter) sequence that includes a minimal regulatory region to initiate transcription, that is a transcription start site. Generally, a core promoter includes a TATA box and a GC rich region associated with a CAAT box or a CCAAT box. These elements act to bind RNA polymerase II to the promoter and assist the polymerase in locating the RNA initiation site. Some promoters may not have a TATA box or CAAT box or a CCAAT box, but instead may contain an initiator element for the transcription initiation site. A core promoter is a minimal sequence required to direct transcription initiation and generally may not include enhancers or other UTRs. Promoters may be derived in their entirety from a native gene, or be composed of different elements derived from different promoters found in nature, or even comprise synthetic DNA segments. It is understood by those skilled in the art that different promoters may direct the expression of a gene in different tissues or cell types, or at different stages of development, or in response to different environmental conditions. Core promoters are often modified to produce artificial, chimeric, or hybrid promoters, and can further be used in combination with other regulatory elements, such as cis-elements, 5'UTRs, enhancers, or introns, that are either heterologous to an active core promoter or combined with its own partial or complete regulatory elements.

[0048] The term “cis-element” generally refers to transcriptional regulatory element that affects or modulates expression of an operably linked transcribable polynucleotide, where the transcribable polynucleotide is present in the same DNA sequence. A cis-element may function to bind transcription factors, which are trans-acting polypeptides that regulate transcription.

[0049] The termination region may be native with the transcriptional initiation region, may be native with the operably linked DNA sequence of interest, may be native with the plant host or may be derived from another source (i.e., foreign or heterologous to the promoter, the sequence of interest, the plant or any combination thereof).

[0050] The sequences include one or more contiguous nucleotides. “Contiguous nucleotides” is used herein to refer to nucleotide residues that are immediately adjacent to one another.

[0051] As used herein non-genomic nucleic acid sequence, nucleic acid molecule or polynucleotide refers to a nucleic acid molecule that has one or more changes in the nucleic acid sequence compared to a native or genomic nucleic acid sequence. In some embodiments, the change to a native or genomic nucleic acid molecule includes but is not limited to: changes in the nucleic acid sequence due to the degeneracy of the genetic code; optimization of the nucleic acid sequence for expression in plants; changes in the nucleic acid sequence to introduce at least one amino acid substitution, insertion, deletion and/or addition compared to the native or genomic sequence; deletion of one or more upstream or downstream regulatory regions associated with the genomic nucleic acid sequence; insertion of one or more heterologous upstream or downstream regulatory regions; deletion of the 5' and/or 3' untranslated region associated with the genomic nucleic acid sequence; insertion of a heterologous 5' and/or 3' untranslated region; and modification of a polyadenylation site. In some embodiments, the non-genomic nucleic acid molecule is a synthetic nucleic acid sequence.

[0052] Provided are polypeptides having at least about or at least 40%, 45%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or greater sequence identity compared to polypeptides referenced in the sequence listing, as well as amino acid substitutions, deletions, insertions, fragments thereof, and combinations thereof. The term “about” when used herein in context with percent sequence identity means $\pm 0.5\%$. These values can be appropriately adjusted to determine corresponding homology of proteins considering amino acid similarity and the like.

[0053] In some embodiments, the sequence identity is against the full-length sequence of a polypeptide disclosed in the sequence listing. In some embodiments, the polypeptide retains activity or shows enhanced or reduced activity.

[0054] As used herein, the term “protein,” “peptide molecule,” or “polypeptide” includes those molecules that undergo modification, including post-translational modifications, such as, but not limited to, disulfide bond formation, glycosylation, phosphorylation or oligomerization.

[0055] The terms “amino acid” and “amino acids” refer to all naturally occurring L-amino acids.

[0056] Variants may be made by making random mutations or the variants may be designed. In the case of designed mutants, there is a high probability of generating variants with similar activity to the native polypeptide when amino acid identity is maintained in critical regions of the polypeptide which account for biological activity or are involved in the determination of three-dimensional configuration which ultimately is responsible for the biological activity. A high probability of retaining activity will also occur if substitutions are conservative. Amino acids may be placed in the following classes: non-polar, uncharged polar, basic, and acidic. Conservative substitutions whereby an amino acid of one class is replaced with another amino acid of the same type are least likely to materially alter the biological activity of the variant.

[0057] Alternatively, alterations may be made to the protein sequence of many proteins at the amino or carboxy terminus without substantially affecting activity. This can include insertions, deletions or alterations introduced by modern molecular methods, such as polymerase chain reaction (PCR), including PCR amplifications that alter or extend the protein coding sequence by inclusion of amino acid encoding sequences in the oligonucleotides utilized in the PCR amplification. Alternatively, the protein sequences added can include entire protein-coding sequences, to generate protein fusions. Such fusion proteins are often used to (1) increase expression of a protein of interest (2) introduce a binding domain, enzymatic activity or epitope to facilitate either protein purification, protein detection or other experimental uses (3) target secretion or translation of a protein to a subcellular organelle, such as the periplasmic space of Gram-negative bacteria, mitochondria or chloroplasts of plants or the endoplasmic reticulum of eukaryotic cells, the latter of which often results in glycosylation of the protein.

[0058] To determine the percent identity of two amino acid sequences or of two nucleic acids, the sequences are aligned for optimal comparison purposes. The percent identity between the two sequences is a function of the number of identical positions shared by the sequences (i.e., percent identity = number of identical positions/total number of positions (e.g., overlapping positions) $\times 100$). In one embodiment, the two sequences are the same length. In another embodiment, the percent identity is calculated across the entirety of the reference sequence. The percent identity between two sequences can be determined using techniques like those described below, with or without allowing gaps. In calculating percent identity, typically exact matches are counted. A gap, (a position in an alignment where a residue is present in one sequence but not in the other) is regarded as a position with non-identical residues.

[0059] The determination of percent identity between two sequences can be accomplished using a mathematical algorithm. A non-limiting example of a mathematical algorithm utilized for the comparison of two sequences is the algorithm incorporated into the BLASTN and BLASTX programs. Karlin and Altschul (1990) Proc. Nat'l. Acad. Sci. USA 87:2264, Altschul et al. (1990) J. Mol. Biol. 215:403, and Karlin and Altschul (1993) Proc. Nat'l. Acad. Sci. USA 90:5873-5877. BLAST nucleotide searches can be performed with the BLASTN program, score=100, word length=12, to obtain nucleotide sequences homologous to nucleic acid molecules disclosed herein. BLAST protein searches can be performed with the BLASTX program, score=50, word length=3, to obtain amino acid sequences

homologous to polypeptides disclosed herein. To obtain gapped alignments for comparison purposes, Gapped BLAST (in BLAST 2.0) can be utilized as described in Altschul et al. (1997) *Nucleic Acids Res.* 25:3389. Alternatively, PSI-Blast can be used to perform an iterated search that detects distant relationships between molecules. See Altschul et al. (1997) *supra*. When utilizing BLAST, Gapped BLAST, and PSI-Blast programs, the default parameters of the respective programs (e.g., BLASTX and BLASTN) can be used. Alignment may also be performed manually by inspection.

[0060] Another non-limiting example of a mathematical algorithm utilized for the comparison of sequences is the ClustalW algorithm (Higgins et al. (1994) *Nucleic Acids Res.* 22:4673-4680). ClustalW compares sequences and aligns the entirety of the amino acid or DNA sequence, and thus can provide data about the sequence conservation of the entire amino acid sequence. The ClustalW algorithm is used in several commercially available DNA/amino acid analysis software packages, such as the ALIGNX module of the Vector NTI Program Suite (Invitrogen Corporation, Carlsbad, Calif.). After alignment of amino acid sequences with ClustalW, the percent amino acid identity can be assessed. A non-limiting example of a software program useful for analysis of ClustalW alignments is GENEDOC™. GENE-DOC™ (Karl Nicholas) allows assessment of amino acid (or DNA) similarity and identity between multiple proteins. Another non-limiting example of a mathematical algorithm utilized for the comparison of sequences is the algorithm of Myers and Miller (1988) *CABIOS* 4 (1): 11-17. Such an algorithm is incorporated into the ALIGN program (version 2.0), which is part of the GCG Wisconsin Genetics Software Package, Version 10 (available from Accelrys, Inc., San Diego, Calif., USA). When utilizing the ALIGN program for comparing amino acid sequences, a PAM120 weight residue table, a gap length penalty of 12, and a gap penalty of 4 can be used. Unless otherwise stated, GAP Version 10, which uses the algorithm of Needleman and Wunsch (1970) *J. Mol. Biol.* 48 (3): 443-453, will be used to determine sequence identity or similarity using the following parameters: % identity and % similarity for a nucleotide sequence using GAP Weight of 50 and Length Weight of 3, and the nws gap-dna.cmp scoring matrix; % identity or % similarity for an amino acid sequence using GAP weight of 8 and length weight of 2, and the BLOSUM62 scoring program. Equivalent programs may also be used. By “equivalent program” is intended any sequence comparison program that, for any two sequences in question, generates an alignment having identical nucleotide residue matches and an identical percent sequence identity when compared to the corresponding alignment generated by GAP Version 10.

[0061] Isolated or recombinant nucleic acid molecules comprising nucleic acid sequences encoding cystathionine gamma-synthase or alpha', alpha, and beta subunits of the 7S storage polypeptides or biologically active portions thereof, as well as nucleic acid molecules sufficient for use as hybridization probes to identify nucleic acid molecules encoding proteins with regions of sequence homology are provided. As used herein, the term “nucleic acid molecule” refers to DNA molecules (e.g., recombinant DNA, cDNA, genomic DNA, plastid DNA, mitochondrial DNA) and RNA molecules (e.g., mRNA) and analogs of the DNA or RNA generated using nucleotide analogs. The nucleic acid mol-

ecule can be single-stranded or double-stranded, but preferably is double-stranded DNA.

[0062] Nucleotide sequences that encode cystathionine gamma-synthase or alpha', alpha, and beta subunits of the 7S storage polypeptides, variants and truncations, may be synthesized and cloned into standard plasmid vectors by conventional means, or may be obtained by standard molecular biology manipulation of other constructs containing the nucleotide sequences.

[0063] In some embodiments, plants, plant parts, plant cells, seeds and methods of making and using thereof include a genome modified to contain an in-frame or frame disruption deletion. Plants, seeds, plant parts and plant cell comprising this deletion and methods of making such plants, seeds, plant parts and plant cells are provided.

[0064] In some embodiments, the nucleic acid molecule is a polynucleotide having the sequence set forth herein and variants, fragments and complements thereof. Nucleic acid sequences that are complementary to a nucleic acid sequence of the embodiments or that hybridize to a sequence of the embodiments are also encompassed. The nucleic acid sequences can be used in DNA constructs or expression cassettes for transformation and expression in organisms, including microorganisms and plants. The nucleotide or amino acid sequences may be synthetic sequences that have been designed for expression in an organism including, but not limited to, a microorganism or a plant.

[0065] In some embodiments, the nucleic acid molecule encoding the polypeptide is a non-genomic nucleic acid sequence.

[0066] In some embodiments, the nucleic acid molecule encoding a polypeptide is a non-genomic polynucleotide having a nucleotide sequence having at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or greater identity, to the nucleic acid sequence herein, wherein the encoded polypeptide is functional to increase protein or total methionine in a seed.

[0067] In some embodiments, the polynucleotide encodes a polypeptide having, or the polypeptide has, at least about 40%, 45%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or greater sequence identity compared to SEQ ID NO: 3, 6, 9, 12, 15, 18, 21, 24, 27, or 30 and optionally has at least one amino acid substitution, deletion, insertion or combination therefore, compared to the native sequence.

[0068] In some embodiments, the nucleic acid molecule encodes a polypeptide comprising, or the polypeptide comprises, an amino acid sequence having at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or greater identity across the entire length of an amino acid sequence disclosed herein.

[0069] In some embodiments, the nucleic acid encodes a polypeptide having or the polypeptide has, at least about 50%, 55%, 60%, 65%, 70%, 75%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or greater sequence

identity compared to a sequence disclosed herein. In some embodiments, the sequence identity is calculated using ClustalW algorithm in the ALIGNX® module of the Vector NTI® Program Suite (Invitrogen Corporation, Carlsbad, Calif.) with all default parameters. In some embodiments, the sequence identity is across the entire length of polypeptide calculated using ClustalW algorithm in the ALIGNX module of the Vector NTI Program Suite (Invitrogen Corporation, Carlsbad, Calif.) with all default parameters.

[0070] The embodiments also encompass nucleic acid molecules encoding cystathionine gamma-synthase or 7S storage polypeptide variants. “Variants” of the polypeptide encoding nucleic acid sequences include those sequences that encode the polypeptides disclosed herein but that differ conservatively because of the degeneracy of the genetic code as well as those that are sufficiently identical as discussed above. Naturally occurring allelic variants can be identified with the use of well-known molecular biology techniques, such as polymerase chain reaction (PCR) and hybridization techniques as outlined below. Variant nucleic acid sequences also include synthetically derived nucleic acid sequences that have been generated, for example, by using site-directed mutagenesis but which still encode the polypeptides disclosed as discussed below.

[0071] Oligonucleotide probes and methods for detecting the polynucleotides described herein are provided. Oligonucleotide probes are detectable nucleotide sequences, such as by an appropriate radioactive label or may be fluorescence as described in, for example, U.S. Pat. No. 6,268,132. As is well known in the art, if the probe molecule and nucleic acid sample hybridize by forming strong base-pairing bonds between the two molecules, it can be reasonably assumed that the probe and sample have substantial sequence homology. Preferably, hybridization is conducted under stringent conditions by techniques well-known in the art, as described, for example, in Keller and Manak (1993). Detection of the probe provides a means for determining in a known manner whether hybridization has occurred. Such a probe analysis provides a rapid method for identifying modified genes of cystathionine gamma-synthase or alpha', alpha, and beta subunits of the 7S storage polypeptides, which modified genes and methods are provided. The nucleotide segments which are used as probes can be synthesized using a DNA synthesizer and standard procedures. These nucleotide sequences can also be used as PCR primers to amplify genes.

[0072] As is well known to those skilled in molecular biology, similarity of two nucleic acids can be characterized by their tendency to hybridize. Provided are nucleic acids that hybridize to those sequences disclosed herein under stringent conditions. As used herein the terms “stringent conditions” or “stringent hybridization conditions” are intended to refer to conditions under which a probe or nucleic acid will hybridize (anneal) to a particular sequence to a detectably greater degree than to other sequences (e.g. at least 2-fold over background).

[0073] Provided are nucleotide constructs comprising sequences described herein. The use of the term “nucleotide constructs” herein is not intended to limit the embodiments to nucleotide constructs comprising DNA. Nucleotide constructs particularly polynucleotides and oligonucleotides composed of ribonucleotides and combinations of ribonucleotides and deoxyribonucleotides may also be employed in the methods disclosed herein. The nucleotide

constructs, nucleic acids, and nucleotide sequences of the embodiments additionally encompass all complementary forms of such constructs, molecules, and sequences. Further, the nucleotide constructs, nucleotide molecules, and nucleotide sequences of the embodiments encompass all nucleotide constructs, molecules, and sequences which can be employed in the methods of the embodiments for transforming plants including, but not limited to, those comprised of deoxyribonucleotides, ribonucleotides, and combinations thereof. Such deoxyribonucleotides and ribonucleotides include both naturally occurring molecules and synthetic analogues. The nucleotide constructs, nucleic acids, and nucleotide sequences of the embodiments also encompass all forms of nucleotide constructs including, but not limited to, single-stranded forms, double-stranded forms, hairpins, stem-and-loop structures and the like.

[0074] Provided are plants, plant cells, plant seeds and plant nuclei that are modified by gene editing. In some embodiments, gene editing may be facilitated through the induction of a double-stranded break (DSB) or single-strand break, in a defined position in the genome near the desired alteration. DSBs can be induced using any DSB-inducing agent available, including, but not limited to, TALENs (transcription activator-like effector nucleases), meganucleases, zinc finger nucleases, Cas9-gRNA systems (based on bacterial CRISPR-Cas systems), guided cpfl endonuclease systems, and the like. In some embodiments, the introduction of a DSB can be combined with the introduction of a polynucleotide modification template. In some embodiments, the methods do not use TALENs enzymes or technology and plants and seeds are produced from methods which do not use TALENs enzymes or technology.

[0075] A polynucleotide modification template can be introduced into a cell by any method known in the art, such as, but not limited to, transient introduction methods, transfection, electroporation, microinjection, particle mediated delivery, topical application, whiskers mediated delivery, delivery via cell-penetrating peptides, or mesoporous silica nanoparticle (MSN)-mediated direct delivery.

[0076] The polynucleotide modification template can be introduced into a cell as a single stranded polynucleotide molecule, a double stranded polynucleotide molecule, or as part of a circular DNA (vector DNA). The polynucleotide modification template can also be tethered to the guide RNA and/or the Cas endonuclease. Tethered DNAs can allow for co-localizing target and template DNA, useful in genome editing and targeted genome regulation, and can also be useful in targeting post-mitotic cells where function of endogenous HR machinery is expected to be highly diminished (Mali et al. 2013 Nature Methods Vol. 10:957-963.) The polynucleotide modification template may be present transiently in the cell or it can be introduced via a viral replicon.

[0077] A “modified nucleotide” or “edited nucleotide” refers to a nucleotide sequence of interest that comprises at least one alteration when compared to its non-modified nucleotide sequence. Such “alterations” include, for example: (i) replacement of at least one nucleotide, (ii) a deletion of at least one nucleotide, (iii) an insertion of at least one nucleotide, or (iv) any combination of (i)-(iii).

[0078] The term “polynucleotide modification template” includes a polynucleotide that comprises at least one nucleotide modification when compared to the nucleotide sequence to be edited. A nucleotide modification can be at least one

nucleotide substitution, addition or deletion. Optionally, the polynucleotide modification template can further comprise homologous nucleotide sequences flanking the at least one nucleotide modification, wherein the flanking homologous nucleotide sequences provide sufficient homology to the desired nucleotide sequence to be edited.

[0079] The process for editing a genomic sequence combining DSB and modification templates generally comprises: providing to a host cell, a DSB-inducing agent, or a nucleic acid encoding a DSB-inducing agent, that recognizes a target sequence in the chromosomal sequence and is able to induce a DSB in the genomic sequence, and at least one polynucleotide modification template comprising at least one nucleotide alteration when compared to the nucleotide sequence to be edited. The polynucleotide modification template can further comprise nucleotide sequences flanking the at least one nucleotide alteration, in which the flanking sequences are substantially homologous to the chromosomal region flanking the DSB.

[0080] The endonuclease can be provided to a cell by any method known in the art, for example, but not limited to transient introduction methods, transfection, microinjection, and/or topical application or indirectly via recombination constructs. The endonuclease can be provided as a protein or as a guided polynucleotide complex directly to a cell or indirectly via recombination constructs. The endonuclease can be introduced into a cell transiently or can be incorporated into the genome of the host cell using any method known in the art. In the case of a CRISPR-Cas system, uptake of the endonuclease and/or the guided polynucleotide into the cell can be facilitated with a Cell Penetrating Peptide (CPP) as described in WO2016073433 published May 12, 2016.

[0081] TAL effector nucleases (TALEN) are a class of sequence-specific nucleases that can be used to make double-strand breaks at specific target sequences in the genome of a plant or other organism. (Miller et al. (2011) *Nature Biotechnology* 29:143-148).

[0082] Endonucleases are enzymes that cleave the phosphodiester bond within a polynucleotide chain. Endonucleases include restriction endonucleases, which cleave DNA at specific sites without damaging the bases, and meganucleases, also known as homing endonucleases (HEases), which like restriction endonucleases, bind and cut at a specific recognition site, however the recognition sites for meganucleases are typically longer, about 18 bp or more (patent application PCT/US12/30061, filed on Mar. 22, 2012). Meganucleases have been classified into four families based on conserved sequence motifs, the families are the LAGLIDADG, GIY-YIG, H-N-H, and His-Cys box families. These motifs participate in the coordination of metal ions and hydrolysis of phosphodiester bonds.

[0083] Zinc finger nucleases (ZFNs) are engineered double-strand break inducing agents comprised of a zinc finger DNA binding domain and a double-strand-break-inducing agent domain. Recognition site specificity is conferred by the zinc finger domain, which typically comprising two, three, or four zinc fingers, for example having a C2H2 structure, however other zinc finger structures are known and have been engineered.

[0084] Genome editing using DSB-inducing agents, such as Cas9-gRNA complexes, has been described, for example in U.S. Patent Application US 2015-0082478 A1, published on Mar. 19, 2015, WO2015/026886 A1, published on Feb.

26, 2015, WO2016007347, published on Jan. 14, 2016, and WO201625131, published on Feb. 18, 2016, all of which are incorporated by reference herein.

[0085] The term “Cas gene” herein refers to a gene that is generally coupled, associated or close to, or in the vicinity of flanking CRISPR loci in bacterial systems. The terms “Cas gene”, “CRISPR-associated (Cas) gene” are used interchangeably herein. The term “Cas endonuclease” herein refers to a protein encoded by a Cas gene. A Cas endonuclease herein, when in complex with a suitable polynucleotide component, is capable of recognizing, binding to, and optionally nicking or cleaving all or part of a specific DNA target sequence. A Cas endonuclease described herein comprises one or more nuclease domains. Cas endonucleases of the disclosure includes those having a HNH or HNH-like nuclease domain and/or a RuvC or RuvC-like nuclease domain. A Cas endonuclease of the disclosure includes a Cas9 protein, a Cpf1 protein, a C2c1 protein, a C2c2 protein, a C2c3 protein, Cas3, Cas 5, Cas7, Cas8, Cas10, or complexes of these.

[0086] As used herein, the terms “guide polynucleotide/Cas endonuclease complex”, “guide polynucleotide/Cas endonuclease system”, “guide polynucleotide/Cas complex”, “guide polynucleotide/Cas system”, “guided Cas system” are used interchangeably herein and refer to at least one guide polynucleotide and at least one Cas endonuclease that are capable of forming a complex, wherein said guide polynucleotide/Cas endonuclease complex can direct the Cas endonuclease to a DNA target site, enabling the Cas endonuclease to recognize, bind to, and optionally nick or cleave (introduce a single or double strand break) the DNA target site. A guide polynucleotide/Cas endonuclease complex herein can comprise Cas protein(s) and suitable polynucleotide component(s) of any of the four known CRISPR systems (Horvath and Barrangou, 2010, *Science* 327:167-170) such as a type I, II, or III CRISPR system. A Cas endonuclease unwinds the DNA duplex at the target sequence and optionally cleaves at least one DNA strand, as mediated by recognition of the target sequence by a polynucleotide (such as, but not limited to, a crRNA or guide RNA) that is in complex with the Cas protein. Such recognition and cutting of a target sequence by a Cas endonuclease typically occurs if the correct protospacer-adjacent motif (PAM) is located at or adjacent to the 3' end of the DNA target sequence. Alternatively, a Cas protein herein may lack DNA cleavage or nicking activity, but can still specifically bind to a DNA target sequence when complexed with a suitable RNA component. (See also U.S. Patent Application US 2015-0082478 A1, published on Mar. 19, 2015 and US 2015-0059010 A1, published on Feb. 26, 2015, both are hereby incorporated in its entirety by reference).

[0087] A guide polynucleotide/Cas endonuclease complex can cleave one or both strands of a DNA target sequence. A guide polynucleotide/Cas endonuclease complex that can cleave both strands of a DNA target sequence typically comprise a Cas protein that has all of its endonuclease domains in a functional state (e.g., wild type endonuclease domains or variants thereof retaining some or all activity in each endonuclease domain). Non-limiting examples of Cas9 nickases suitable for use herein are disclosed in U.S. Patent Appl. Publ. No. 2014/0189896, which is incorporated herein by reference.

[0088] Other Cas endonuclease systems have been described in PCT patent applications PCT/US16/32073,

filed May 12, 2016 and PCT/US16/32028 filed May 12, 2016, both applications incorporated herein by reference.

[0089] “Cas9” (formerly referred to as Cas5, Csn1, or Csx12) herein refers to a Cas endonuclease of a type II CRISPR system that forms a complex with a crNucleotide and a tracrNucleotide, or with a single guide polynucleotide, for specifically recognizing and cleaving all or part of a DNA target sequence. Cas9 protein comprises a RuvC nuclease domain and an HNH (H-N-H) nuclease domain, each of which can cleave a single DNA strand at a target sequence (the concerted action of both domains leads to DNA double-strand cleavage, whereas activity of one domain leads to a nick). In general, the RuvC domain comprises subdomains I, II and III, where domain I is located near the N-terminus of Cas9 and subdomains II and III are located in the middle of the protein, flanking the HNH domain (Hsu et al., Cell 157:1262-1278). A type II CRISPR system includes a DNA cleavage system utilizing a Cas9 endonuclease in complex with at least one polynucleotide component. For example, a Cas9 can be in complex with a CRISPR RNA (crRNA) and a trans-activating CRISPR RNA (tracrRNA). In another example, a Cas9 can be in complex with a single guide RNA.

[0090] Any guided endonuclease can be used in the methods disclosed herein. Such endonucleases include, but are not limited to Cas9 and Cpf1 endonucleases. Many endonucleases have been described to date that can recognize specific PAM sequences (see for example—Jinek et al. (2012) Science 337 p 816-821, PCT patent applications PCT/US16/32073, filed May 12, 2016 and PCT/US16/32028 filed May 12, 2016 and Zetsche B et al. 2015. Cell 163, 1013) and cleave the target DNA at a specific position. It is understood that based on the methods and embodiments described herein utilizing a guided Cas system one can now tailor these methods such that they can utilize any guided endonuclease system.

[0091] The guide polynucleotide can also be a single molecule (also referred to as single guide polynucleotide) comprising a crNucleotide sequence linked to a tracrNucleotide sequence. The single guide polynucleotide comprises a first nucleotide sequence domain (referred to as Variable Targeting domain or VT domain) that can hybridize to a nucleotide sequence in a target DNA and a Cas endonuclease recognition domain (CER domain), that interacts with a Cas endonuclease polypeptide. By “domain” it is meant a contiguous stretch of nucleotides that can be RNA, DNA, and/or RNA-DNA-combination sequence. The VT domain and/or the CER domain of a single guide polynucleotide can comprise a RNA sequence, a DNA sequence, or a RNA-DNA-combination sequence. The single guide polynucleotide being comprised of sequences from the crNucleotide and the tracrNucleotide may be referred to as “single guide RNA” (when composed of a contiguous stretch of RNA nucleotides) or “single guide DNA” (when composed of a contiguous stretch of DNA nucleotides) or “single guide RNA-DNA” (when composed of a combination of RNA and DNA nucleotides). The single guide polynucleotide can form a complex with a Cas endonuclease, wherein said guide polynucleotide/Cas endonuclease complex (also referred to as a guide polynucleotide/Cas endonuclease system) can direct the Cas endonuclease to a genomic target site, enabling the Cas endonuclease to recognize, bind to, and optionally nick or cleave (introduce a single or double strand break) the target site.

[0092] The term “variable targeting domain” or “VT domain” is used interchangeably herein and includes a nucleotide sequence that can hybridize (is complementary) to one strand (nucleotide sequence) of a double strand DNA target site. In some embodiments, the variable targeting domain comprises a contiguous stretch of 12 to 30 nucleotides. The variable targeting domain can be composed of a DNA sequence, a RNA sequence, a modified DNA sequence, a modified RNA sequence, or any combination thereof.

[0093] The terms “single guide RNA” and “sgRNA” are used interchangeably herein and relate to a synthetic fusion of two RNA molecules, a crRNA (CRISPR RNA) comprising a variable targeting domain (linked to a tracr mate sequence that hybridizes to a tracrRNA), fused to a tracrRNA (trans-activating CRISPR RNA). The single guide RNA can comprise a crRNA or crRNA fragment and a tracrRNA or tracrRNA fragment of the type II CRISPR/Cas system that can form a complex with a type II Cas endonuclease, wherein said guide RNA/Cas endonuclease complex can direct the Cas endonuclease to a DNA target site, enabling the Cas endonuclease to recognize, bind to, and optionally nick or cleave (introduce a single or double strand break) the DNA target site.

[0094] The terms “guide RNA/Cas endonuclease complex”, “guide RNA/Cas endonuclease system”, “guide RNA/Cas complex”, “guide RNA/Cas system”, “gRNA/Cas complex”, “gRNA/Cas system”, “RNA-guided endonuclease”, “RGEN” are used interchangeably herein and refer to at least one RNA component and at least one Cas endonuclease that are capable of forming a complex, wherein said guide RNA/Cas endonuclease complex can direct the Cas endonuclease to a DNA target site, enabling the Cas endonuclease to recognize, bind to, and optionally nick or cleave (introduce a single or double strand break) the DNA target site. A guide RNA/Cas endonuclease complex herein can comprise Cas protein(s) and suitable RNA component(s) of any of the four known CRISPR systems (Horvath and Barrangou, 2010, Science 327:167-170) such as a type I, II, or III CRISPR system. A guide RNA/Cas endonuclease complex can comprise a Type II Cas9 endonuclease and at least one RNA component (e.g., a crRNA and tracrRNA, or a gRNA).

[0095] The guide polynucleotide can be introduced into a cell transiently, as single stranded polynucleotide or a double stranded polynucleotide, using any method known in the art such as, but not limited to, particle bombardment, *Agrobacterium* transformation or topical applications. The guide polynucleotide can also be introduced indirectly into a cell by introducing a recombinant DNA molecule (via methods such as, but not limited to, particle bombardment or *Agrobacterium* transformation) comprising a heterologous nucleic acid fragment encoding a guide polynucleotide, operably linked to a specific promoter that can transcribe the guide RNA in said cell. The specific promoter can be, but is not limited to, a RNA polymerase III promoter, which allow for transcription of RNA with precisely defined, unmodified, 5'- and 3'-ends (DiCarlo et al., Nucleic Acids Res. 41:4336-4343; Ma et al., Mol. Ther. Nucleic Acids 3: e161) as described in WO2016025131, published on Feb. 18, 2016, incorporated herein in its entirety by reference.

[0096] Provided are plants, plant cells, plant seeds and plant nuclei that are transformed with sequences described herein. Transformation may be stable or transient. “Stable

transformation” as used herein means that the nucleotide construct introduced into a plant integrates into the genome of the plant and is capable of being inherited by the progeny thereof. “Transient transformation” as used herein means that a polynucleotide is introduced into the plant and does not integrate into the genome of the plant or a polypeptide is introduced into a plant. “Plant” as used herein refers to whole plants, plant organs (e.g., leaves, stems, roots, etc.), seeds, plant cells, propagules, embryos and progeny of the same. Plant cells can be differentiated or undifferentiated (e.g. callus, suspension culture cells, protoplasts, leaf cells, root cells, phloem cells and pollen).

[0097] Transformation methods include introduction of a recombinant DNA construct comprising an expression cassette. Provided are constructs which include one or more heterologous promoter sequences operably connected to one or more polynucleotides encoding polypeptides disclosed herein and appropriate transcription termination sequences and plants, seeds, cells and nuclei containing the recombinant DNA construct or expression cassette.

[0098] Transformation methods include introduction of a suppression DNA construct or a construct that results in increased expression of a target gene, such as encoding the cystathionine gamma-synthase or alpha', alpha, and beta subunits of the 7S storage polypeptides. “Suppression DNA construct” is a recombinant DNA construct which when transformed or stably integrated into the genome of the plant, results in “silencing” of a target gene in the plant. The target gene may be endogenous or transgenic to the plant. “Silencing,” as used herein with respect to the target gene, refers generally to the suppression of levels of mRNA or protein/enzyme expressed by the target gene, and/or the level of the enzyme activity or protein functionality. The term “suppression” includes lower, reduce, decline, decrease, inhibit, eliminate and prevent. “Silencing” or “gene silencing” does not specify mechanism and is inclusive, and not limited to, anti-sense, cosuppression, viral-suppression, hairpin suppression, stem-loop suppression, RNAi-based approaches and small RNA-based approaches.

[0099] The embodiments further relate to plant-propagating material of a transformed plant of the embodiments including, but not limited to, seeds, tubers, corms, bulbs, leaves and cuttings of roots and shoots. Methods of plant breeding by crossing a modified plant described herein with a second different plant are provided. Progeny plants, plant cells, seeds and plant nuclei from such breeding methods are provided, such as F1 progeny plants, plant cells, seeds and plant nuclei.

[0100] Transformation of any plant species can be carried out, including, but not limited to, monocots and dicots. Examples of plants of interest include, but are not limited to, corn (*Zea mays*), *Brassica* spp. (e.g., *B. napus*, *B. rapa*, *B. juncea*), particularly those *Brassica* species useful as sources of seed oil, alfalfa (*Medicago sativa*), rice (*Oryza sativa*), rye (*Secale cereale*), sorghum (*Sorghum bicolor*, *Sorghum vulgare*), millet (e.g., pearl millet (*Pennisetum glaucum*), proso millet (*Panicum miliaceum*), foxtail millet (*Setaria italica*), finger millet (*Eleusine coracana*)), sunflower (*Helianthus annuus*), safflower (*Carthamus tinctorius*), wheat (*Triticum aestivum*), soybean (*Glycine max*), tobacco (*Nicotiana tabacum*), potato (*Solanum tuberosum*), peanuts (*Arachis hypogaea*), cotton (*Gossypium barbadense*, *Gossypium hirsutum*), sweet potato (*Ipomoea batatas*), cassava (*Manihot esculenta*), coffee (*Coffea* spp.),

coconut (*Cocos nucifera*), pineapple (*Ananas comosus*), *Citrus* trees (*Citrus* spp.), cocoa (*Theobroma cacao*), tea (*Camellia sinensis*), banana (*Musa* spp.), avocado (*Persea americana*), fig (*Ficus casica*), guava (*Psidium guajava*), mango (*Mangifera indica*), olive (*Olea europaea*), papaya (*Carica papaya*), cashew (*Anacardium occidentale*), macadamia (*Macadamia integrifolia*), almond (*Prunus amygdalus*), sugar beets (*Beta vulgaris*), sugarcane (*Saccharum* spp.), oat (*Avena sativa*), barley (*Hordeum vulgare*), vegetables, ornamentals, and conifers.

[0101] Plants of interest include grain plants that provide seeds of interest, oil-seed plants, and leguminous plants. Seeds of interest include grain seeds, such as corn, wheat, barley, rice, sorghum, rye, millet, etc. Oil-seed plants include cotton, soybean, safflower, sunflower, *Brassica*, maize, alfalfa, palm, coconut, flax, castor, olive, etc. Leguminous plants include beans and peas. Beans include guar, locust bean, fenugreek, soybean, garden beans, cowpea, mung bean, lima bean, fava bean, lentils, chickpea, etc.

[0102] The methods comprise providing a plant or plant cell expressing a polynucleotide encoding the polypeptide sequence disclosed herein and growing the plant or a seed thereof in a field. In some embodiments, the expression of the modified polypeptide results in a plant producing increased yield or biomass.

[0103] As defined herein, the “yield” of the plant refers to the quality and/or quantity of biomass produced by the plant. “Biomass” as used herein refers to any measured plant product. An increase in biomass production is any improvement in the yield of the measured plant product. Increasing plant yield has several commercial applications. An increase in yield can comprise any statistically significant increase including, but not limited to, at least a 1% increase, at least a 3% increase, at least a 5% increase, at least a 10% increase, at least a 20% increase, at least a 30%, at least a 50%, at least a 70%, at least a 100% or a greater increase in yield compared to a plant not expressing the modified sequence.

[0104] Methods of detecting the modified polynucleotides are provided. Methods of extracting modified DNA from a sample or detecting the presence of DNA corresponding to the modified genomic sequences comprising deletions, can be carried out. Sequences may contain a feature following which feature can be any can be A, G, C or T. For example, an “n” at a position can be entirely absent or a length of 0 to 1, 0 to 2, 0 to 3, 0 to 4, 0 to 5, 0 to 10, 0 to 15, 0 to 20, 0 to 25, 0 to 30, 0 to 35, 0 to 40, or 0 to 45 nucleotides, and when present may contain any combination of C, T, G or A. Such methods of detecting polynucleotides comprise contacting a sample comprising soybean genomic DNA with a DNA primer set, that when used in a nucleic acid amplification reaction, such as the polymerase chain reaction (PCR), with genomic DNA extracted from soybeans produces an amplicon that is diagnostic for either the presence or absence of the deleted sequence, cystathionine gamma-synthase coding sequence or alpha', alpha, and beta subunits of the 7S storage protein coding sequences. The methods include the steps of performing a nucleic acid amplification reaction, thereby producing the amplicon and detecting the amplicon.

[0105] In some embodiments one of the pair of DNA molecules comprises the wild type sequence where the modification occurs with the second of the pair being upstream or downstream as appropriate and suitably in proximity to the wild type sequence where the modification

occurs, such that an amplicon is produced when the wild type sequence is present, but no amplicon is produced when the deletion is present. In the context of the methods, in proximity means sufficiently close such that the distance between the first and second of the pair of DNA molecules facilitates the production of an amplicon when included in a DNA amplification reaction comprising soybean genomic DNA. For example, the second primer may bind at a location beginning at, within or less than 1, 2, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 150, 200, 250, 300, 350, 400, 450, 500, 600, 700, 800, 900, 1000, 1500, 16, 1700, 1800, 1900, 2000, 2100, 2200, 2300, 2400, 2500, 2600, 2700, 2800, 2900, 3000, 3500, 4000, 4500, 5000, 5500, 6000, 6500, 7000, 7500, 8000, 8500, 9000, 9500 or 10000 nucleotides upstream or downstream of the end of the binding site of the first DNA primer molecule.

[0106] Probes and primers are provided which are of sufficient nucleotide length to bind specifically to the target DNA sequence under the reaction or hybridization conditions. Suitable probes and primers are at least 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29 or 30 nucleotides in length, and less than 35, 34, 33, 32, 31, 30, 29, 28, 27, 26, 2, 5, 2, 4, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, or 12 nucleotides in length. Such probes and primers can hybridize specifically to a target sequence under high stringency hybridization conditions. Preferably, probes and primers have complete or 100% DNA sequence similarity of contiguous nucleotides with the target sequence, although probes which differ from the target DNA sequence but retain the ability to hybridize to target DNA sequence may also be used. Reverse complements of the primers and probes disclosed herein are also provided and can be used in the methods and compositions described herein.

[0107] In some embodiments, one of the pair of DNA molecules comprises the modification or traverses the modification junction, such as the deletion junction, with the second DNA molecule of the pair being upstream or downstream of the genomic sequence as appropriate, such that an amplicon is produced when the modified allele is present, but no amplicon is produced when the wild type allele is present. Suitable primers for use in reactions to detect the presence of the modified alleles can be designed based on the junction sequences described herein. In some embodiments, the first or second primer molecule binds to a sequence corresponding to the position, or a complement thereof. In some embodiments, the primers bind to the target sequence to produce an amplicon of a length described herein. The amplicon molecule produced can be at least 5, 10, 15, 20, 25, 30, 35, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 60, 70, 80, 90, 100, 125, 150, 175, 200, 225, 250, 275, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900, 950, 1000, 1500 or 2000 nucleotides in length and less than about 10000, 9000, 8000, 7500, 7000, 6500, 6000, 5500, 5000, 4500, 4000, 3500, 3000, 2500, 2000, or 1500 nucleotides in length.

[0108] Gene-specific primers for use in genotyping soybean cystathionine gamma-synthases and 7S storage proteins are provided in Table 2.

TABLE 2

F-Glyma.20G146200-geno	ACCTGGCAGATATGATGTATGCAC (SEQ ID NO: 115)
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TABLE 2-continued

R-Glyma.20G146200-geno	GACAGCCAGCCACATCAGCCATG (SEQ ID NO: 116)
F-Glyma.20G148200-geno	ACCTGGCAGATATGATGTATGCAC (SEQ ID NO: 117)
R-Glyma.20G148200-geno	GAACATGTATCATGATCACACC (SEQ ID NO: 118)
F-Glyma.20G148400-geno	TGATGCCCTGAGAGTCCCCTCAG (SEQ ID NO: 119)
R-Glyma.20G148400-geno	CTTCAGAGTAAATTACTTCTGCGT (SEQ ID NO: 120)
F-Glyma.10G246500-geno	CTACCGTGTGTAGAGCTCATGGC (SEQ ID NO: 121)
R-Glyma.10G246500-geno	ACCCAATCAGCATGCTTCCACATG (SEQ ID NO: 122)
F-Glyma.10G028300-geno	CGTGATCTGGGACTGATCTTGAC (SEQ ID NO: 123)
R-Glyma.10G028300-geno	ACATTCCTCCGTGATCTTCAGCC (SEQ ID NO: 124)
F-Glyma.02G145700-geno	CTTATTGTATGGATGGTATAGTC (SEQ ID NO: 125)
R-Glyma.02G145700-geno	TGTGTACTCTCACTGCAACG (SEQ ID NO: 126)
F-Glyma.20G148300-geno	ATCCAAGTGTACACTACACTGGC (SEQ ID NO: 127)
R-Glyma.20G148300-geno	AAGCATCATATGCTGTCTCTGTCG (SEQ ID NO: 128)
F-Glyma.10G246300-geno	GCGATGCCCTAAGAGTCCCTGCAG (SEQ ID NO: 129)
R-Glyma.10G246300-geno	TAGCTCAGTCAGCTATGTTTCAC (SEQ ID NO: 130)
F-GmCGS1-geno	GACAAAGGAGTGACGCGTCACG (SEQ ID NO: 131)
R-GmCGS1-geno	CAACATATGAAGCAGTTTCCATGC (SEQ ID NO: 132)
F-GmCGS2-geno	ACCCCATCGAAATCGAAGAGTAC (SEQ ID NO: 133)
R-GmCGS2-geno	GACGTCTCTCGGCGGCGGAGGT (SEQ ID NO: 134)

[0109] Unless expressly stated or apparent from the context of usage, the methods and compositions of the present disclosure can be used with any plant species including, for example, monocotyledonous plants and dicotyledonous plants. Examples of plant species of interest include, but are not limited to, corn (*Zea mays*), *Brassica* sp. (e.g., *B. napus*, *B. rapa*, *B. juncea*), alfalfa (*Medicago sativa*), rice (*Oryza sativa*), rye (*Secale cereale*), triticale (x*Triticosecale* or *Triticum*x*Secale*) sorghum (*Sorghum bicolor*, *Sorghum vulgare*), teff (*Eragrostis tef*), millet (e.g. pearl millet (*Pennisetum glaucum*), proso millet (*Panicum miliaceum*), foxtail millet (*Setaria italica*), finger millet (*Eleusine coracana*), switchgrass (*Panicum virgatum*), sunflower (*Helianthus annuus*), safflower (*Carthamus tinctorius*), wheat (*Triticum aestivum*), soybean (*Glycine max*), tobacco (*Nicotiana tabacum*), potato (*Solanum tuberosum*), peanut (*Arachis hypo-*

gaea), cotton (*Gossypium barbadense*, *Gossypium hirsutum*), strawberry (e.g. *Fragaria xananassa*, *Fragaria vesca*, *Fragaria moschata*, *Fragaria virginiana*, *Fragaria chiloensis*), sweet potato (*Ipomoea batatas*), yam (*Dioscorea* spp., *D. rotundata*, *D. cayenensis*, *D. alata*, *D. polystachya*, *D. bulbifera*, *D. esculenta*, *D. dumetorum*, *D. trifida*), cassava (*Manihot esculenta*), coffee (*Coffea* spp.), coconut (*Cocos nucifera*), okra (*Abelmoschus esculentus*), oil palm (e.g. *Elaeis guineensis*, *Elaeis oleifera*), pineapple (*Ananas comosus*), citrus trees (*Citrus* spp.), cocoa (*Theobroma cacao*), tea (*Camellia sinensis*), banana (*Musa* spp.), avocado (*Persea americana*), fig (*Ficus casica*), grape (*Vitis vinifera*), guava (*Psidium guajava*), mango (*Mangifera indica*), olive (*Olea europaea*), papaya (*Carica papaya*), cashew (*Anacardium occidentale*), macadamia (*Macadamia integrifolia*), almond (*Prunus amygdalus*), date (*Phoenix dactylifera*), cultivated forms of *Beta vulgaris* (sugar beets, garden beets, chard or spinach beet, mangelwurzel or fodder beet), sugarcane (*Saccharum* spp.), oat (*Avena sativa*), barley (*Hordeum vulgare*), cannabis (*Cannabis sativa*, *C. indica*, *C. ruderalis*), poplar (*Populus* spp.), eucalyptus (*Eucalyptus* spp.), *Arabidopsis thaliana*, *Arabidopsis rhizogenes*, *Nicotiana benthamiana*, *Brachypodium distachyon*, tomato (*Solanum lycopersicum*), eggplant (*Solanum melongena*), pepper (*Capsicum annuum*), lettuce (e.g. *Lactuca sativa*), bean (*Phaseolus vulgaris*), lima bean (*Phaseolus limensis*), pea (*Lathyrus* spp.), chickpea (*Cicer arietinum*), and members of the genus *Cucumis* such as cucumber (*C. sativus*), cantaloupe (*C. cantalupensis*), and musk melon (*C. melo*), and ornamentals. Ornamentals include azalea (*Rhododendron* spp.), hydrangea (*Macrophylla hydrangea*), hibiscus (*Hibiscus rosasansensis*), roses (*Rosa* spp.), tulips (*Tulipa* spp.), daffodils (*Narcissus* spp.), petunias (*Petunia hybrida*), carnation (*Dianthus caryophyllus*), poinsettia (*Euphorbia pulcherrima*), and *Chrysanthemum*. In certain embodiments, the plant is a *Brassica*, wheat, maize, potato, soybean, or cotton plant.

[0110] Also provided are progeny plants and seeds thereof comprising a polynucleotide or modification of the present disclosure. The present disclosure also provides fruits, seeds, tubers, leaves, stems, roots, and other plant parts produced by the plants and/or progeny plants of the disclosure as well as biological samples comprising, or produced or derived from, the plants or any part or parts thereof including, but not limited to, fruits, tubers, leaves, stems, roots, and seed. In certain embodiments, the biological sample is a commodity plant product. As used herein, a “commodity plant product” refers to any composition or product that is comprised of material derived from a plant, seed, plant cell, or plant part of the present disclosure. Commodity plant products may be sold to consumers and can be viable or nonviable. Nonviable commodity products include but are not limited to nonviable seeds and grains; processed seeds, seed parts, and plant parts; dehydrated plant tissue, frozen plant tissue, and processed plant tissue; seeds and plant parts processed for animal feed for terrestrial and/or aquatic animal consumption, oil, meal, flour, flakes, bran, fiber, paper, tea, coffee, silage, crushed or whole grain, and any other food for human or animal consumption; and biomasses and fuel products; and raw material in industry. Industrial uses of oils derived from the agricultural plants described herein include ingredients for paints, plastics, fibers, detergents, cosmetics, lubricants, and biodiesel fuel. Commodity plant products also include industrial com-

pounds, such as a wide variety of resins used in the formulation of adhesives, films, plastics, paints, coatings and foams. It is recognized that such commodity plant products can be consumed or used by humans and other animals including, but not limited to, pets (e.g. dogs and cats), livestock (e.g. pigs, cows, chickens, turkeys, and ducks), and animals produced in freshwater and marine aquaculture systems (e.g. fish, shrimp, prawns, crayfish, and lobsters).

EMBODIMENTS

[0111] The following numbered embodiments also form part of the present disclosure:

[0112] 1. A plant, plant part, or seed comprising one or more modifications that increase seed protein content, the modification selected from a deletion, insertion, or substitution of a nucleotide in (i) a genomic sequence encoding a cystathionine gamma-synthase (CGS), the modification resulting in suppression of the feedback regulation of the same while retaining enzymatic activity, and/or (ii) a genomic sequence encoding a low methionine content seed storage protein, the modification resulting in suppression of the activity of the same.

[0113] 2. The plant, plant part, or seed of embodiment 1, wherein the modification comprises a deletion, insertion, or substitution of a nucleotide at or near the MTO1 region of the genomic sequence encoding the CGS.

[0114] 3. The plant, plant part, or seed of embodiment 1 or embodiment 2, wherein the modification is an in-frame modification of the genomic sequence encoding the CGS.

[0115] 4. The plant, plant part, or seed of any one of embodiments 1-3, wherein the deletion or insertion results in a frameshift mutation of the genomic sequence encoding the low methionine content seed storage protein.

[0116] 5. The plant, plant part, or seed of any one of embodiments 1-4, wherein the plant, plant part, or seed comprises a modification selected from a deletion, insertion, or substitution of a nucleotide in a genomic sequence encoding a beta-conglycinin storage protein.

[0117] 6. The plant, plant part, or seed of any one of embodiments 1-5, wherein the plant, plant part, or seed comprises a modification selected from a deletion, insertion, or substitution of a nucleotide in a genomic sequence encoding an alpha', alpha, and beta subunit of the 7S storage protein.

[0118] 7. The plant, plant part, or seed of any one of embodiments 1-6, further comprising a heterologous nucleic acid sequence selected from a reporter gene, a selection marker, a disease resistance gene, a herbicide resistance gene, an insect resistance gene, a gene involved in carbohydrate metabolism, a gene involved in fatty acid metabolism, a gene involved in amino acid metabolism, a gene involved in plant development, a gene involved in plant growth regulation, a gene involved in yield improvement, a gene involved in drought resistance, a gene involved in increasing nutrient utilization efficiency, a gene involved in cold resistance, a gene involved in heat resistance, and a gene involved in salt resistance in plants.

[0119] 8. The plant, plant part, or seed of any one of embodiments 1-7, wherein protein content of the seed is increased relative to a control seed not comprising the modification.

- [0120] 9. The plant, plant part, or seed of any one of embodiments 1-8, wherein methionine content of the seed is increased relative to a control seed not comprising the modification.
- [0121] 10. The plant, plant part, or seed of any one of embodiments 1-9, wherein proteome rebalancing results in an increase in high methionine seed storage proteins in the seed.
- [0122] 11. The plant, plant part, or seed of any one of embodiments 1-10, wherein 11S glycinin polypeptide content of the seed is increased relative to a control seed not comprising the modification as a result of proteome rebalancing.
- [0123] 12. The plant, plant part, or seed of any one of embodiments 1-11, wherein the modification is homozygous.
- [0124] 13. The plant, plant part, or seed of any one of embodiments 1-12, wherein the genomic sequence encoding the CGS comprises a nucleotide sequence having at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% sequence identity to SEQ ID NO: 1 or 4.
- [0125] 14. The plant, plant part, or seed of any one of embodiments 1-13, wherein the genomic sequence encoding the low methionine content seed storage protein comprises a nucleotide sequence having at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% sequence identity to SEQ ID NO: 7, 10, 13, 16, 19, 22, 25, or 28.
- [0126] 15. A method of making a modified plant with high seed protein content compared to the seed protein content of a nonmodified plant, the method comprising: (a) crossing a first plant comprising a deletion, insertion, or substitution of a nucleotide in (i) a genomic sequence encoding cystathionine gamma-synthase (CGS), the modification resulting in suppression of the feedback regulation of the same while retaining enzymatic activity, and/or (ii) a genomic sequence encoding a low methionine content seed storage protein, the modification resulting in suppression of the activity of the same, with a second plant; and (b) selecting a progeny plant, or plant part thereof, with high protein content for further breeding.
- [0127] 16. The method of embodiment 15, wherein the modified plant is an F1 progeny plant.
- [0128] 17. A method of producing high protein seed meal comprising: harvesting the seed of any one of embodiments 1-14 from a plant comprising the modification and processing the seed to form meal.
- [0129] 18. A modified CGS polynucleotide comprising an in-frame insertion or deletion in the MTO1 region so that feedback inhibition of expression is suppressed.
- [0130] 19. The modified CGS polynucleotide of embodiment 18, wherein the polynucleotide comprises the sequence of SEQ ID NO: 44, 45, 46, 48, 49, 50, or 51.
- [0131] 20. The modified CGS polynucleotide of embodiment 18 or embodiment 19, wherein the polynucleotide comprises a sequence having at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% sequence identity to SEQ ID NO: 52, 53, 54, 56, 58, 60, or 61.
- [0132] 21. A modified 7S polynucleotide comprising an insertion, deletion, or substitution that suppresses activity of the polynucleotide.
- [0133] 22. The modified 7S polynucleotide of embodiment 21, wherein the polynucleotide comprises a sequence having at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% sequence identity to one or more of SEQ ID NOs: 62-108.
- [0134] 23. A plant cell comprising the polynucleotide of any one of embodiments 18-22.
- [0135] 24. The plant cell of embodiment 23, wherein the plant cell is a soybean plant cell.
- [0136] 25. A plant comprising the plant cell of embodiment 23 or embodiment 24.
- [0137] 26. A seed of the plant of embodiment 25, wherein the seed has increased protein content compared to a seed comprising a comparable polynucleotide without the modification.
- [0138] 27. A seed of the plant of embodiment 25, wherein the seed has increased methionine content compared to a seed comprising a comparable polynucleotide without the modification.
- [0139] 28. A modified polypeptide encoded by the polynucleotide of any one of embodiments 18-22.
- [0140] 29. A method of producing a plant cell having increased protein content, the method comprising: introducing a double-strand-break-inducing (DSB-inducing) agent to a plant cell, wherein the DSB-inducing agent creates a double-strand break in an endogenous CGS gene and/or a low methionine content seed storage protein gene in the genome of the plant, wherein the repair of the DSB results in the production of a modified polynucleotide, and wherein the production of the modified polynucleotide produces increased protein content in the seed.
- [0141] 30. The method of embodiment 29, wherein the modified polynucleotide includes a deletion, insertion, or substitution in the MTO1 region of the CGS gene.
- [0142] 31. The method of embodiment 30, wherein the deletion or insertion is an in-frame deletion or insertion.
- [0143] 32. The method of any one of embodiments 29-31, wherein the modified polynucleotide is a deletion, insertion, or substitution of a 7S beta-conglycinin.
- [0144] 33. The method of any one of embodiments 29-32, wherein the endogenous CGS gene comprises a nucleotide sequence having at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% sequence identity to SEQ ID NO: 1 or 4.
- [0145] 34. The method of any one of embodiments 29-33, wherein the endogenous low methionine content seed storage protein gene comprises a nucleotide sequence having at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% sequence identity to SEQ ID NO: 7, 10, 13, 16, 19, 22, 25, or 28.
- [0146] 35. A method of editing the genome of a plant cell to produce a plant cell having increased protein, the method comprising: introducing a guide RNA, a polynucleotide modification template, and a Cas endonuclease to a plant cell; and regenerating a plant from the plant cell wherein the genome of the regenerated plant comprises the modified polynucleotide of any one of embodiments 18-22.

- [0147] 36. The method of embodiment 35, wherein the seed of the plant has increased protein content compared with a comparable seed not comprising the modified polynucleotide.
- [0148] 37. The method of embodiment 35 or embodiment 36, wherein the seed of the plant has increased methionine content compared with a comparable seed not comprising the modified polynucleotide.
- [0149] 38. The method of any one of embodiments 35-37, wherein the genome is modified by a site-directed endonuclease.
- [0150] 39. The method of any one of embodiments 35-38, wherein the plant cell is a soybean cell.
- [0151] 40. A method of producing a plant cell having increased seed protein content, the method comprising: transforming a plant cell with the modified polynucleotide of any one of embodiments 18-22, wherein a plant regenerated from the transformed plant cell has increased seed protein content compared with a comparable plant cell not comprising the modified polynucleotide.
- [0152] 41. A polynucleotide construct comprising a sequence encoding a guide RNA which targets an endogenous CGS gene and/or a low methionine content storage polypeptide gene when expressed in a plant cell and thereby generates a modified endogenous CGS gene and/or a low methionine content storage polypeptide gene.
- [0153] 42. The polynucleotide construct of embodiment 41, wherein the guide RNA comprises the spacer sequence of SEQ ID NO: 109, 110, 111, 112, 113, or 114.
- [0154] 43. A plant cell comprising the polynucleotide construct of embodiment 41.
- [0155] 44. The plant cell of embodiment 43, wherein the plant cell is a soybean cell.
- [0156] 45. A method of producing a soybean seed comprising: a) sexually crossing a first soybean line comprising the polynucleotide of any one of embodiments 18-22 with a second soybean line not comprising the polynucleotide; and b) harvesting the seed produced thereby.
- [0157] 46. The method of embodiment 45 further comprising the step of backcrossing a second generation progeny plant that comprises the polynucleotide to the parent plant that lacks the polynucleotide, thereby producing a backcross progeny plant that produces seed with increased protein content.
- [0158] 47. A method of screening for the presence or absence of the polynucleotide of any one of embodiments 18-22 in a plurality of genomic soybean DNA samples, the method comprising the steps of (a) contacting a plurality of genomic soybean DNA samples, at least some of which comprise the polynucleotides, with a first DNA primer molecule and a second DNA primer molecule; (b) providing a plurality nucleic acid amplification reaction conditions; (c) performing the nucleic acid amplification reactions, thereby producing a DNA amplicon molecule indicating the presence of the polynucleotide or a wild-type CGS and/or a low methionine content storage polypeptide gene nucleotide sequence; and (d) detecting the DNA amplicon molecules, wherein the presence, absence or size of the DNA amplicon molecule indicates the presence or

absence of the polynucleotide in the at least one of the plurality of genomic soybean DNA samples.

- [0159] 48. The method of embodiment 47, wherein the first DNA primer molecule or the second DNA primer molecule comprises the sequence of one or more of SEQ ID NOs: 115-134.

- [0160] 49. An animal feed comprising the plant, plant part, or seed of any one of embodiments 1-14.

[0161] The foregoing invention has been described in detail by way of illustration and example for purposes of clarity and understanding. As is readily apparent to one skilled in the art, the foregoing disclosures are only some of the methods and compositions that illustrate the embodiments of the foregoing invention. It will be apparent to those of ordinary skill in the art that variations, changes, modifications, and alterations may be applied to the compositions and/or methods described herein without departing from the true spirit, concept, and scope of the invention.

[0162] All publications, patents, and patent applications mentioned in the specification are incorporated by reference herein for the purpose cited to the same extent as if each was specifically and individually indicated to be incorporated by reference herein.

[0163] The following examples illustrate particular aspects of the disclosure and are not intended in any way to limit the disclosure.

[0164] Although the foregoing invention has been described in some detail by way of illustration and example for purposes of clarity of understanding, it will be obvious that certain changes and modifications may be practiced within the scope of the appended claims.

[0165] The following examples are offered by way of illustration and not by way of limitation.

EXAMPLES

Example 1. In-Frame Mutations in *Arabidopsis* CGS1 Using CRISPR/Cas9 to Enhance Free Met Pool

[0166] In *Arabidopsis*, the AtCGS1 gene encodes protein which catalyzes the first committed step in Met biosynthesis, and its expression is regulated by a negative feedback-inhibition in response to S-adenosyl-L-methionine (AdoMet), a direct metabolite of methionine. Several single-base mutations within this MTO1 region relieve feedback inhibition, leading to over-accumulation of free Methionine content in vegetative tissues. Therefore, it was hypothesized that in-frame mutations within the MTO1 region could also create feedback-insensitive CGS, a phenotypic mimic of EMS mutants, and increase free Met content. A CRISPR/Cas9 vector designed to create mutations within the MTO1 was constructed. More than a hundred T1 *Arabidopsis* transgenic lines were screened on plates containing half-strength MS supplemented with 100 μ M DL-ethionine, a toxic analogue of Met, to identify the putative over-accumulate Met thionine CRISPR lines. Two independent lines (At_CRS3-2 and At_CRS4-2) showing normal growth were identified (FIG. 1A). The phenotypes of two CRISPR mutant lines were similar to mto1-1, an AtCGS1 EMS mutant line. In contrast, the growth of wild-type Col-0 was completely inhibited in the presence of ethionine (FIG. 1A). PCR and Sanger sequencing of PCR amplicons of the AtCGS1 gene from two CRISPR mutant lines showed that At_CRS3-2 and At_CRS4-2 harbored CRISPR/Cas9-in-

duced in-frame deletions of 39 and 42 base pairs, respectively, within and extended MTO1 region (FIG. 1B-C). The transcript levels of AtCGS1 from leaves of two weeks post germination on the ½ MS media of the CRISPR lines were quantified and compared with wildtype Col-0 and mto1-1 genotypes. While the mRNA level of AtCGS1 of the CRISPR lines were no difference to the mto1-1, they were 3-fold higher than wild-type plants under the ½ MS media (FIG. 1D). Free amino acid analysis was performed 3-week-old leaves siliques and mature seeds; whereas, total amino acid contents were quantified on mature seeds of wild-type, mto1-1 and two in-frame CRISPR mutants. Free Met content in the leaves of the EMS, mto1-1 mutant and the in-frame CRISPR mutants were 38 to 60 fold higher than the wild-type. A significant increase in Free Met was found in siliques and mature seeds of mutant lines (FIG. 1E-G). Although Free Met content increased in leaves, siliques and mature seeds, soluble amino acids that are metabolically related to Methionine did not significantly differ between the wild-type and mutant lines. Total amino acids were measured in mature seed; however, there was no difference in total Met content and other amino acids among genotypes (FIG. 1H). Taken together, the data showed that induced in-frame mutations within the MTO1 region abolished the negative feedback inhibition on AtCGS1 expression and led to over-accumulated free Met in leaves and seeds. However, mutation in MTO1 region alone did not improve the total Met content in *Arabidopsis* seeds.

Example 2. Multiplex CRISPR/Cas9 Based Genome Editing for Improving Methionine and Protein Content in Soybean Seeds

[0167] Soybean (*Glycine max* (L.) Merr) is one of the most important crops grown worldwide, providing 70% and 28% of protein meal and vegetable oil consumption, respectively. Soy meal contains high protein content and well-balanced amino acid profile; however, the content of sulfur containing amino acids cysteine (Cys) and methionine (Met) in soybean seed is not found at adequate levels for optimal growth and development of monogastric animals. Supplementation of synthetic amino acids is added to meet the nutritional requirements in soy-based animal feed, though it significantly increases the production cost. In this example, multiplex CRISPR/Cas9 based editing was used to generate mutant soybean lines with in-frame deletions in the MTO1 region of CGS1 and 7S knock-out to provide increased Met and protein in soybean seed. Creating soybean lines with mutations of CGS and Cupin gene family by CRISPR/Cas9

[0168] The MTO1 region is highly conserved in plants and in-frame mutation of the MTO1 region led to a significant increase in free Met in *Arabidopsis* (Example 1). In addition, knock down of seed storage proteins in soybean greatly changes soybean seed protein profiles. Therefore, it was hypothesized that the combination of null mutation of seed storage proteins (Cupin gene family) and an in-frame mutant of soybean CGS would increase the free Met and total Met in soybean seed.

[0169] A CRISPR/Cas9 construct with five single guide RNAs (sgRNAs) to target the MTO1 region of two soybean CGS genes [GmCGS1 (Glyma.09g235400) and GmCGS2 (Glyma.18g261600)] and the conserved exon region of eight members of soybean seed specific Cupin gene family [Glyma.20G148300, Glyma.20G148400 encoding 7S α subunits; Glyma.10G246500, Glyma. 10G246300 and

Glyma. 10g028300 encoding 7S α' subunits; Glyma. 20G146200, Glyma.20G148200 encoding 7S β -subunits; Glyma.02g145700 encoding SBP1 (Sucrose Binding Protein 1)] was constructed and transformed into soybean cultivar Maverick (FIG. 2A-C). Mutant lines that segregated from Cas9 transgenic lines were used in this example to avoid further gene edits.

[0170] To identify feedback insensitive CGS in soybean, seeds were sown on media supplemented with ethionine. Ten days after sowing, soybean root length was 6-fold shorter than those sown in normal conditions (FIG. 3A-B). Therefore, this screening approach was used to identify mutant lines that possessed in-frame mutations in the MTO1 region of GmCGS genes. Out of 16 CRISPR lines, certain T1 seedlings from four independent lines (7S-CGS-4, -6, -13, and -36) exhibited normal growth on ½ MS media supplemented with 50 μ M ethionine, and they segregated for ethionine resistant phenotype. A Cleaved Amplified Polymorphic Sequences (CAPS) marker was developed for genotyping based on the insensitivity of the DNA sequence to *AscI* restriction enzyme. Sanger sequencing and genotyping showed that these four T0 lines were heterozygous mutations (either frameshift (KO) or in-frame mutations) for GmCGS1 and GmCGS2: 7S-CGS-4 gmcgs1 (1-bp deletion, frameshift mutation)/gmcgs2 (27-bp deletion, in-frame mutation); 7S-CGS-6 gmcgs1 (3-bp deletion, in-frame mutation)/gmcgs2 (2-bp insertion, KO mutation); 7S-CGS-13 gmcgs1 (4-bp deletion, KO mutation)/gmcgs2 (12-bp deletion, in-frame mutation); 7S-CGS-36 gmcgs1 (3-bp deletion, in-frame mutation)/gmcgs2 (1-bp insertion, KO mutation) (FIG. 3C, Table 3).

TABLE 3

CGS modification	MTO1 region sequence	Genomic sequence	Amino acid sequence
CGS1 (-1 deletion)	SEQ ID NO: 44	SEQ ID NO: 52	—
CGS1 (-4 deletion)	SEQ ID NO: 45	SEQ ID NO: 53	—
CGS1 (-3 deletion)	SEQ ID NO: 46	SEQ ID NO: 54	SEQ ID NO: 55
CGS2 (-27 deletion)	SEQ ID NO: 48	SEQ ID NO: 56	SEQ ID NO: 57
CGS2 (-12 deletion)	SEQ ID NO: 49	SEQ ID NO: 58	SEQ ID NO: 59
CGS2 (+2 insertion)	SEQ ID NO: 50	SEQ ID NO: 60	—
CGS2 (+1 insertion)	SEQ ID NO: 51	SEQ ID NO: 61	—

[0171] To identify 7S β -conglycinin mutations, the soluble protein from pooled T1 seed samples of the 16 T0 lines were analyzed with SDS-PAGE. 7S β -conglycinin and SBP1 fractions from five CRISPR lines (7S-CGS-4, 7S-CGS-6, 7S-CGS-13, 7S-CGS-15, and 7S-CGS-36) were absent on the SDS-PAGE as compared to the wild-type. The CRISPR lines 7S-CGS-4, 7S-CGS-6, 7S-CGS-13, and 7S-CGS-36 were chosen for further analysis because they were also resistant to ethionine as discussed above. Specific PCR amplicons of 7S genes which amplified DNA samples of T0 generation of the 4 CRISPR lines were sequenced. These 4 lines were homozygous and biallelic for small, large deletions or insertions as well as inversions of 7S genes (SEQ ID NOs: 62-108).

[0172] Ethionine screening, CAPS marker analysis, Sanger sequencing, and SDS-PAGE to obtain desired genotypes. Two independent lines per genotype at the T4 generation were obtained: $7S^{KO}-1$; $7S^{KO}-2$; $7S^{KO}/cgs1^{KO}-1$; $7S^{KO}/cgs1^{KO}-2$; $7S^{KO}/cgs2^{KO}$; $7S^{KO}/cgs1^{inf}-1$; $7S^{KO}/cgs1^{inf}-2$; $7S^{KO}/cgs2^{inf}-1$; $7S^{KO}/cgs2^{inf}-2$. The double in-frame GmCGS1/GmCCGS2 in $7S^{KO}$ background, $7S^{KO}/cgs1^{inf}/cgs2^{inf}$, was obtained by crossing between line $7S^{KO}/cgs1^{inf}-1$ and line $7S^{KO}/cgs2^{inf}-1$ (FIG. 3A-B). This mutant was strongly resistant to DL-ethionine treatment at 300 μ M compared to its parental lines and wild-type control. The same strategy was applied to generate the double KO $gmcs1/gmcs2$ in $7S^{KO}$ background; however, the line was not able to be obtained.

Free Amino Acids Profiles and Protein Seed Profiles in Soybean

[0173] To investigate the effect of in-frame and frameshift mutations of CGS on methionine production, the transcript level of GmCGS1 and GmCGS2 and free methionine content in different tissues of all seven genotypes were examined. The expression level of GmCGS1 and GmCGS2 in leaves and green seeds were increased two- to three-fold in the in-frame mutants and undetectable in the frameshift mutants compared to the wild-type (FIG. 3F-I). The single GmCGS mutants showed slight increases or no difference in free methionine content compared to the wild-type control. However, three- to forty-six-fold increases in free methionine were obtained in leaves, roots, green seeds, and mature seed of the double in-frame $7S^{KO}/cgs1^{inf}/cgs2^{inf}$ mutant (FIG. 3J-L). These data indicated that GmCGS1 and GmCGS2 are functionally redundant for methionine biosynthesis in soybean. Besides increasing free Met, free asparagine, histidine, serine, and cysteine were also significantly increased in the mature seeds of the $7S^{KO}/cgs1^{inf}/cgs2^{inf}$ compared to the wild-type and other CRISPR lines.

[0174] Next, a global quantitative proteomics approach was employed to investigate the effect of conglycinin and methionine on seed protein profiles in the mature soybean seeds. At a false discovery rate (FDR) of <1%, 1592 proteins passed these criteria: presence in at least three of five biological replicates with more than two spectrum counts per protein and identified in mature seeds of seven genotypes. The shared proteins among groups were then analyzed with UpSetR. About 70% of identified proteins were quantified in all seven genotypes while a few qualitative alterations were found in each genotype (FIG. 4A).

[0175] A total of 20 proteins were found only in the mature seed of the $7S^{KO}/cgs1^{inf}/cgs2^{inf}$ genotype, and they were involved in glutathione metabolisms pathway (FIG. 4A). Standard one-way ANOVA analysis controlled by a Benjamini-Hochberg FDR-corrected $p < 0.05$ was performed and revealed 173 differential modulated proteins among different genotypes. Hierarchical clustering of these proteins based on abundance levels across different genotypes yielded a heatmap with clearly distinct protein clusters, especially compared between the wild-type and the $7S^{KO}/cgs1^{inf}/cgs2^{inf}$ CRISPR line. Gene ontology enrichment analysis ($p < 0.05$) of the 173 significantly differential proteins revealed strong links to proteolysis, peptide biosynthesis, amide metabolic process and nutrient reservoir activity. The relative contribution of major protein groups to the total seed proteome showed untraceable 7S proteins in the seed proteome of all the CRISPR lines and greatly increased glycinins and other

proteins (FIG. 4B-C). Intriguingly, the contribution of Kunitz trypsin inhibitors (KTI) and the Bowman-Birk protease inhibitors (BBI), other major soybean seed proteins, were strongly regulated by the excess of methionine content in seed proteome (FIG. 4D). KTI and BBI conventionally take ~5% and 7% in seed proteome of wild-type, Maverick; while, KTI was accumulated to 7% and BBI was 2% reduction in the seed proteome of the 7S null mutant, $7S^{KO}$. These effects were reversed in the seed proteome of the $7S^{KO}/cgs1^{inf}/cgs2^{inf}$ CRISPR line of which KTI and BBI were 5% and 7%, no difference to wild-type seed proteome, respectively (FIG. 4E).

[0176] The mRNA levels of the major seed storage proteins among seven genotypes were investigated using qRT-PCR analysis on seed development samples when soybean seed storage proteins start to accumulate. The transcript levels of 7S, which were highly expressed in the wild-type, were undetectable in all CRISPR lines. Contrast to mRNA levels of glycinins which were not difference among genotypes, the expression levels of major KIT (Glyma.08g341500) and BBI (Glyma.16g208900) were significantly correlated with the present of methionine in the CRISPR lines (FIG. 4E-F). The data suggests, along with rebalancing, that the expression level of KIT and BBI are regulated by methionine content.

Increasing Methionine and Protein Content without Growth and Yield Defect in Soybean

[0177] A standard protein hydrolysate analysis was performed to quantify and compare amino acid content in mature seeds of six genotypes with wild-type under field condition during 2-year period, 2020 and 2021. The null 7S alone or combined with single in-frame GmCGS mutant showed a little impact on the accumulation of total methionine content in seeds; however, they were not significant throughout two years. However, the total content of methionine of the null 7S and double GmCGS mutant, $7S^{KO}/cgs1^{inf}/cgs2^{inf}$ was ~30% higher than wild-type in the 2 years (FIG. 5A). In addition, the $7S^{KO}/cgs1^{inf}/cgs2^{inf}$ produced higher amounts of threonine, alanine, glycine, tyrosine, and arginine with little impact on the other amino acids in mature seed, resulting in high accumulation of total amino acids in seed of this line (FIG. 5B). Spearman analysis indicated a positive correlation of the increasing methionine content with other amino acids in the mature seed of $7S^{KO}/cgs1^{inf}/cgs2^{inf}$ compared with the wild-type.

[0178] Protein and oil content were also analyzed in these samples. Seed protein content of the $7S^{KO}/cgs1^{inf}/cgs2^{inf}$ CRISPR line, in correlated with higher amino acid content, increased up to 3% as compared to wild-type control; whereas, no significant difference in seed protein contents was found in other CRISPR lines during 2 years analysis (FIG. 5C). Oil content was only slightly reduced or unchanged in the $7S^{KO}/cgs1^{inf}/cgs2^{inf}$ and 7S null mutants; however, the 7S-CGS1KO mutant lines produced ~1.5% higher oil content in seed compared other lines (FIG. 5D). Fatty acid profiling revealed that oleic acid content was increased up to 8% higher than in of the $7S^{KO}/cgs1^{inf}/cgs2^{inf}$ compared to wild-type while the linoleic acid was reduced 7%. However, there was no difference in the transcript levels of GmFAD2-1, a major gene encoding for delta12 fatty acid desaturase 2 (FAD2) enzyme that catalyzes the conversion of oleic acid to linoleic acid.

[0179] No differences in germination rates between the CRISPR lines and wild-type seeds were observed under

growth chamber conditions (FIG. 6A). Lastly, the plant morphology, yield, and germination rate of the CRISPR lines were compared to wild-type. The CRISPR mutant lines displayed no growth penalties in terms of plant height, 100 seed weight, or grain yield under field conditions (FIG. 6B-D).

DISCUSSION

[0180] Increasing methionine and protein content in seed with no growth defects is one of the major efforts in crop biotechnology and breeding in recent years. In this example, soybean lines with high methionine and protein in seed were generated in a short time period and with a single transformation event using the CRISPR/Cas9 editing system. In-frame mutations within and next to the MTO1 region of CGS (up to 27-base pair deletion in soybean) resulted in higher transcript abundance and free methionine in all tissues, which is a phenotypic mimic to the EMS mutants. The findings indicated that the in-frame GmCGS1 and GmCGS2 mutations interact synergistically in synthesizing methionine in soybean. The results also provide a new approach to generate high free methionine content crops through CRISPR/Cas9 approach other than expressing the feedback insensitive CGS mutant form into crops.

[0181] Increasing free Met alone is not sufficient to improve the total methionine in seeds.

[0182] Combining seed proteome rebalancing with high free Met through multiplex CRISPR/Cas9 editing boosted the total methionine and protein contents up to 30% and 3% in seeds, respectively. Null 7S mutants significantly changed the soybean seed proteome profile with increases in the accumulation of glycinins, lipoxygenase, KIT, and other seed proteins to compensate for the lack of 7S proteins. However, there is no impact on total methionine content. The results indicate soybean seeds have the capacity to compensate for a protein accumulation shortage by accumulating additional quantities of other seed proteins. Importantly, the data showed the availability of free methionine plays a significant role on regulating protein components in the rebalanced proteome in soybean seeds. While the GY were not strongly impacted by the free methionine at transcript and protein levels, methionine regulated the transcript levels of BBI, a major sulfur amino acid containing protein. The data suggests that regulation of free methionine at the transcription and post-transcriptional regulation of sulfur-rich amino acid containing proteins during proteome rebalancing are required to increasing methionine and protein content in soybean seeds. The approach developed from this example could be widely applied for nutritional quality crop improvement beyond methionine and protein content in seeds.

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SEQ ID NO: 3      moltype = AA length = 536
FEATURE          Location/Qualifiers
source           1..536
                 mol_type = protein
                 organism = Glycine max

SEQUENCE: 3
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RQLSTKARN CSNIGVAQIV AASWSNNSDN SPAAGAPAPP AATATDAATV PLPVVVAANE 120
DVVSAAADE NGAVQLNSSS YSFLKSDAS KTIHAAERLG RGIETDGIIT PVVNTSAYFF 180
KKTADLIDFK ENRQVSYEYG RYGNPTTVVL EEKISALEGA ESTVIMASGM CASVVLFMAL 240
VPAGGHLVTT TDCYRKTRIF IETFLPKMGI TTTVIDPADV GALESALEQH NVSLFFTESP 300
TNPFLRCVDI KLVSELCHKK GTLLCIDGTF ATPLNQKALA LGADLILHSL TKYMGGHHDV 360
LGGCISGSIK VVSQIRTLHH VLGGLNPNNA AYLFIKGMKT LHLRVQQNS TGMRAKLLLE 420
AHPKVVRVY PGLPSHPHE LAKRQMTGFG GVSFEIDGD LHTTIKFIDS LKIPYIAASF 480
GGCESIVDQP AILSYWDLPO SERAKYKIYD NLVRFSPGVE DFEDLKADVL QALEAI 536

SEQ ID NO: 4      moltype = DNA length = 5135
FEATURE          Location/Qualifiers
source           1..5135
                 mol_type = genomic DNA
                 organism = Glycine max

SEQUENCE: 4
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SEQ ID NO: 5      moltype = DNA length = 1608
FEATURE          Location/Qualifiers
source            1..1608
                  mol_type = other DNA
                  organism = synthetic construct

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tcggcgcgcg cgtttcacgg catctcctcc ctcatcctcc gattccctcc caacttcacg 180
cgccagctca gccaccaaggc gcgcgcgaat tgtagcaaca tcggcgctcg gcaaattgtc 240
gcgcttcgtt ggtcgacaac cagcgacaac tctccggccg ccggggctcc ggcgcgcgcc 300
gcgccacagc cagcggagcg cgccacggtg cctctccccg tcgacctcgc cgccgcgcag 360

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gacgtcgtcg tctctgccgc ggcgcgcgag aacggggactg tacagttaaa cagtagttct 420
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SEQ ID NO: 6      moltype = AA length = 535
FEATURE          Location/Qualifiers
source           1..535
                  mol_type = protein
                  organism = Glycine max

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SEQUENCE: 6
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RQLSTKARRN CSNTGVQIV AASWSNNSDN SPAAGAPAPP AATAADAAATV PLPVDLAAAE 120
DVVVSAAAEE NGTVQLNSSS YSLKSDASK TIHAERLGR GIETDGITTP VLNTSAYFFK 180
KTADLIDFKE NRQVSIEYGR YGNPTTVVLE EKISILEGAE STVIMASGMC ASVVLPMALV 240
PAGGHLVTTT DCYRKTRIFI ETFLPKMGIT TTVIDPADVG ALESALEQHN VSLFFTESPT 300
NPFLRCVDIK LVSELCHKKG TLLCIDGTFA TPLNQKALAL GADLILHSLT KYMGHHHDAL 360
GGCISGSTKV VSQIRTLHHV LGGTLNPNAA YLFIRGMKTL HLRVQQQNST GMRMAELLEA 420
HPKVKRVYYP GLPSHPHEHL AKRQMTGFGG VVSFEIDGDL HTTIKFIDSL KIPYIAASFG 480
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SEQ ID NO: 7      moltype = DNA length = 3103
FEATURE          Location/Qualifiers
source           1..3103
                  mol_type = genomic DNA
                  organism = Glycine max

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SEQUENCE: 7
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cccgcaact tgcatgccac cccatgcatg caagttaaca agagctatat ctcttctatg 540
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SEQ ID NO: 8      moltype = DNA length = 1818
FEATURE          Location/Qualifiers
source           1..1818
                 mol_type = other DNA
                 organism = synthetic construct

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SEQ ID NO: 9      moltype = AA length = 605
FEATURE          Location/Qualifiers
source           1..605
                 mol_type = protein
                 organism = Glycine max

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SEQUENCE: 9
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LKVEKEECE GEIPRPRPRP QHPEREPQQP GEKEEDEDEQ PRPIPPRPQ PRQEEHEQR 120
EEQEWPRKEE KRGEKGSEEE DEDEDEEQDE RQPPFPRPPH QKEERKQEE EDEEQQRESE 180
ESEDSELRH KNKNPFLFGS NRFETLFPKQ YGRIRVLQRF NQRSPQLQNL RYRILEFNS 240
KPNTLLLPNH ADADYLVIL NGTAILSLVN NDDRDSYRLQ SGDALRVPSG TTYVVVNP DN 300
NENLRLITLA IPVKNKGRFE SFPLSSTEAQ QSYLQGFERN ILEASYDTKF EEINKVLFNR 360
EEGQQQGEQR LQESVIVEIS KEQIRALSKR AKSSSRKTIS SEDKPFNLRS RDPYISNKL 420
KFFEITPEKN PQLRDLDFL SIVDMNEGAL LLPHFNSKAI VILVINEGDA NIELVLKEQ 480
QQEQQQEQP LEVRKYRAEL SEQDIFVIPA GYPVVVNATS NLNFFAIGIN AENNQRNFLA 540
GSQDNVISQI PSQVQELAFP GSAQAVEKLL KNQRESYFVD AQPKKKEEGN KGRKGPLSSI 600
LRAFY 605

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SEQ ID NO: 10      moltype = DNA length = 3038
FEATURE           Location/Qualifiers
source            1..3038
                  mol_type = genomic DNA
                  organism = Glycine max

SEQUENCE: 10
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cacataaaagt atccttttga aagcaatgtc caagtccatc aaataaaatt gagacaaaat 180
gcaacotcac cccacttcac tatccatggc tgatcaagat cgccgcgtcc atgtagggtct 240
aaatgccatg cacatcaaca cgtactcaac atgcagccca aattgctcac catcgctcaa 300
cacattttctt gttaatttctt aagtacactg cctatgagac tctaactcga tcacaacccat 360
cttcgcgtcac atcaattttg ttcaattcaa caccgcgtcaa cttgcatgcc accccatgca 420
tgcaagttaa caagagctat atctcttcta tgactataaa taccgcgaat ctcggtccag 480
gttttcatca tcgagaacta gttcaatcct ctagtatacc ttaataaata atttaagata 540
ctatgatgag agcacgggttc ccattactgt tgctgggact tgttttcctg gtttcagttt 600
ctgctctcatt tggcattgct tactgggaaa aagagaaccc caaacacaa aagtgtctcc 660
agagttgcaa tagcgagaga gactcgtaca ggaaccaagc atgccacgct cgttgcaacc 720
tccttaagggt ggagaaagaa gaattgtgaag aaggtgaaat tccacgaca cgaccacgac 780
cacacacccc ggagagggaa cctcagcaac ccggtgagaa ggaggaagac gaagatgagc 840
aaccacgtcc aatcccatc cccagccccc aacctcgtca agaagaagag cacgagcaga 900
gagaggaaca ggaatggcct cgcaaggagg aaaaacgcgg agaaaaggga agtgaagagg 960
aagatgagga tgaggatgag gaacaagatg aacgtcaatt cccattccca cgcccacctc 1020
atcagaagga agagcgaag caagaggaag atgaggatga ggagcagcag cgagagagcg 1080
aagaaagtga agattctgag ttacgaagac ataagaataa gaacctttt ctcttcggct 1140
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tcaaccaacg ctccccacaa cttcagaatc tccgagacta cgcattttg gagttcaact 1260
ccaaacccaa caccctcctt ctccccaaac atgtgcagcg tgattacctc atcggtatcc 1320
ttaacgggtg gtattattta cttaataaac atcatcactt gcttctgctt ctgtgaaaat 1380
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gacgacagag actcctacag acttcaatct ggtgatgccc tgagagtcct ctccaggaac 1500
acatactatg tggccaaccc tgacaacaac gaaaatctca gattaataac actcgccata 1560
cccgtaaca agcctggtag atttgaggtc ctgccccttc ttataaata attaatatt 1620
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tccgtgcttg attttacaga ccaaatcga ggagataaac aaggttctgt tttagtaga 1920
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taaataaact gacagcatat gatgcttttg ttttgtgt 3038

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SEQ ID NO: 11      moltype = DNA length = 1818
FEATURE           Location/Qualifiers
source            1..1818
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 11
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gtctcatttg gcattgtcta ctgggaaaaa gagaacccca aacacacaa gtgtctccag 120
agttgcaata cggagagaga ctctgacagg aaccaagcat gccacgctcg ttgcaacctc 180
cttaagggtg agaaagaaga atgtgaagaa ggtgaaattc cagcaccacg accacgacca 240
caacacccgg agagggaaac tcagcaaccc ggtgagaagg aggaagacga agatgagcaa 300
ccacgtccaa tcaccattcc acgcccacaa cctcgtcaag aagaagagca cgagcagaga 360
gaggaacagg aatggcctcg caaggaggaa aaacgaggag aaaagggaag tgaagaggaa 420
gatgaggatg aggatgagga acaagatgaa cgtcaattcc cattcccacg cccacctcat 480
cagaaggaag agcgaaagca agaggaagat gaggatgagg agcagcagcg agagagcgaa 540
gaaagtgaag attctgagtt acgaagacat aagaataaga accctttctt cttcggtctc 600
aacaggttcg aaactctctt caaaaacaa tatggtcgca ttcgcgtcct ccagaggttc 660

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aaccaacgct cccacaaact tcagaatctc cgagactacc gcatthttgga gttcaactcc 720
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aacgggactg ccattcttct cttgggtgaac aacgacgaca gagactccta cagacttcaa 840
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ttgagggctt ttactga 1818

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SEQ ID NO: 12      moltype = AA length = 605
FEATURE           Location/Qualifiers
source            1..605
                  mol_type = protein
                  organism = Glycine max

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SEQUENCE: 12
MMRARFPLLL LGLVFLASVS VSPGIAYWEK ENPKHNKCLQ SCNSERDSYR NQACHARNCL 60
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EEQEWPRKEE KRGEKGSEEE DEDEDEQDE RQFPFPRPH QKEERKQED EDEEQRESE 180
ESEDSELRH KKNPFLLFS NRFTLKNQ YGRIRVLQRF NQSPQLNL RDYRILEFNS 240
KPNLLLLPNH ADADYLIVIL NGTAILSLVN NDDRDSYRLQ SGDALRVPSG TTYVVNPDN 300
NENLRLITLA IPVKNKGRFE SFPLSSTEAQ QSYLQGFERN ILEASYDTKF EEINKVLFSR 360
EEGQQQGEQR LQESVIVEIS KEQIRALSKR AKSSSRKTIS SEDKPFNLRS RDPIYSNKL 420
KFFEITPEKN PQLRDLDFL SIVDMNEGAL LLPHFNSKAI VILVINEGDA NIELVLKQEQ 480
QQEQQQEEEP LEVRKYRAEL SEQDIFVIPA GYPVVVNATS NLNFFAIGIN AENNQRNFLA 540
GSQDNVSIQI PSQVQELAFP GSAQAVEKLL KNQRESYFVD AQPKKKEBGN KGRKGLSSI 600
LRAFY 605

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SEQ ID NO: 13      moltype = DNA length = 2679
FEATURE           Location/Qualifiers
source            1..2679
                  mol_type = genomic DNA
                  organism = Glycine max

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SEQUENCE: 13
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cgagcgcaac attattcgag tacgttagct aacaatattt taagcaattg ctgctacaat 240
gatagcgagg tttctttgtg tgcggttgc gttgctggga attgtttttc tagcatcagt 300
ttctgtctca ttacagtgga agccgcccag tcaccacagag tgctccaga gttgcgagag 360
tgaaacggat ccatacaagc ggaagcatg caagcttagt tgcaagttcg ttcttgagca 420
cggggagcaa caacatgaag aagaagaaa agaaagaaa catccacaac cacacccacc 480
acacgaggag agaggacgta gagaaaaggg acgccaggaa ggtgagaagg aggaaaagga 540
acaagacgag cgtgaattac aaccacgtcc atttccacga ccagaggagc atgaaggaga 600
acagtggaaa ggaaggcatc gtcataaaga cccggaagag agggcgagaa tgagactgag 660
ggaagcagaa agatagaga gggaaaggga ggaacagagg gtgagggagg aggaagagaa 720
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tcgattatga accattctac ttgagaggct cccaaatctc ttccaacaac ttggaaagt 1800
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ctgtggatat	caacgaggta	agtgcaaaaa	ccatgtagat	ctagacaaag	tgaactgcac	1920
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aaatacaggg	agggtcttct	ctgccacact	acaattcaaa	ggccatagtg	atactaatgg	2040
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aaacaaggga	agtgcgaaag	tatagagccg	agttgtctga	agacgatata	tttgaatcc	2160
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caagatatcg	agaatctaata	aaagaaccag	agggaaatcct	attttgctga	tgctcagcct	2520
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ctaggagttc	acagcgagcg	aggaaggtat	agggatatgt	aactagaata	acttgagatc	2640
cacctttcta	ctactagtaa	ataaataaaa	taaatcgag			2679

SEQ ID NO: 14 moltype = DNA length = 1755
 FEATURE Location/Qualifiers
 source 1..1755
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 14

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agtgaaacgg	atccatacaa	gcggaaagca	tgcaagctta	gttgcaagtt	cgttccctgag	180
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ccacacgagg	agagaggag	tagagaaaga	ggacgccagg	aaggtgagaa	ggaggaaaag	300
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SEQ ID NO: 15 moltype = AA length = 584
 FEATURE Location/Qualifiers
 source 1..584
 mol_type = protein
 organism = Glycine max

SEQUENCE: 15

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EQWGRHRHE	DPEERARMRL	REAERIERER	EEQVRREEEE	KLRRREKKEQ	REEKEEGQG	180
SEDSHSKRQN	NPPHFSSNRF	QTLFKNQHG	LRVLQRFDR	SPQLENLRDY	RVVELMAKPN	240
TLFLPHHADA	DFLLLVLSGR	ALINLVKPD	RDYYLDRGY	AQRIPAGTTV	YLVNPKKKD	300
LRVIKLAIPV	NKPGNFEDFF	LSSTQDQSY	LQGFSENILE	ASFNTKFEEI	NRVLFGGEGR	360
RHQEGVILE	LSKEQIRELS	KRAKSSSRST	NSFDYEPFYL	RGSQISSNNF	GKIFYEITPEK	420
NPQLRDFDIL	LNTVDINEGG	LLLPHYNSKA	IVILMVTEGE	ANIELVLKE	QQQGEETREV	480
RKYRAELSED	DIFVIPAAYP	FVVNATSNLN	FVAFGINAEN	NQRNFLAGSK	DNVIRQIQKQ	540
VKELAFPAGS	AQDIENLIKN	QRESYFADAQ	PLQKEEGKKG	SFVF		584

SEQ ID NO: 16 moltype = DNA length = 2787
 FEATURE Location/Qualifiers
 source 1..2787
 mol_type = genomic DNA
 organism = Glycine max

SEQUENCE: 16

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SEQ ID NO: 18 moltype = AA length = 621
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 mol_type = protein
 organism = Glycine max

SEQUENCE: 18

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SEQ ID NO: 19 moltype = DNA length = 5224
 FEATURE Location/Qualifiers
 source 1..5224
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 organism = Glycine max

SEQUENCE: 19

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source             Location/Qualifiers
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SEQ ID NO: 21 moltype = AA length = 560
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 mol_type = protein
 organism = Glycine max

SEQUENCE: 21

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SEQ ID NO: 22 moltype = DNA length = 2725
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SEQUENCE: 22

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acttggatat	cttctcagtt	tctgtggata	tcaacgaagt	aagcataaaa	ttagaaaaaa	1380
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ggagatgcaa	acattgaact	tggtggcatt	aaagaacaac	aacagaagca	gaaacaggaa	1620
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aatggcagag	ttttgatgtg	aaatggccac	tgtaggtgtg	aaggttgatg	taggaattta	2640
ccaagggtag	aaggcaggat	actatgtata	tgggttgga	tatccttgta	cctattttata	2700
aaatatattt	taatatgtgc	tgatg				2725

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SEQ ID NO: 23 moltype = DNA length = 1320
 FEATURE Location/Qualifiers
 source 1..1320
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 23

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agcttccaaa	ctctctttga	gaacccaaa	ggtcgcattc	gtctctcca	gagattcaac	180
aaacgctccc	cacaacttga	gaaccttcga	gactaccgga	ttgtccagtt	tcagtcaaaa	240
cccaacacaa	tccttctccc	ccaccatgct	gacgcccatt	tcctcctctt	tgctcttagc	300
gggagagcca	tacttacctt	ggtgaacaac	gacgacagag	actcctacaa	ccttcaccct	360
ggcgatgcc	agagaatccc	agctggaacc	acttactatt	tggttaaccc	tcacgaccac	420
cagaatctca	aaataatcaa	acttgccata	cccgccaaca	aacctggcag	atatgatgat	480
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gaggagcaga	ggcagcaaga	gggagtgtac	gtggaactct	caaagggaac	aattcggcaa	660
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gaaggagctc	ttcttctacc	acacttcaat	tcaaaggcca	tagtgatact	agtgattaat	900
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gaaggagAAC	ctttggaagt	gcaaaagtac	agagctgaat	tgtctgaaga	cgatgtattt	1020
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gtaaggcaga	tagaaagaca	agtgcaggag	cttgcgttcc	ctgggtctgc	acaagatgtt	1200
gagaggctat	taaagaagca	gagggaaatc	tactttgttg	atgctcagcc	tcagcagaag	1260
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SEQ ID NO: 24 moltype = AA length = 439
 FEATURE Location/Qualifiers
 source 1..439
 mol_type = protein
 organism = Glycine max

SEQUENCE: 24

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KRSPQLENLR	DYRIVQFQSK	PNTILLPHHA	DADFLFLVLS	GRAILTLVNN	DDRDSYNLHP	120
GDAQRIIPAGT	TYIIVNPHDH	QNLKIKLAI	PVNKPGRYDD	FFLSSTQAQQ	SYLQGFSHNI	180
LETSPHSEFE	EINRVLFGE	EEQRQQEGVI	VELSKEQIRQ	LSRRAKSSSR	KTISSEDEPF	240
NLRSRNPIYS	NNPGKFFEIT	PEKNPQLRDL	DIFLSSVDIN	EGALLLPHFN	SKAIVILVIN	300
EGDANIELVG	IKEQQQKQKQ	EEEPLEVQRY	RAELSEDDVF	VIPAAYPFV	NATSNLNFLA	360
FGINAENNQR	NFLAGKDNV	VRQIERQVQE	LAPPGSAQDV	ERLLKKQRES	YFVDAQPQQK	420
EEGSKGRKGP	PPSILGALY					439

SEQ ID NO: 25 moltype = DNA length = 2311
 FEATURE Location/Qualifiers
 source 1..2311
 mol_type = genomic DNA
 organism = Glycine max

SEQUENCE: 25

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ctacttgaga	agctctaaca	gcttccaaac	tctctttgag	aacccaaacg	gtcgcattcg	180
tctctccag	agattcaaca	aacgctcccc	acaacttgag	aaccttcgag	actaccggat	240
tgtccagttt	cagtcacaa	ccaacacaa	ccttctcccc	caccatgctg	acgcccattt	300
cctcctcttt	gtccttagcg	gtttgtattt	actcaaatgt	tgatcagtag	tgtttttagg	360
acattgatta	agaaacccaa	aaaataatta	tttttattga	aacgcataaa	tttatactag	420
cogtgactgt	ttttatgttc	ttatatgata	ttcgcaatat	atattttcta	ttataagttt	480
cttaaccaat	gcactaaact	actgttaaca	agaccttatt	attaaacatc	atctatcact	540
tggttaattg	tattcatttg	atgcattgta	atgcattaca	tatatcacag	gagagccata	600
cttacccttg	tgaacaacga	cgacagagac	tcctacaacc	ttcaccctgg	cgatgcccag	660
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tcgaggagat	aaacaggggt	ttgttgggag	aggaagaggga	gcagaggcag	caagaggggag	1140
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caaggaaaaa	catttctctc	gaagatgaac	cattcaactt	gagaagccgc	aaccccatct	1260
attccaacaa	ctttggaag	ttctttgaga	tcacccctga	gaaaaaccca	cagcttcggg	1320
acttgatata	cttctcagtc	tctgtggata	tcaacgaagt	aagcataaaa	tttaagaaaa	1380
ccatatactc	aaactcgtga	actggacaat	tcagttctat	tcactctctc	cttagtttagc	1440
gaacctaat	attcagttct	atcaacaaat	aatgatttct	tctttgtgtc	acaaatatag	1500

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ggagctcttc ttctaccaca cttcaattca aaggccatag tgatactagt gattaatgaa 1560
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gaggaaacct tggaaagtca aaggtacaga gctgaattgt ctgaagacga tgtatttcta 1680
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caggagcttg cgttccctgg gtctgcacaa gatgttgaga ggctattaaa gaagcagagg 1980
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agtaatttac ccgctcaagc taaattatct g 2311

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SEQ ID NO: 26      moltype = DNA length = 1320
FEATURE
source            Location/Qualifiers
                  1..1320
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 26
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agcttccaaa ctctctttga gaacccaaac ggctgcattc gtctcctcca gagattcaac 180
aaacgctccc cacaacttga gaaccttcga gactaccgga ttgtccagtt tcagtcaaaa 240
cccaacacaa tctctctccc ccaccatgct gacgcccatt tctcctctt ttgctctagc 300
gggagagcca tacttacctt ggtgaacaac gacgcagag actcctacaa ccttcaccct 360
ggcgatgccc agagaatccc agctggaacc acttactatt tggtaaccc tcacgaccac 420
cagaatctca aataatcaaa acttgccata cccgtcaaca aacctggcag atatgatgat 480
ttctcttata cttagcactca agcccaacag tctactctgc aaggcttcag ccataatatt 540
ctagagacct ccttccatag cgaattcgag gagataaaca gggttttgtt gggagaggaa 600
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gagaggctat taaagaagca gaggggaatc tactttgttg atgctcagcc tcagcagaag 1260
gaggagggga gtaagggaag aaagggtcct ttctcttcaa tcttaggtgc tctctactga 1320

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SEQ ID NO: 27      moltype = AA length = 439
FEATURE
source            Location/Qualifiers
                  1..439
                  mol_type = protein
                  organism = Glycine max

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SEQUENCE: 27
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KRSPQLENLR DYRIVQFQSK PNTILLPHHA DADFLFLVLS GRAILTLVNN DDRDSYNLHP 120
GDAQRIPIAGT TYLVNPHDH QNLKIIKLAI PVNKPGRYDD FFLSSTQAQQ SYLQGFHSHNI 180
LETSPHSEFE EINRVLLGEE EEQRQEGVI VELSKQIRQ LSRRAKSSSR KTISSEDEPF 240
NLRSRNPIYS NNFQKFFFEIT PEKNPQLRDL DIFLSSVDIN EGALLLPHFN SKAIVILVIN 300
EGDANIELVG IKEQQQKQKQ EEEPLEVQRY RAELSEDDVF VIPAAYPFV NATSNLNLFLA 360
FGINAENNQR NFLAGEDNV VRQIERQVQE LAFPGSAQDV ERLKKQRES YFVDAQPPQK 420
EEGSKGRKGP FPSILGALY 439

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SEQ ID NO: 28      moltype = DNA length = 4137
FEATURE
source            Location/Qualifiers
                  1..4137
                  mol_type = genomic DNA
                  organism = Glycine max

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SEQUENCE: 28
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caagagttag tttgtgttgc tctactgacc tcaccatggc gaccagagcc aagctttctt 240
tagctatctt ccttttcttt cttttagcct tgatttcaaa cctagccttg ggcaaaccta 300
aagaaaccca ggtcgaagaa gatcccagc tcgtaacatg caaacaccag tgccaacagc 360
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ccagagtcga aacagaaggt ggcagcattc ggggtctcaa gaagttcact gagaaatcca 600
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tcgtgtcccc acgccacttt gattccgagg ttgtcttgggt caacattaag ggtgcgtata 720
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tagtgattac actctaattt tatatcatta tccaatcata cattaatatt tatatgattt 840
ttagaataat tattTTTTTT aatgaaactt tatgataggt tttgtctta gatgacaata 900
tatcaattaa tttattgatg cctTTTTTTT tattaatttg ttggttattc acacatctac 960
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SEQ ID NO: 29      moltype = DNA length = 1521
FEATURE            Location/Qualifiers
source              1..1521
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 29
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gccttgggca aacttaaa gaaccagggtc gaagaagatc ccgagctcgt aacatgcaaa 180
caccagtgcc aacagcaacg gcaatacact gagagtgaca agcgaacatg cttgcaacaa 240
tgtgacagta tgaagcaaga gcgagagaaa caagtcgaa aggaaactcg cgagaaggaa 300
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ttcactgaga aatccaagct tcttcaaggc attgagaatt tccgtttggc catcttagaa 480
gctagagcac acacgttcgt gtccccacgc cactttgatt ccgaggttgt cttgttcaac 540
atgaaggga gagccgtact tgggttgggt agggaaagt aaacagaaaa aatcacccca 600
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SEQ ID NO: 30 moltype = AA length = 506
 FEATURE Location/Qualifiers
 source 1..506
 mol_type = protein
 organism = Glycine max

SEQUENCE: 30
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 IKGRAVLGLV RESETEKITL EPGDmiHIPA GTPLYIVNRD ENKLLLLAML HIPVSTPGKF 240
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SEQ ID NO: 31 moltype = DNA length = 63
 FEATURE Location/Qualifiers
 source 1..63
 mol_type = genomic DNA
 organism = Arabidopsis thaliana

SEQUENCE: 31
 agaaactgta gcaacatcgg tgttgacacag atcgtggcgg ctaagtggtc caacaaccca 60
 tcc 63

SEQ ID NO: 32 moltype = AA length = 21
 FEATURE Location/Qualifiers
 source 1..21
 mol_type = protein
 organism = Arabidopsis thaliana

SEQUENCE: 32
 RNCSNIGVAQ IVAAKWSNNP S 21

SEQ ID NO: 33 moltype = DNA length = 63
 FEATURE Location/Qualifiers
 source 1..63
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 33
 agaaactgta gcaacatcag tgttgacacag atcgtggcgg ctaagtggtc caacaaccca 60
 tcc 63

SEQ ID NO: 34 moltype = AA length = 21
 FEATURE Location/Qualifiers
 source 1..21
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 34
 RNCSNISVAQ IVAAKWSNNP S 21

SEQ ID NO: 35 moltype = DNA length = 24
 FEATURE Location/Qualifiers
 source 1..24
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 35
 agaaactgta gcaacaaccc atcc 24

SEQ ID NO: 36 moltype = DNA length = 21
 FEATURE Location/Qualifiers

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source                1..21
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 36
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SEQ ID NO: 37        moltype = DNA length = 50
FEATURE              Location/Qualifiers
source               1..50
                      mol_type = genomic DNA
                      organism = Glycine max

SEQUENCE: 37
agcaccaagg cgcgccgcaa ctgcagcaac atcggcgctcg cgcaaatcgt    50

SEQ ID NO: 38        moltype = DNA length = 50
FEATURE              Location/Qualifiers
source               1..50
                      mol_type = genomic DNA
                      organism = Glycine max

SEQUENCE: 38
agcaccaagg cgcgccgcaa ttgtagcaac atcggcgctcg cgcaaattgt    50

SEQ ID NO: 39        moltype = DNA length = 22
FEATURE              Location/Qualifiers
source               1..22
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 39
acgtgccaaa tctagttcaa gg                                22

SEQ ID NO: 40        moltype = DNA length = 22
FEATURE              Location/Qualifiers
source               1..22
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 40
ccagaggaac ttccttgca g t                                22

SEQ ID NO: 41        moltype = DNA length = 22
FEATURE              Location/Qualifiers
source               1..22
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 41
catgcaacga agatagcact gg                                22

SEQ ID NO: 42        moltype = DNA length = 22
FEATURE              Location/Qualifiers
source               1..22
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 42
cctggaatgg tgttgttgt cc                                22

SEQ ID NO: 43        moltype = DNA length = 38
FEATURE              Location/Qualifiers
source               1..38
                      mol_type = genomic DNA
                      organism = Glycine max

SEQUENCE: 43
cagctaagca ccaaggcgcg ccgcaactgc agcaacat              38

SEQ ID NO: 44        moltype = DNA length = 37
FEATURE              Location/Qualifiers
source               1..37
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 44
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SEQ ID NO: 45        moltype = DNA length = 34
FEATURE              Location/Qualifiers
source               1..34
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 45

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SEQ ID NO: 46 moltype = DNA length = 35 FEATURE Location/Qualifiers source 1..35 mol_type = other DNA organism = synthetic construct	
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SEQ ID NO: 47 moltype = DNA length = 38 FEATURE Location/Qualifiers source 1..38 mol_type = genomic DNA organism = Glycine max	
SEQUENCE: 47	
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SEQ ID NO: 48 moltype = DNA length = 11 FEATURE Location/Qualifiers source 1..11 mol_type = other DNA organism = synthetic construct	
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SEQUENCE: 49	
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SEQ ID NO: 50 moltype = DNA length = 40 FEATURE Location/Qualifiers source 1..40 mol_type = other DNA organism = synthetic construct	
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SEQ ID NO: 51 moltype = DNA length = 39 FEATURE Location/Qualifiers source 1..39 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 51	
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SEQ ID NO: 52 moltype = DNA length = 4990 FEATURE Location/Qualifiers source 1..4990 mol_type = other DNA organism = synthetic construct	
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FEATURE            Location/Qualifiers
source              1..4987
                    mol_type = other DNA
                    organism = synthetic construct

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SEQ ID NO: 54 moltype = DNA length = 4988
 FEATURE Location/Qualifiers
 source 1..4988
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 54

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SEQ ID NO: 55 moltype = AA length = 535
 FEATURE Location/Qualifiers
 source 1..535
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 55

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SEQ ID NO: 56 moltype = DNA length = 5108
 FEATURE Location/Qualifiers
 source 1..5108
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 56

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FEATURE           Location/Qualifiers
source            1..526
                  mol_type = protein
                  organism = synthetic construct

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PGLPSHPHEH LAKRQMTGFG GVSFEIDGD LHHTIKFIDS LKIPYIAASF GGCEIVDQP 480
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SEQ ID NO: 58      moltype = DNA length = 5123
FEATURE           Location/Qualifiers
source            1..5123
                  mol_type = other DNA
                  organism = synthetic construct

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 organism = synthetic construct

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SEQ ID NO: 61      moltype = DNA length = 5136
FEATURE            Location/Qualifiers
source              1..5136
                    mol_type = other DNA
                    organism = synthetic construct

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Location/Qualifiers

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FEATURE Location/Qualifiers
source 1..4134
 mol_type = other DNA
 organism = synthetic construct

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SEQ ID NO: 70 moltype = DNA length = 4135
FEATURE Location/Qualifiers
source 1..4135
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 70

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SEQ ID NO: 71 moltype = DNA length = 4132
 FEATURE Location/Qualifiers
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 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 71

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SEQ ID NO: 72 moltype = DNA length = 4136
FEATURE Location/Qualifiers
source 1..4136
 mol_type = other DNA
 organism = synthetic construct

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SEQ ID NO: 73 moltype = DNA length = 4123
FEATURE Location/Qualifiers
source 1..4123
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 73

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SEQ ID NO: 74 moltype = DNA length = 4136
FEATURE Location/Qualifiers
source 1..4136
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 74

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caaagactca	acctcatgct	tacctttacc	aacatcacc	aggtatatat	attaccaaca	2280
tgcttatctt	aatttttttt	acttgatagt	cataaggaaa	agattgtgag	tttgatttca	2340
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SEQ ID NO: 75 moltype = DNA length = 4133
FEATURE Location/Qualifiers
source 1..4133
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 75

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SEQ ID NO: 76 moltype = DNA length = 2240
FEATURE Location/Qualifiers
source 1..2240
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 76

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SEQ ID NO: 77 moltype = DNA length = 2781

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FEATURE	Location/Qualifiers					
source	1..2781					
	mol_type = other DNA					
	organism = synthetic construct					
SEQUENCE: 77						
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SEQ ID NO: 79 moltype = DNA length = 2781
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 source 1..2781
 mol_type = other DNA
 organism = synthetic construct

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FEATURE           Location/Qualifiers
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                  mol_type = other DNA
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FEATURE           Location/Qualifiers
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SEQ ID NO: 83      moltype = DNA length = 2145
FEATURE           Location/Qualifiers
source            1..2145
                  mol_type = other DNA
                  organism = synthetic construct

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cgagcgcaac attattcgag tacgtagctt aacaatatTT taagcaattg ctgctacaat 240
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SEQ ID NO: 84      moltype = DNA length = 2677
FEATURE           Location/Qualifiers
source            1..2677
                  mol_type = other DNA
                  organism = synthetic construct

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cgagcgcaac attattcgag tacgtagctt aacaatatTT taagcaattg ctgctacaat 240
gatagcgcggt tttcctttgt tgcggttgc gttgctggga attgtttttc tagcatcagt 300
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acaagacgag cgtgaattac aaccacgtcc atttccacga ccagaggagc atgaaggaga 600
acagtggaaa ggaaggtcat gtcattgaaga cccggaagag agggcgagaa tgagactgag 660
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SEQ ID NO: 85 moltype = DNA length = 2158
 FEATURE Location/Qualifiers
 source 1..2158
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 85

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SEQ ID NO: 86      moltype = DNA length = 2721
FEATURE           Location/Qualifiers
source            1..2721
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 86
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SEQ ID NO: 87      moltype = DNA length = 2720
FEATURE           Location/Qualifiers
source            1..2720
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 87
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SEQ ID NO: 88      moltype = DNA length = 2718
FEATURE           Location/Qualifiers
source            1..2718
                  mol_type = other DNA
                  organism = synthetic construct

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 organism = synthetic construct

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 organism = synthetic construct

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 organism = synthetic construct

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SEQ ID NO: 93 moltype = DNA length = 1923
 FEATURE Location/Qualifiers
 source 1..1923
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 93

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tgtccagttt	cagtcacaaac	ccaacacaa	ccttctcccc	caccatgctg	acgccgattt	300
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acattgatta	agaaacccaa	aaaataatta	tttttattga	aacgcataaa	tttatactag	420
cgtgactgt	ttttatgtcc	ttatatgac	ttcgcaatat	atattttcta	ttataagttt	480
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SEQ ID NO: 94      moltype = DNA length = 2301
FEATURE           Location/Qualifiers
source            1..2301
                  mol_type = other DNA
                  organism = synthetic construct

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cctctctctt gtccttagcg gtttgtattt actcaaatgt tgatcagtag tgtttttagg 360
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ccgtgactgt ttttatgtcc ttatatgatc ttgcgaatat atattttcta ttataagttt 480
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SEQ ID NO: 95      moltype = DNA length = 2299
FEATURE           Location/Qualifiers
source            1..2299
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 95

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SEQ ID NO: 96

moltype = DNA length = 1160

FEATURE

Location/Qualifiers

source

1..1160

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 96

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SEQ ID NO: 97

moltype = DNA length = 2303

FEATURE

Location/Qualifiers

source

1..2303

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 97

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SEQ ID NO: 98      moltype = DNA length = 2951
FEATURE            Location/Qualifiers
source              1..2951
                    mol_type = other DNA
                    organism = synthetic construct

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 organism = synthetic construct

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                    mol_type = other DNA
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SEQ ID NO: 103 moltype = DNA length = 3034
 FEATURE Location/Qualifiers
 source 1..3034
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 103

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SEQ ID NO: 104 moltype = DNA length = 2814
 FEATURE Location/Qualifiers
 source 1..2814
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 104

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SEQ ID NO: 105 moltype = DNA length = 3100
 FEATURE Location/Qualifiers
 source 1..3100
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 105

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SEQ ID NO: 106      moltype = DNA length = 3099
FEATURE             Location/Qualifiers
source               1..3099
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 106
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 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 107

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SEQUENCE: 120
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SEQ ID NO: 121 moltype = DNA length = 24
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SEQUENCE: 121
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SEQ ID NO: 122 moltype = DNA length = 24
FEATURE Location/Qualifiers
source 1..24
 mol_type = other DNA
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SEQUENCE: 122
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SEQ ID NO: 123 moltype = DNA length = 23
FEATURE Location/Qualifiers
source 1..23
 mol_type = other DNA
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SEQUENCE: 123
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SEQ ID NO: 124 moltype = DNA length = 22
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 organism = synthetic construct

SEQUENCE: 124
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SEQ ID NO: 125 moltype = DNA length = 23
FEATURE Location/Qualifiers
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SEQUENCE: 125
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SEQUENCE: 126
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SEQ ID NO: 127 moltype = DNA length = 24
FEATURE Location/Qualifiers
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 organism = synthetic construct

SEQUENCE: 127
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SEQ ID NO: 128 moltype = DNA length = 23
FEATURE Location/Qualifiers
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SEQUENCE: 128
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SEQ ID NO: 129 moltype = DNA length = 24

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	mol_type = other DNA	
	organism = synthetic construct	
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	mol_type = other DNA	
	organism = synthetic construct	
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SEQUENCE: 134		
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What is claimed is:

1. A plant, plant part, or seed comprising one or more modifications that increase seed protein content, the modification selected from a deletion, insertion, or substitution of a nucleotide in (i) a genomic sequence encoding a cystathionine gamma-synthase (CGS), the modification resulting in suppression of the feedback regulation of the same while retaining enzymatic activity, and/or (ii) a genomic sequence encoding a low methionine content seed storage protein, the modification resulting in suppression of the activity of the same.

2. The plant, plant part, or seed of claim 1, wherein the modification comprises a deletion, insertion, or substitution of a nucleotide at or near the MTO1 region of the genomic sequence encoding the CGS.

3. The plant, plant part, or seed of claim 1, wherein the modification is an in-frame modification of the genomic sequence encoding the CGS.

4. The plant, plant part, or seed of claim 1, wherein the deletion or insertion results in a frameshift mutation of the genomic sequence encoding the low methionine content seed storage protein.

5. The plant, plant part, or seed of claim 1, wherein the plant, plant part, or seed comprises a modification selected from a deletion, insertion, or substitution of a nucleotide in a genomic sequence encoding a 7S storage protein.

6. The plant, plant part, or seed of claim 5, wherein the plant, plant part, or seed comprises a modification selected from a deletion, insertion, or substitution of a nucleotide in a genomic sequence encoding an alpha', alpha, and beta subunit of the 7S storage protein.

7. The plant, plant part, or seed of claim 1, further comprising a heterologous nucleic acid sequence selected from a reporter gene, a selection marker, a disease resistance gene, a herbicide resistance gene, an insect resistance gene, a gene involved in carbohydrate metabolism, a gene involved in fatty acid metabolism, a gene involved in amino acid metabolism, a gene involved in plant development, a gene involved in plant growth regulation, a gene involved in yield improvement, a gene involved in drought resistance, a gene involved in increasing nutrient utilization efficiency, a gene involved in cold resistance, a gene involved in salt resistance, and a gene involved in resistance in plants.

8. The plant, plant part, or seed of claim 1, wherein protein content of the seed is increased relative to a control seed not comprising the modification.

9. The plant, plant part, or seed of claim 1, wherein methionine content of the seed is increased relative to a control seed not comprising the modification.

10. The plant, plant part, or seed of claim 1, wherein proteome rebalancing results in an increase in high methionine seed storage proteins in the seed.

11. The plant, plant part, or seed of claim 1, wherein 11S glycinin polypeptide content of the seed is increased relative to a control seed not comprising the modification as a result of proteome rebalancing.

12. The plant, plant part, or seed of claim 1, wherein the modification is homozygous.

13. The plant, plant part, or seed of claim 1, wherein the genomic sequence encoding the CGS comprises a nucleotide sequence having at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% sequence identity to SEQ ID NO: 1 or 4.

14. The plant, plant part, or seed of claim 1, wherein the genomic sequence encoding the low methionine content seed storage protein comprises a nucleotide sequence having at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% sequence identity to SEQ ID NO: 7, 10, 13, 16, 19, 22, 25, or 28.

15. A method of making a modified plant with high seed protein content compared to the seed protein content of a nonmodified plant, the method comprising: (a) crossing a first plant comprising a deletion, insertion, or substitution of a nucleotide in (i) a genomic sequence encoding cystathionine gamma-synthase (CGS), the modification resulting in suppression of the feedback regulation of the same while retaining enzymatic activity, and/or (ii) a genomic sequence encoding a low methionine content seed storage protein, the modification resulting in suppression of the activity of the same, with a second plant; and (b) selecting a progeny plant, or plant part thereof, with high protein content for further breeding.

16. The method of claim 15, wherein the modified plant is an F1 progeny plant.

17. A method of producing high protein seed meal comprising: harvesting the seed of claim 1 from a plant comprising the modification and processing the seed to form meal.

18. A modified CGS polynucleotide comprising an in-frame insertion or deletion in the MTO1 region so that feedback inhibition of expression is suppressed.

19. The modified CGS polynucleotide of claim 18, wherein the polynucleotide comprises the sequence of SEQ ID NO: 44, 45, 46, 48, 49, 50, or 51.

20. The modified CGS polynucleotide of claim 18, wherein the polynucleotide comprises a sequence having at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% sequence identity to SEQ ID NO: 52, 53, 54, 56, 58, 60, or 61.

21. A modified 7S polynucleotide comprising an insertion, deletion, or substitution that suppresses activity of the polynucleotide.

22. The modified 7S polynucleotide of claim 21, wherein the polynucleotide comprises a sequence having at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% sequence identity to one or more of SEQ ID NOs: 62-108.

23. A plant cell comprising the polynucleotide of claim 18 or claim 21 or both.

24. The plant cell of claim 23, wherein the plant cell is a soybean plant cell.

25. A plant comprising the plant cell of claim 23.

26. A seed of the plant of claim 25, wherein the seed has increased protein content compared to a seed comprising a comparable polynucleotide without the modification.

27. A seed of the plant of claim 25, wherein the seed has increased methionine content compared to a seed comprising a comparable polynucleotide without the modification.

28. A modified polypeptide encoded by the polynucleotide of claim 18 or claim 21.

29. A method of producing a plant cell having increased protein content, the method comprising: introducing a double-strand-break-inducing (DSB-inducing) agent to a plant cell, wherein the DSB-inducing agent creates a double-strand break in an endogenous CGS gene and/or a low methionine content seed storage protein gene in the genome of the plant, wherein the repair of the DSB results in the production of a modified polynucleotide, and wherein the production of the modified polynucleotide produces increased protein content in the seed.

30. The method of claim 29, wherein the modified polynucleotide includes a deletion, insertion, or substitution in the MTO1 region of the CGS gene.

31. The method of claim 30, wherein the deletion or insertion is an in-frame deletion or insertion.

32. The method of claim 29, wherein the modified polynucleotide is a deletion, insertion, or substitution of a 7S beta-conglycinin.

33. The method of claim 29, wherein the endogenous CGS gene comprises a nucleotide sequence having at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% sequence identity to SEQ ID NO: 1 or 4.

34. The method of claim 29, wherein the endogenous low methionine content seed storage protein gene comprises a nucleotide sequence having at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% sequence identity to SEQ ID NO: 7, 10, 13, 16, 19, 22, 25, or 28.

35. A method of editing the genome of a plant cell to produce a plant cell having increased protein, the method comprising: introducing a guide RNA, a polynucleotide modification template, and a Cas endonuclease to a plant cell; and regenerating a plant from the plant cell wherein the genome of the regenerated plant comprises the modified polynucleotide of claim 18 or claim 21 or both.

36. The method of claim 35, wherein the seed of the plant has increased protein content compared with a comparable seed not comprising the modified polynucleotide.

37. The method of claim 35, wherein the seed of the plant has increased methionine content compared with a comparable seed not comprising the modified polynucleotide.

38. The method of claim 35, wherein the genome is modified by a site-directed endonuclease.

39. The method of claim 35, wherein the plant cell is a soybean cell.

40. A method of producing a plant cell having increased seed protein content, the method comprising: transforming a plant cell with the modified polynucleotide of claim 18 or claim 21 or both, wherein a plant regenerated from the transformed plant cell has increased seed protein content compared with a comparable plant cell not comprising the modified polynucleotide.

41. A polynucleotide construct comprising a sequence encoding a guide RNA which targets an endogenous CGS

gene and/or a low methionine content storage polypeptide gene when expressed in a plant cell and thereby generates a modified endogenous CGS gene and/or a low methionine content storage polypeptide gene.

42. The polynucleotide construct of claim **41**, wherein the guide RNA comprises the spacer sequence of SEQ ID NO: 109, 110, 111, 112, 113, or 114.

43. A plant cell comprising the polynucleotide construct of claim **41**.

44. The plant cell of claim **43**, wherein the plant cell is a soybean cell.

45. A method of producing a soybean seed comprising: a) sexually crossing a first soybean line comprising the polynucleotide of claim **18** or claim **21** or both with a second soybean line not comprising the polynucleotide; and b) harvesting the seed produced thereby.

46. The method of claim **45** further comprising the step of backcrossing a second generation progeny plant that comprises the polynucleotide to the parent plant that lacks the polynucleotide, thereby producing a backcross progeny plant that produces seed with increased protein content.

47. A method of screening for the presence or absence of the polynucleotide of claim **18** or claim **21** or both in a

plurality of genomic soybean DNA samples, the method comprising the steps of (a) contacting a plurality of genomic soybean DNA samples, at least some of which comprise the polynucleotides, with a first DNA primer molecule and a second DNA primer molecule; (b) providing a plurality nucleic acid amplification reaction conditions; (c) performing the nucleic acid amplification reactions, thereby producing a DNA amplicon molecule indicating the presence of the polynucleotide or a wild-type CGS and/or a low methionine content storage polypeptide gene nucleotide sequence; and (d) detecting the DNA amplicon molecules, wherein the presence, absence or size of the DNA amplicon molecule indicates the presence or absence of the polynucleotide in the at least one of the plurality of genomic soybean DNA samples.

48. The method of claim **47**, wherein the first DNA primer molecule or the second DNA primer molecule comprises the sequence of one or more of SEQ ID NOs: 115-134.

49. An animal feed comprising the plant, plant part, or seed of claim **1**.

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